Skills-2.1 Web Development

HTML / CSS / JavaScript / Backend / React

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1_HTML

1_Web

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Mine</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <h1>Photonics</h1>
        <h2>Optical fibre</h2>
            <h3>Numerical Aperture</h3>
                <p>Lorem ipsum dolor sit, amet consectetur adipisicing elit. Ab, consequuntur?</p>

             <h3>V- number</h3>
                <p>Lorem ipsum dolor sit, amet consectetur adipisicing elit. Voluptatem, dignissimos quas? Alias possimus, aspernatur, neque cupiditate praesentium corporis nostrum commodi assumenda vel, debitis modi. Possimus numquam provident vero illum perspiciatis!
                 </p>
    <h1>Links</h1>
        <p><a href="https://google.com">Open google</a></p>

    <h1>Images</h1>
    <img width="230" height="" src="image1.png" alt="Hack image">
    <br> <!--  brings to next line -->
    <img src="if_not_uploaded_properly" alt="This_will_be_shown image">

    <h1>Tables</h1>
        <p>
            <table>

                <thead><!--thead - header, tbody and tfoot- footer , it is optional but makes easy while styling in css-->
                    <caption><h2>Sem I result</h2></caption>
                    <tr>
                        <th>Subject</th>
                        <th>Grade</th>
                        <th>Credits</th>
                        <th>Marks</th>
                        <th>Grade points</th>
                    </tr>
                </thead>

                <tbody>
                    <tr>
                        <td>Maths</td>
                        <td>87</td>
                        <td rowspan="3">4</td><!--Rowspan combines the number of rows mentioned-->
                        <td>9</td>
                        <td>36</td>
                    </tr>
                    <tr>
                        <td>Chemistry</td>
                        <td>75</td>
                        <td>8</td>
                        <td>32</td>
                    </tr>
                    <tr>
                        <td>Phython</td>
                        <td>75</td>
                        <td>8</td>
                        <td>32</td>
                    </tr>
                    <tr>
                        <td>Ai</td>
                        <td>73</td>
                        <td rowspan="2">3</td>
                        <td>8</td>
                        <td>24</td>
```

```
                    </tr>
                    <tr>
                        <td>Electrical</td>
                        <td>90</td>
                        <td>10</td>
                        <td>30</td>
                    </tr>
                    <tr>
                        <td>English</td>
                        <td>97</td>
                        <td rowspan="2">1</td>
                        <td colspan="2">10</td>
                    </tr>
                    <tr>
                        <td>IDEA</td>
                        <td>96</td>
                        <td colspan="2">10</td>
                    </tr>
                </tbody>
                <tfoot>
                    <tr>
                        <td>Total</td>
                        <td>-</td>
                        <td>20</td>
                        <td>-</td>
                        <td>174</td>
                    </tr>
                </tfoot>
                </table>
            </p>


        <h1>Lists</h1>
            <h2>Unordered Lists</h2>
                <ul><!--Default type = disc-->
                    <li>Maths</li>
                    <li>Chemistry</li>
                    <li>Phython</li>
                    <li>Ai</li>
                    <li>Electrical</li>
                    <li>English</li>
                    <li>IDEA</li>
                </ul>
                <ul type="Square">
                    <li>Square</li>
                </ul>
                <ul type="Circle">
                    <li>circle</li>
                </ul>

            <h2>Ordered Lists</h2>
                <ol><!--Default type = 1-->
                    <li>Maths</li>
                    <li>Chemistry</li>
                    <li>Phython</li>
                    <li>Ai</li>
                    <li>Electrical</li>
                    <li>English</li>
                    <li>IDEA</li>
                </ol>
                <ol type="i">
                    <li>Roman small</li>
                </ol>
                <ol type="I">
                    <li>Roman capital</li>
                </ol>
                <ol type="A">
                    <li>capital letter</li>
                </ol>
                <ol type="a">
                    <li>small letter</li>
                </ol>

            <h2>Definition Lists</h2>
                <dl>
                    <dt>Html</dt><!--Definition Title-->
                    <dd>Skeleton of a webpage</dd><!--Definition Data-->

```

```
                    <dt>CSS</dt>
                    <dd>For Styling</dd>
                </dl>



    </body>
    </html>
```

script.js

```
    (empty file)
```

style.css

```
body {
    background-color: rgba(28, 121, 118, 0.504);
    color: rgb(50, 17, 73);
}

th{
    border: 3px solid ;
}

td{
    border: 2px solid black;
}
```

2_Forms

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Forms</title>
</head>
<body>
    <h1>Forms</h1>

    <form action="post">
        <div>
            <label for="un">Username</label>
            <input type="text" id="un" name="username" placeholder="Enter Your Username" required autofo
              cus><!--if there is required then the form will not be submitted until it is filled-->

            <!--When there is autofocus it will be automatically typed in the box whenever you type afte
              r a reload-->
        </div>

        <div>
            <label for="m">Male</label>
            <input type="radio" id="m" name="gender" value="male">

            <label for="f">Female</label>
            <input type="radio" id="f" name="gender" value="female">
        </div>
        <div>
            <input type="checkbox" id="criteria1" name="criteria1" value="yes">
            <label for="criteria1">1st Criteria</label>

            <input type="checkbox" id="criteria2" name="criteria2" value="yes">
            <label for="criteria2">2nd Criteria</label>

            <input type="checkbox" id="criteria3" name="criteria3" value="yes">
            <label for="criteria3">3rd Criteria</label>
        </div>

        <div>
            <label for="comment">Enter your comment</label><br>
            <textarea name="comment" id="comment" rows="10" cols="70"></textarea>
        </div>

        <div>
            <label for="Fruits">Select the Fruits</label>
            <select name="Fruits" id="Fruits">
                <option value="Apple">Apple</option>
                <option value="Banana">Banana</option>
                <option value="Coconut">Coconut</option>
            </select>
        </div>
    </form>

</body>
</html>
```

3_Inline and Block

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Inline and Block</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <p>A block element is the one which takes the entire width</p>
    <p>Which means that if the para is cooured then it will be coloured for the entire width</p>

    <a href="https://google.com">Google</a><!--Anchor is a inline element which means that it takes spac
      e only how much it is there-->

    <!--A block element will always come in a new line as it requires etire width-->
    <div>It is a block element</div>

    <!--A inline element continues will come in the same line if the previous element wasnt a block elme
      nt-->
    <a href="https://youtube.com">YouTube</a>
    <span>It is a inline elment</span>
    <a href="">Demo</a>

    <!--Inline and block elements can be interchanged in css-->
</body>
</html>
```

style.css

```css
p{
    background-color: bisque;
}

a{
    background-color: aquamarine;
}

div{
    background-color: brown;
}

span{
    background-color: blueviolet;
}
```

4_ID and classes

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>ID amd Classes</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div id="odiv" class="red bg-yellow"><!--an element can have only one id but can have multiple classes like here 1div is in both red and bg-yellow-->
    <!--A class can also have multiple elements-->
        First
    </div>
    <div id="sdiv">
        Second
    </div>
    <span class="red bg-yellow">
        Span
    </span>
</body>
</html>

<!--id amd classes are attributes of the element
using id of a particular division and typing it as "link_of_main_webpage#id" typing this will directly l
  and you on that particular division in the webpage-->
```

style.css

```css
.red{
    color: red;
}

.bg-yellow{ /* classes are accessed using '.' and id using '#' .and # are selectors in css*/
    background-color: yellow;
}

/* #2div{
    background-color: aqua;
    color: blueviolet;
} ids having numbers in their beginning cannot be styled here */

#sdiv{
    background-color: aqua;
    color: blueviolet;
}

#odiv{
    background-color: beige;
}
```

5_VIdeo_images

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Video and Audio</title>
</head>
<body>
    <h1>Videos</h1>
    <ol type="i">
        <li><video src="" alt="v1" controls autoplay loop muted poster="poster.png" height="196"></video
          ></li>
        <li><video src="" alt="v2" controls ></video></li>
        <li><video src="" alt="v3" controls ></video></li>
        <li><video src="" alt="v4" controls ></video></li>
        <li><video src="" alt="v5" controls ></video></li>
        <li><video src="" alt="v6" controls ></video></li>
    </ol>

    <h1>Audios</h1>
    <ol>
        <li><audio src="" alt="a1" controls preload="auto"></audio></li><!--preload attribute its defaul
          t value is none it can be set to metadata or auto-->
        <li><audio src="" alt="a2" controls></audio></li>
        <li><audio src="" alt="a3" controls></audio></li>
        <li><audio src="" alt="a4" controls></audio></li>
        <li><audio src="" alt="a5" controls></audio></li>
        <li><audio src="" alt="a6" controls></audio></li>
    </ol>

    <svg height="200" width="200">
        <circle cx="90" cy="100" stroke="black" stroke-width="5" r="30" fill="red"/>
    </svg>

    <svg height="200" width="200">
        <circle cx="90" cy="100" stroke="black" stroke-width="5" r="50" fill="blue"/>
     </svg>

    <!--Video audio images and svgs are inline elements-->

    <iframe src="https://www.amazon.in" height="200" width="300"></iframe>

    <iframe width="560" height="315" src="https://www.youtube.com/embed/XZwBNDGuWGU?si=oZU-F0TDcIUGRuLb"
       title="YouTube video player" frameborder="0" allow="accelerometer; autoplay; clipboard-write; enc
      rypted-media; gyroscope; picture-in-picture; web-share" referrerpolicy="strict-origin-when-cross-o
      rigin" allowfullscreen></iframe>
</body>
</html>
```

6_HTML_Entities

index.html

```
<!--if you want to display some symbols which collide with the syntax of html the entities are used-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML Entities</title>
</head>
<body>
    <!--if you want to display <p><\p> as it is then &lt; and &gt; are used-->
    &lt;p&gt; this is a para &lt;\p&gt;

    <pre>
        Everything written within this will appear as it
        is including



        sapaces                    and



        newlines
    </pre>

    <h1>Others</h1>
    <ul>
        <li>&amp;</li>
        <li>&copy;</li>
    </ul>

    <h1>Spaces</h1>
    <p>When you give too many spaces                    html considers only one apaces and ignores rest</p>
    <p>So When you want many spaces use                How many times it is used those many times spaces will be printed</p>

    <h1>Quotation</h1>
    <blockquote cite="https://google.com">google</blockquote>
    <q cite="https://google.com">google</q>

    <h1>Code</h1>
    <pre><code>
&lt;!DOCTYPE html&gt;
&lt;html lang="en"&gt;
&lt;head&gt;
    &lt;meta charset="UTF-8"&gt;
    &lt;meta name="viewport" content="width=device-width, initial-scale=1.0"&gt;
    &lt;title>Document$lt;/title&gt;
&lt;/head&gt;
&lt;body&gt;

&lt;/body&gt;
&lt;/html&gt;

    </code></pre>
</html>
```

2_CSS

1_Selectors

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Css</title>
    <style>/* One way of adding css to html file is using style tag */

        /* element selector */

        div{ /* Herer div is the selector */
            color: red;/*This is the declaration*/

            /*colour is the key / property and red is the value*/

            /*this will change the colours of all the divs in the page*/
        }
        /* you can consider multiple selectors at a time*/

        /* Class selector */
        .green{
            color: brown;
            background-color: rgba(61, 182, 126, 0.937);
        }

        /* ID selector*/
        #para{
            color: white;
            background-color: black;
        }

        /* Child Selector */
        div > p{
            background-color: beige;
        }
        div>article>p{
            color:blue;
        }

        /* Descendent Selector */
        div p{
            color: rgb(199, 27, 110);
            background-color: rgb(241, 80, 12);
        }

        /* Universal Selector */
        *{
            margin: 0;
            padding: 0;
        }

        /* Pseudo Selector */
        a:visited{
            color: yellow;
        } /*Once you click on the link and visit the website the link color changes to yellow*/
        a:link{
            color:green;
        }/*If the lin not yet visited then it remains green*/
        a:hover{
            background-color: aquamarine;
        }/*When you move the cursor on to the link*/
        a:active{
            background-color: crimson;
        }/*When you click on the link the bg turns crimpson
        if at all this was above the hover then it would not be applied if clicked and there are some ot
          her elements also like if the decendent selector was below a child selector and both refered t
          o the same element then the below one would be given preference
        => Therefore the below one is given preference*/
        p:first-child{
```

```
            background-color: aqua;
        }

        /*Attribute selector*/
        [div-x=a]{
            background-color: blueviolet;
        }
    </style>
</head>
<body>
    <div>
        This is css
        hello
    </div>
    <div class="green">
        This is the second one
    </div>

    <span class="green">Lorem ipsum dolor sit amet consectetur adipisicing elit. Dignissimos odit quaera
      t repellat quod asperiores laudantium repudiandae atque assumenda ipsam, mollitia ad, odio sequi?<
      /span>

    <p id="para">Lorem ipsum dolor sit amet consectetur adipisicing elit. Esse, et? Cupiditate quam atqu
      e repudiandae. Ad tempore reiciendis, similique suscipit blanditiis quo asperiores sit aliquam at.
      </p>

    <div>
        <p>This is a para under div</p>

        <article>
            <p>this is under article</p>
        </article>
    </div>

    <div>
        <span>
            <p>Though this is inside sapn it is a descendent of div</p>
        </span>
    </div>

    <a href="https://google.com">Go to google</a>
    <br>
    <a href="khbhkbjhb">Something</a>
    <div>
        <div>
            <p>I am first</p>
            <p>I am second</p>
            <p>I am third</p>
        </div>
    </div>
    <div div-x="a">This belongs to attribute</div>
</body>
</html>
```

2_Box-model

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Box Model</title>
    <style>

        *{
            margin: 0;
            padding: 0;
        }

        .box{
            background-color: blanchedalmond;
        }

        .box1{
            padding: 10px;
            margin:25px;
            /*bg is also applied to padding but not to margin*/
            border: 8px solid green;

            height: 200px;/*When only height is mentioned then it adds 200px only to the main content box*/
            box-sizing: border-box;/*But when i write this line then the height and width mentioned mentioned will be considered including padding and border and content box total willbe 200px excluding margin*/
        }
        .box2{
            padding: 10px;
            margin: 25px;
            border: 8px solid blue;
            height: 200px;
            box-sizing: border-box;
        }

        /*Here both have margins 25 25 so the distance between the two should be 50px but the actual distance bw these is 25 this is margin collapse.
        >If margins of 2 elements are overlapping then the max margin between the two is applied between the two.
        >Here if the box1 had 35 and box2 had 25 margin then the margin between the two would be 35 the max of the two.*/
    </style>
</head>
<body>
    <div class="box box1">
        I am First div
    </div>
    <div class="box box2">
        I am Second div
    </div>
</body>
</html>
```

3_Fonts

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Fonts</title>
    <style>
        @import url('https://fonts.googleapis.com/css2?family=Quintessential&display=swap');
        h1{
            font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
        }
        p{
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; /*here a set of fonts arre use
              d so that if the fonts are not available in the users device then the next font will be ap
              plied*/

            font-style: italic;
            line-height: 5;

        }
        #t2{
            font-family: "Quintessential", serif;
            font-style: normal;
            font-weight: 500;
            font-size: 20px;
            text-decoration: underline;
        }

        #t3{
            text-decoration: overline;
        }

        #t4{
            letter-spacing: 4px;
        }

        h2{
            text-transform: capitalize;
            /* text-transform: uppercase; */
            text-decoration: underline;
            text-decoration-color: blue;
            text-decoration-style:wavy ;
            text-decoration-thickness: 5px;
            text-indent: 23px;
            text-align:center;
        }

        #lorem{
            line-height: normal;
            width: 50px;height: 500px;
            border: 3px solid green;
            text-overflow:ellipsis;
            overflow:hidden;
            /* word-break: break-all; */
            text-align:center;

        }
    </style>
</head>
<body>
    <h1>Fonts</h1>
    <p>This is a font</p>
    <p id="t2">This is a font</p>
    <p id="t3">This is a font</p>
    <p id="t4">This is a font</p>

    <h2>transform</h2>

    <p id="lorem">Lorem ipsum dolor sit amet consectetur adipisicing elit. Laboriosam officia ab facilis
       recusandae laborum, eligendi sed cum blanditiis voluptatem commodi aut excepturi omnis?</p>
</body>
</html>
```

4_Colours

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Colours</title>
    <style>
        *{
            background-color: rgb(53, 53, 53);
            color: azure;
            border: 2px solid white;
            text-align: center;
        }
        h1{
            color: bisque;
        }
        #l1{
            background-color: brown;
        }
        #l2{
            background-color: #76DEA1;
        }
        #l3{
            background-color: rgb(150, 39, 184);
        }
        #l4{
            background-color: rgba(139, 60, 31, 0.699);
        }
        #l5{
            background-color: hsl(174, 56%, 27%);
        }
        #l6{
            background-color: hsla(0, 56%, 27%, 0.6) ;
        }

    </style>

</head>
<body>
    <h1>Colours</h1>
    <h2>
        Colours can be represented as
    </h2>
    <p>
        <ol>
            <li id="l1">Color Keywords </li>
            <li id="l2">Hex Color code </li>
            <li id="l3">RGB </li>
            <li id="l4">RGBA </li>
            <li id="l5">HSL</li>
            <li id="l6">HSLA</li>
        </ol>
    </p>
</body>
</html>
```

5_Exercise_2

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Exercise 2</title>
    <style>

        div>p{
            background-color: blue;
            color: white;
        }
        body>:first-child>:first-child{
            background-color: yellow;
            color: red;
        }/*not(:first-child) is the opposite of the first-child refers to all other childs other than th
          e first one*/

        /* div>p:nth-child(2){
            background-color: yellow;
            color: red;
        } */
    </style>
</head>
<body>
    <div>
        <p>Lorem ipsum dolor sit, amet consectetur adipisicing elit. Esse excepturi corporis similique?
          Consequuntur assumenda, repellendus ipsa sed illo exercitationem magnam aperiam non odio.</p>
        <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Cupiditate, quaerat.</p>
        <p>Lorem, ipsum dolor sit amet consectetur adipisicing elit. Tempore, placeat!</p>
    </div>

    <div>
        <p>Lorem ipsum dolor sit, amet consectetur adipisicing elit. Esse excepturi corporis similique?
          Consequuntur assumenda, repellendus ipsa sed illo exercitationem magnam aperiam non odio.</p>
        <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Cupiditate, quaerat.</p>
        <p>Lorem, ipsum dolor sit amet consectetur adipisicing elit. Tempore, placeat!</p>
    </div>

</body>
</html>
```

6_Specificity

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h1>Specificity</h1>
    <h2>Position and order of appearence</h2>
    <p>If there are many selectors belonging to same group then the one at the last will be considered</p>

    <h2>Specificity</h2>
    <p>The one which specifies an element more precisely will be considered</p>
    <p>Like Inline selectors are the most specific(so if at all they are there then no other selectors are checked) id is more specific than class or attribute selector and class is more specific than element selector which is more specific than universal selectors</p>
    <p>Inline > Id > Class or attribute> Element > Universal</p>

    <h2>Origin</h2>
    <p>If you have not used any style sheet then the styling will happen from the browesr's default style sheet which user agent stylesheet if you have applied any then it will be used</p>

    <h2>Important</h2>
    <p>The < color: "blue" !important > this important zpecificity will override inline styles also</p>
</body>
</html>
```

7_Sizing_Units

1_px_vw_vh.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sizing Units</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }
        #d1{
            border: 2px solid black;
            /* width: 2048px; If you fix in pixels the problem you will face is that is youe set px is l
              arger than the users screen px then the user has to scroll to see the end contents */
            box-sizing: border-box;
            background-color: aquamarine;
        }

        #d2{
            border: 2px solid black;
            width: 100vw;/*vw will adjust itself to the users screen (view port) to fit it perfectly*/
            box-sizing: border-box;
            background-color: aquamarine;
        }

        #p1{
            border: 2px solid black;
            width: 80vw;/*80 means it will cover only 80% of screen you can give more than 100 also but
              then scroll bar will appear*/
            box-sizing: border-box;
            background-color: rgb(209, 47, 221);
        }
        #p2{
            border: 2px solid black;
            width: 80vw;/*80 means it will cover only 80% of screen you can give more than 100 also but
              then scroll bar will appear*/
            margin: 20px auto;/*Whem this is applied then it will apply margins automatically on both si
              des equally and the element will still occupy 80% of viewport
            > Here 20px is for top bottom anuto is for left nd right*/
            box-sizing: border-box;
            background-color: rgb(209, 47, 221);
        }/*The above applicable only to block element and not to a inline element like span etc you cant
           apply width , margin auto etc*/

        #d3{
            border: 2px solid black;
            width: 80vw;
            height: 50vh;
            margin: 20px auto;
            box-sizing: border-box;
            background-color: rgb(209, 47, 221);

        }
        </style>

</head>
<body>
    <div id="d1">This is a box</div>
    <br>
    <div id="d2">
        This is for vw
    </div>
    <br>

    <p id="p1">This an example</p>
    <br>
    <p id="p2">This is vw centre allign</p>
    <br>

    <div id="d3">vw - viewport width and vh - viewport height</div>
```

```
</body>
</html>
```

2_em_rem.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>em and rem</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }

        html{
            font-size: 10px;
        }/*By default the root font size is 16px but here i changed the root to 10 so the below em examp
          les will have 2em as 20 and 2em of 2em as 40px
        > The root font size is 10px*/

        .container{
            border: 2px solid black;
            width: 80vw;
            font-size: 2em;
            box-sizing: border-box;
            margin:20px auto;
            background-color: aqua;
        }
        .container #p1{
            font-size: 2em;
        }
        /*em means that whatever it inherits make that many times of it
        > Here the default font size is 16px which the .container inherits so 2em of that means 32px
        > And as p inherits from container 2em of that means 2 times os 32px so 64px*/

        .container #p2{
            font-size: 1.5rem;/*rem means that many times of the root property here root font size is 10
               1.5 times of 10 is 15px */
        }
        </style>

</head>
<body>
    <div class="container">Lorem ipsum dolor sit amet consectetur adipisicing elit.</div>
    <div class="container">
        <p id="p1">
            Lorem ipsum dolor sit amet consectetur adipisicing elit. Voluptatibus tempore omnis expedita
               illum eligendi incidunt blanditiis placeat cumque alias culpa! Fugit, molestias. Officia,
               provident ea pariatur corrupti suscipit aperiam. Assumenda eligendi placeat blanditiis li
              bero aut animi deleniti voluptatibus molestiae eius, doloribus veniam doloremque maiores q
              uam.
        </p>
    </div>

    <div class="container">
        <p id="p2">Lorem ipsum dolor sit amet consectetur adipisicing elit. Molestiae, fugit illum? Aspe
          riores exercitationem minus est vitae similique, adipisci dolore voluptatum sequi a cumque, ev
          eniet culpa porro consequuntur! Excepturi incidunt tempora cupiditate autem, molestiae at?</p>

    </div>
</body>
</html>
```

3_vmin_vmax.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>vmin and vmax</title>
    <style>
        div{
```

```
            border: 2px solid black;
            background-color: bisque;
            box-sizing: border-box;
            width: 100vmin;/*in laptop vmin will be the height therfore the width will be the 100% of th
              e screen h hence it is same as 100vh here*/
        }
        /*vmax also works in the same way*/
    </style>
</head>
<body>
    <div>
        In mobile vmin width will be minimum and height will be max(height > width) therefore for mobile
           screen vmin will be the width but in computer screen its the opposite(width > height) therefo
          re here the height will be vmin
    </div>
</body>
</html>
```

4_min_width_and_min_height.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Min Width and Min Height</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }
        .class{
            border: 2px solid black;
            background-color: azure;
            min-height: 80vh;
            width: 80vw;
            box-sizing: border-box;
            margin: auto;
        }
        /*min height specifies the minimum height wt should be there even if the content is small and th
          e max height can be anything
        > Here both the boxes have same min height but the content of the second one is more due to whic
          h it is large
        >here in the same thing if you change the min to max then the content will overflow the box*/
    </style>
</head>
<body>
    <div class="class">
        Lorem, ipsum dolor sit amet consectetur adipisicing elit. Similique suscipit nesciunt atque, vit
          ae cumque consequatur. Expedita incidunt consequuntur placeat, fugit quo minus animi!
    </div>
    <br>
    <div class="class">
        Lorem ipsum dolor sit amet consectetur adipisicing elit. Praesentium itaque aliquam iste quia qu
          ae atque minima ullam optio, eius quo illum ipsum labore incidunt doloribus. Maxime odio accus
          antium consequuntur dolorum, dolore perferendis? Iure id, in porro ipsa maiores expedita obcae
          cati aliquam atque tempore ea! Quo autem velit cumque ipsa recusandae non voluptas ab sapiente
           vitae sint earum maxime in doloremque atque accusamus nobis exercitationem, blanditiis evenie
          t praesentium assumenda id odio a consectetur veritatis. Reiciendis eius praesentium doloremqu
          e est, dignissimos placeat eligendi ex sit similique dolor eveniet, iste commodi vel voluptate
           dolorem voluptas ab. Officia illum, corrupti exercitationem eaque minima, quod modi quam vita
          e non optio incidunt aspernatur! Adipisci tenetur fuga harum animi. Ad doloremque laudantium q
          uos eos magni iusto porro dicta ab totam recusandae, esse saepe illo sequi aspernatur! Error f
          ugiat repellat quasi eveniet. Soluta omnis nisi ex alias rerum, qui consectetur ea culpa volup
          tatum reiciendis hic temporibus ducimus, officia itaque. Explicabo dignissimos dolore culpa. S
          int deserunt tempore corrupti consectetur sit totam architecto odit libero at. Ex repellendus
          doloribus rerum ea ipsum aliquid minus aspernatur, repudiandae voluptatem distinctio id evenie
          t adipisci repellat placeat labore saepe accusamus et! Nesciunt optio harum sapiente autem sit
           natus, quae ratione voluptatem dolorem odio, inventore esse asperiores quas, atque quos simil
          ique. Corrupti minus, corporis tempora quisquam doloremque est dolores temporibus ullam blandi
          tiis recusandae modi delectus praesentium, expedita quaerat. Nostrum repudiandae animi quis nu
          lla aliquam sunt ea facere molestiae laudantium facilis cupiditate aut debitis sit illo quas,
          iste error, neque, labore enim illum nam? Atque quo obcaecati nemo debitis deserunt cupiditate
           quae reprehenderit quasi. Reprehenderit incidunt minima recusandae! Fuga nam excepturi necess
          itatibus? Voluptatibus natus magni cum, saepe eum cupiditate aliquid quae assumenda, adipisci,
           aspernatur ducimus libero! Repellat nobis dolorem provident itaque maiores atque ratione unde
           cum cupiditate, molestias voluptatum doloremque cumque aliquam odit doloribus et eveniet? Rec
```

usandae incidunt delectus nam, quis illum modi atque eum placeat adipisci molestiae ipsam natus illo, autem harum est sapiente at! Illum tempore amet nihil consequatur incidunt ducimus eveniet, quae recusandae facilis voluptatibus mollitia, sapiente explicabo dolores cupiditate optio exercitationem, reiciendis quo quos voluptatum ipsa quisquam! Nostrum voluptas deserunt magnam aliquid repellat corporis nobis voluptatum error commodi quo. Ipsum exercitationem alias iusto quis esse hic vero, corporis voluptatem itaque, sapiente explicabo facere nam delectus impedit aspernatur? Impedit rerum tempora distinctio, maxime autem sapiente! Assumenda vero architecto hic deserunt atque in magni enim repellendus voluptates, totam, nemo voluptatibus suscipit! Deserunt reiciendis commodi dolorum numquam facilis vel! Delectus magnam iusto nisi quisquam ipsum qui possimus, velit ipsam culpa quam accusantium mollitia tenetur. Quaerat dolore at reiciendis animi fugit, laborum molestias soluta aliquid nemo consequuntur optio exercitationem quasi, voluptas rerum distinctio illo architecto, expedita corporis nihil rem quo quidem. Repudiandae, odio consequuntur voluptatem inventore perferendis delectus provident assumenda? Beatae nesciunt et, voluptates saepe architecto ex. Ad blanditiis eligendi dolorem quia dolore rerum, architecto cumque. Modi totam laboriosam, nisi quisquam temporibus quis facilis molliti a autem doloremque voluptatibus assumenda reprehenderit dolorem unde perferendis sequi possimus officia commodi natus ipsa aspernatur dolorum ut maxime beatae accusantium. Ipsam consequatur voluptatibus deserunt vero dignissimos, laborum quaerat fugit, id molestias soluta perferendis sed atque facere minus. Dolor quaerat ea omnis maxime. Voluptatum, eveniet unde? Perferendis, accusantium aut quis, necessitatibus temporibus quidem ipsa illo maxime libero, qui voluptatibus distinctio est quasi sed dolorem non quod saepe provident at illum rem sint. Eius iusto repudiandae quam cum ipsa ipsam non, id minus ratione error consectetur neque quo laborum eum odit sapiente praesentium maiores excepturi quaerat possimus dolores impedit velit totam optio! Libero, velit quaerat quae harum deleniti, temporibus quibusdam accusamus laborum magnam qui a illum dolore veritatis odit hic quam enim id suscipit natus ad tenetur ullam voluptates deserunt aperiam sapiente! Quas soluta cum accusamus sit enim velit reprehenderit! Veniam obcaecati id qui modi molestiae, eos quis fugiat delectus corporis nam omnis ipsa sequi perferendis voluptates possimus ratione aliquam ea nesciunt! Recusandae voluptatem explicabo numquam, asperiores quaerat, nam deserunt fuga facere consequuntur natus nesciunt temporibus, molestias nisi modi assumenda consequatur fugit. Ea quas aut itaque praesentium fugiat fugit, rem repellat, corrupti accusamus dolores eaque doloremque perspiciatis eius facere quisquam voluptatem sed voluptate nobis deserunt quis. Voluptate saepe quae delectus, quasi excepturi neque omnis reiciendis cum nihil ducimus totam sapiente, vero facilis a, voluptatibus necessitatibus praesentium! Voluptatum, molestiae aliquid earum dolor accusantium, culpa officia cum, reiciendis eligendi nihil quibusdam similique rem consectetur iste vitae aperiam excepturi dicta ad! Porro vitae laborum aperiam quaerat eum vel cupiditate. Reprehenderit veniam officiis sapiente assumenda adipisci magni accusamus amet neque quod, delectus vitae excepturi nam aperiam in tenetur aspernatur quisquam dignissimos, repellendus quia ratione architecto et provident. Ullam ad possimus aperiam rerum! Dolor dolore eius unde animi nostrum consequuntur. Fugiat explicabo possimus est molestias, sequi labore assumenda deleniti libero, optio quibusdam similique illum. Expedita velit libero sed iste optio iure officia et. Consequatur illum impedit aperiam incidunt vel repellendus magnam sit, porro eum provident! Quaerat, aliquid suscipit nemo est, sunt voluptas perspiciatis repellat debitis obcaecati, enim ex ut aliquam dolorem asperiores nisi. Laboriosam cum vitae eum officia. Dicta quam qui dolore. Asperiores quas ratione esse, mollitia sint voluptatem? Quae veritatis debitis ex, deserunt iste voluptatibus obcaecati vel error natus ipsum unde at optio consequuntur quibusdam recusandae ducimus iusto sequi earum. Esse rerum nemo error obcaecati quisquam iste. Dolorum voluptas veritatis unde officia, mollitia quo dolore! Consequatur, beatae. Doloremque iusto reprehenderit delectus sit, qui tempore ut tenetur rerum deleniti culpa eum voluptatibus natus quia cupiditate accusantium dolor, quo fugiat reiciendis quam voluptatem repellat. Ipsa minus nihil odit tempore itaque quam sunt! Illum nemo consequatur expedita fuga adipisci officiis quos iusto mollitia hic incidunt nesciunt quis minima, nihil unde cumque, consequuntur qui. Dolore, illum dignissimos. Doloribus tenetur neque blanditiis commodi veniam autem accusamus molestiae nihil quidem rerum velit perferendis pariatur illo labore hic cumque nisi iste aperiam odio laboriosam modi fugit, sequi aliquam corporis. Veniam ad nostrum illum qui molestias voluptatibus dolore quo. Expedita, provident nulla doloremque voluptas veritatis cum atque mollitia dolorem reiciendis dignissimos, itaque consequatur distinctio facilis accusantium ipsam consectetur nostrum. Eligendi obcaecati, tempore eius atque delectus quidem sapiente sit molestiae ratione non! Magnam quisquam nesciunt sit facilis quam tempore maiores distinctio saepe fugit, in eius nulla exercitationem at veniam, provident voluptatibus possimus incidunt ea sunt. Obcaecati odit neque molestiae dolore numquam cum, quas similique distinctio accusamus laudantium! Dignissimos doloribus architecto eum quasi assumenda tempora at modi? Fuga, sint beatae corporis explicabo quo, quisquam ex voluptatum necessitatibus voluptate molestiae non ipsam, ducimus doloribus tempore possimus! Earum, in. Porro nihil tenetur quia perspiciatis minus cumque pariatur qui nesciunt tempore ullam, vitae natus perferendis quam. Iure ullam perferendis officia qui, minus possimus natus, praesentium delectus asperiores neque nihil odit amet fugit? Amet perferendis commodi, voluptatibus quo optio numquam adipisci animi exercitationem iure? Nemo deleniti ut distinctio quis tempore quidem vel deserunt omnis placeat eveniet! Ipsa nam corrupti reiciendis eius repudiandae, rerum magni sit dicta, consequatur cumque eum! Libero fuga molestiae, sapiente eaque facilis sunt, cumque eligendi, aspernatur tenetur cupiditate nesciunt doloribus totam iure amet quod modi. Adipisci corporis iste veritatis vitae praesentium, molestiae tempora, dolorum, eos repudiandae eveniet est consequatur! Eligendi consequatur culpa quae illo blanditiis quasi eum vitae voluptatem repudiandae, iusto explicabo repellat molestias cupiditate, officiis dignissimos voluptatibus! Dolorum illo enim voluptatibus, ratione aut, sequi nisi dignissimos temporibus officiis a quisquam vero sed quia ut voluptate modi commodi. Voluptatum, pariatur quibusdam laborum neque assumenda dolor, porro unde ea dignissimos beatae accusantium in, aperiam quis. Ipsam eveniet velit voluptatum libero voluptatem, delectus amet.

```
32      </div>
```

```
</body>
</html>
```

5_percentage_units.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>% units</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }
        .outer{
            background-color: antiquewhite;
            width: 80vw;
            height: 80vh;
            box-sizing: border-box;
            margin: auto;
            border: 3px solid black;
        }

        .outer div{
            border: solid greenyellow;
            width: 50%;/*Its width will be 50% of the entire outer one*/
        }

        .outer >div>p{
            border: solid rebeccapurple;
            width: 50%;/*It will be 50% of the div which is in fact the 50% of the outer one
            >Therefore the p will be 25% of the outer one*/
        }
    </style>
</head>
<body>
    <div class="outer">
        <div>
            Lorem ipsum dolor sit amet consectetur, adipisicing elit. Minima sapiente aperiam ducimus re
              rum placeat quod repudiandae deserunt facilis tenetur voluptatum, exercitationem nobis nis
              i, necessitatibus mollitia? Ut assumenda, a voluptates optio commodi quo velit esse cupidi
              tate odit. Corporis in a mollitia adipisci sequi? Officia, minima modi!

            <p>
                Lorem ipsum dolor sit amet consectetur adipisicing elit. Inventore dolore iure doloremqu
                e ullam ex sit iusto, cumque laboriosam deleniti suscipit dignissimos, quaerat ducimus?
            </p>
        </div>
    </div>
</body>
</html>
```

8_Display_Properties

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Display Properties</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }

        .box{
            background-color: aqua;
            border: 3px solid brown;
            margin: 30px;
            padding: 20px;
            width: 100px;
            height: 100px;
            /* display: inline; if you mention only inline then you cannot apply width and when you appl
              y margin and padding they will not be applied on the top so inline block is used*/
            display: inline-block;
        }

        .box1{
            /* display: none; When you do this the box will dissapear and the space occupied by it will
              also dissapear */
            visibility: hidden;/*When you do this the box will dissapear but its space will remain*/
        }
    </style>
</head>
<body>
    <div class="box box1">
        This is a box
    </div>
    <div class="box">
        This is another
    </div>
    <span>
        This is a span
    </span>
</body>
</html>
```

9_Shadows_Borders

Shadows.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Shadows</title>
    <style>
        .box-shadow{
            border: 3px solid brown;
            padding: 30px;
            box-shadow: 5px 5px 10px 8px grey;
            outline: 5px solid black;
            outline-offset: 2px;

        }

        .text-shadow{
            text-shadow: 1px 3px 7px rgb(188, 52, 90);
            padding: 10px;
            border: 3px solid rgb(121, 0, 202);
            outline: 5px solid black;/*Outline is different than the border outline appears outside bord
              er.
            > Outline is not a part of the box model.
            > Outline is applied outside the border
            > Outline offset is distance between the border and outline.
            outline cannot be curved directly but it takes the shape of the border when it is curved*/
            border-radius: 7px;
            outline-offset: 5px;
            /*Borders can have varying width on different sides but outline has a uniform width*/
        }
    </style>
</head>
<body>
    <div class="box-shadow">This is a box</div>
    <br>
    <br>
    <div class="text-shadow">Lorem, ipsum dolor sit amet consectetur adipisicing elit. In veniam, numqua
      m tenetur quibusdam, earum et sed voluptas adipisci ab ipsa repellendus atque illum.</div>

</body>
</html>
```

10_Lists

lists.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CSS Lists</title>
    <style>
        /* nav ul or nav ul li{
            list-style:armenian; you can change the marker with this
        } */

        nav ul li{
            background-color: aquamarine;/*This is applied only to the items not marker but if you want to include markers also then do*/

            /* list-style-position: inside; */
            border: 2px solid ;/*If the above line wasnt there then the border also would be applied only to the the items*/

            /* list-style:disc inside url("Screenshot 2026-03-23 200852.png"); though disc is there first url will be considered as it is at last*/
            /*All can be included in one line itself here the image in the url will be used as the marker*/
            /* list-style-image: url("Screenshot\ 2026-03-23\ 200852.png"); */

            /* list-style: inside "□"; or */
            list-style-type: "□";
            list-style-position: inside;

        }
    </style>
</head>
<body>
    <nav>
        <ul>
            <li>Home</li><!--By default the display of a list will be list-item where each element will have a marker and a item but for a navigation bar you do not want it like that so you must change-->
            <li>About</li>
            <li>User</li>
        </ul>
    </nav>
</body>
</html>
```

nav.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        /* nav ul{
            list-style: none;
            display: flex;
            padding: 5px;
        }
            > Here padding will be applied to entire ul as a whole not to each li element so as a result there will be no individual padding between the li elements hence no space between the elements*/
        *{
            margin: 0;
            padding: 0;
        }
        nav ul {
            list-style: none;
            display: flex;
        }

```

```
        nav ul li {
            padding: 5px;
        }
    </style>
</head>

<body>
    <nav>
        <ul>
            <li>Home</li>
            <li>About</li>
            <li>User</li>
        </ul>
    </nav>
</body>

</html>
```

11_Overflow

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Overflow</title>
    <style>
        .box {
            width: 40vw;
            height: 30vh;
            border: 2px solid black;
        }

        /*Here in this the contents are overflowing to avoid this*/

        /* One way is to */
        /* .box{
            overflow: scroll;
        } */

        /* another way is  */
        .box1 {
            overflow: auto;
            /*Automatically adds scroll if the content is overflowing otherwise not*/
            /*Problem with only scroll is that even if the content is not overflowing scroll bars will a
              ppear*/
        }

        .box2{
            overflow: auto;

            white-space: nowrap;/*Content will be displayed in one line only*/

            /* text-overflow:ellipsis;Adds three dots at last */

            /* overflow: clip; is this is added then you can never scroll even programatically
            > Even hidden does the same thing*/
        }

        .box3{
            /*overflow in x and y can be controlled separately*/

            /* overflow-x: hidden;
            overflow-y: auto;
            this one can be written in one line as*/

            overflow: hidden auto;
        }
    </style>
</head>

<body>
    <div class="box box1">Lorem ipsum dolor sit amet consectetur adipisicing elit. Asperiores est autem
      quidem totam nisi qui
        doloribus assumenda fugit, rerum eius soluta tempore reiciendis! Vel, quasi? Minima tenetur pers
          piciatis,
        voluptatum, iste sed commodi qui illum labore pariatur beatae provident ex? Fuga officia laborio
          sam voluptatem
        impedit rem, aliquam ipsum, atque ea facilis modi dolor pariatur accusantium quo sunt.</div>
    <br>
    <div class="box box2">Lorem ipsum dolor sit amet, consectetur adipisicing elit. Dolorem autem distin
      ctio placeat eaque
        soluta odit. Nulla reprehenderit, esse error in cupiditate placeat quasi at debitis ut assumenda
           ea eos
        exercitationem. Laborum voluptatibus labore delectus, explicabo ut odio itaque reprehenderit ips
          um sint earum
        quaerat quo molestiae! Magnam, ducimus magni? Suscipit officia aut excepturi distinctio perferen
          dis nihil?
        Magni.</div>

    <br>
```

```
65      <div class="box box3">Lorem ipsum dolor sit amet, consectetur adipisicing elit. Dolorem autem distin
          ctio placeat eaque
66          soluta odit. Nulla reprehenderit, esse error in cupiditate placeat quasi at debitis ut assumenda
              ea eos
67          exercitationem. Laborum voluptatibus labore delectus, explicabo ut odio itaque reprehenderit ips
              um sint earum
68          quaerat quo molestiae! Magnam, ducimus magni? Suscipit officia aut excepturi distinctio perferen
              dis nihil?
69          Magni.</div>
70  </body>
71
72  </html>
```

12_Position

1_Relative.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CSS Position</title>
    <style>

        *{
            margin: 0;
            padding: 0;
        }

        .box{
            height: 25vh;
            width: 25vw;
            border: 2px solid black;
            margin: 5px;
        }

        /*By default the position property will be static which means that you cant alter any position p
          roperty of elements*/

        .box1{
            background-color: aquamarine;
        }
        .box2{
            background-color: cyan;
            position: relative;/*When you do relative you can change the position of the box*/
            top: 56px;
            left: 23px;
            /*Now this element will be top on the third element but if you want the other to be on top t
              hen you should set the z-index of the third more than 0 but before that you should set its
               position also as relative*/
        }
        .box3{
            background-color: orange;
            position: relative;
            /*The box comes above 2nd without setting z-index as the one below is given more precedence
              than above so once you set it relative than it automatically comes above*/
        }
        .box4{
            background-color: lightgray;
        }

    </style>
</head>
<body>
    <div class="box box1">1</div>
    <div class="box box2">2</div>
    <div class="box box3">3</div>
    <div class="box box4">4</div>
</body>
</html>
```

2_Absolute.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Absolute</title>
    <style>

        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .box{
```

```
            margin: 10px;
            padding: 10px;
            height: 15vh;
            width: 70vw;
            border: 2px solid black;
            margin: 5px;
        }

        .box1{
            background-color: rgb(189, 192, 164);

        }

        .box2{
            background-color: aqua;
            border: 2px solid black;
            position: absolute;/*When absolute is applied then the applied position will be applied in r
              elation to the nearest positioned parent(whose position is not static) so here for box1 in
              side parent1 the nearest positioned one is the webpage(html) as parent1,body are static so
               therefore the position is applied in relative with the webpage

            > but for box1 inside parent2 the nearest positioned one is parent2 so all its position is a
              pplied in relative with the parent2

            > There are some other cases where even if the parent is not set to absolute/sticky/relative
              /fixed still it appears positioned when properties like transform filter or perspective ar
              e used these can make the parent positioned and the element will be set relative to the pa
              rent*/
            top: 0px;
            left: 25px;
        }

        .box3{
            background-color:bisque;
        }

        .box4{
            background-color: brown;
        }

        .parent1{
            margin: 10px;
            padding: 10px;
            height:90vh;
            width: 90vw;
            border: 5px solid red;
        }

        .parent2{
            margin: 10px;
            padding: 10px;
            position: absolute;
            height:90vh;
            width: 90vw;
            border: 5px solid rgb(202, 12, 97);

        }

    </style>
</head>
<body>

    <div class="parent1">

        <div class="box box1"></div>
        <div class="box box2"></div>
        <div class="box box3"></div>
        <div class="box box4"></div>
    </div>
    <br>
    <br>
    <div class="parent2">

        <div class="box box1"></div>
        <div class="box box2"></div>
        <div class="box box3"></div>
        <div class="box box4"></div>
    </div>
```

```
</body>
</html>
```

3_Fixed.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Fixed</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .box{
            margin: 10px;
            padding: 10px;
            height: 15vh;
            width: 70vw;
            border: 2px solid black;
            margin: 5px;
        }

        .box1{
            background-color: rgb(189, 192, 164);

        }

        .box2{
            background-color: aqua;
            position: fixed;
            bottom: 0;
            right: 0;/*Its position will be fixed now even if you scroll*/
        }

        .box3{
            background-color:bisque;
        }

        .box4{
            background-color: brown;
        }

        .parent1{
            margin: 10px;
            padding: 10px;
            height:150vh;
            width: 90vw;
            border: 5px solid red;
        }
    </style>
</head>
<body>
    <div class="parent1">

        <div class="box box1"></div>
        <div class="box box2"></div>
        <div class="box box3"></div>
        <div class="box box4"></div>
    </div>
</body>
</html>
```

4_Sticky.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Fixed</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .box{
            margin: 10px;
            padding: 10px;
            height: 15vh;
            width: 70vw;
            border: 2px solid black;
            margin: 5px;
        }

        .box1{
            background-color: rgb(189, 192, 164);

        }

        .box2{
            background-color: aqua;
            position: sticky;
            height: 10vh;
            top: 0;
            right: 0;/*Its position will be same until you scroll and it reaches the top or bottom once
              it reaches it doesnt move up or down it will be fixed at the top or bottom but not applica
              ble when moved right or left */
        }

        .box3{
            background-color:bisque;
        }

        .box4{
            background-color: brown;
        }

        .parent1{
            margin: 10px;
            padding: 10px;
            height:150vh;
            width: 150vw;
            border: 5px solid red;
        }
    </style>
</head>
<body>
    <div class="parent1">

        <div class="box box1"></div>
        <div class="box box2"></div>
        <div class="box box3"></div>
        <div class="box box4"></div>
    </div>
</body>
</html>
```

13_Exercise_3

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Card</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }
        body{
            background-color: rgba(168, 167, 167, 0.477);
        }

        .card{
            margin: 10px auto;
            background-color: white;
            width: 314px;
            height: 460px;
            border: 2px solid rgba(128, 128, 128, 0.486);
            border-radius: 15px;
        }

        .image{
            padding: 7px;
        }

        .image img{
            border-radius: 15px;
        }

        .buttons{
            padding: 5px 20px;/*padding: top_and_bottom left_and_right can also be written as
            > paddinkg: top right bottom left*/
        }

        .buttons button{
            font-size: 12px;
            margin: 0 3px;
            border-radius: 18px;
            padding: 6px 14px;
            color: rgba(105, 105, 105, 0.516);
            border: 1px solid rgba(158, 157, 157, 0.401);
            background-color: rgba(186, 186, 186, 0.126);
            cursor: pointer;/*If cursor pointer is applied universally then the cursor will be pointer e
              verywhere

            >So it should be applied only inside the button for which the cursor should turn as a pointe
              r*/
        }

        .buttons button:hover{
            color: rgb(255, 248, 248);
            background-color: rgba(105, 105, 105, 0.516);
        }

        .content{
            margin-top: 20px;
            color: rgb(51, 51, 51);
            padding: 0 20px;
        }

        .content p{
            margin-top: 10px;
            color: rgb(110, 110, 110);
        }
        .read{
            text-align: center;
            margin: 30px;
        }

```

```
        .read button{
            border: none;
            font-size: 12px;
            border-radius: 18px;
            padding: 10px 15px;
            color: rgb(40, 143, 226);
            background-color: rgba(40, 142, 226, 0.2);
            cursor: pointer;
        }

        .read button:hover{/*There should be no space between the button :hover like this or button: hov
          er like this then it wouldnt be implemented*/
            color: rgb(255, 255, 255);
            background-color: rgb(79, 152, 212);
        }

    </style>
</head>
<body>
    <div class="card">
        <div class="image">
            <img width="300" height="220" src="lake.jpeg" alt="">
        </div>
        <div class="buttons">
            <button>Nature</button>
            <button>Lake</button>
        </div>

        <div class="content">
            <h2>Heading</h2>
            <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Aliquam sed eos laborum?</p>
        </div>
        <div class="read">
            <button>Read More</button>
        </div>
    </div>
</body>
</html>
```

14_Variables

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Variables</title>
    <style>
        :root{
            --color: aquamarine;
            --seccolor: rgba(51, 246, 181, 0.201);
        }/*Anything declared within the root are global variables*/

        .nav ul{
            list-style: none;
            display: flex;
            background-color: var(--color);
        }

        .nav ul li{
            padding: 10px;
        }

        .d1{
            background-color: var(--seccolor);
        }

        .d2{
            --color: rgb(75, 187, 122);/*Here color has a local value so inside this block the global co
              lor value will be overiden by the local value for this block*/
            background-color: var(--color);
        }

        .d3{
            background-color: var(--notfound,lightblue);/*the second one is a fallback variable(You cant
               give another variable you can just give value ) so that if at all the first one is not fo
              und then the second one will be used*/
        }
    </style>
</head>
<body>
    <nav class="nav">
        <ul>
            <li>Home</li>
            <li>About</li>
            <li>Contact</li>
        </ul>
    </nav>
    <div class="d1">
        Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eos, ducimus numquam voluptatum cum no
          strum explicabo suscipit earum assumenda ipsum, magni voluptates omnis! Adipisci assumenda ull
          am mollitia odit dolor molestias asperiores.
    </div>
    <br>
    <div class="d2">
        Lorem ipsum dolor sit amet consectetur adipisicing elit. Deleniti veritatis perferendis fuga tem
          pora placeat optio eos consequuntur illum doloremque dolorum autem nam cum totam excepturi inv
          entore voluptas, dignissimos incidunt eveniet?
    </div>
    <br>
    <div class="d3">
        Lorem ipsum dolor sit, amet consectetur adipisicing elit. Quaerat vero rerum eligendi sit pariat
          ur? Necessitatibus, tempore quidem minus vero commodi enim deleniti unde! Quibusdam id exercit
          ationem quo? Perferendis, corrupti nesciunt.
    </div>
</body>
</html>
```

15_Media_Queries

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Media Queries</title>
    <style>
        body{
            background-color: aquamarine;
        }

        @media screen and (max-width:500px){
            body{
                background-color: antiquewhite;/*So here as soon as the width goes below 500px the backg
                round color changes to antiquewhite (here)
                > If min width was given then it would be applied after the screen goes beyond 500px*/
            }
        }

        .boxes{
            display: flex;
        }

        .box{
            background-color: aqua;
            height: 233px;
            width: 233px;
            margin: 10px;
            padding: 5px;
        }

        @media screen and (orientation: portrait){/*Gets applied when the screen is in portrait*/
            .boxes{
                flex-direction: column;
            }
        }
    </style>
</head>
<body>
    <div>
        Hello there
    </div>
    <br>

    <div class="boxes">
        <div class="box">1</div>
        <div class="box">2</div>
        <div class="box">3</div>
    </div>
</body>
</html>
```

16_Exercise_4

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Basic website</title>
    <style>
        :root {
            --pc: rgb(178, 178, 178);
            --sc: rgba(211, 211, 211, 0.826);
            --nav: rgba(62, 62, 62, 0.786);
            --text: black;
            --hover: black;

        }

        /* :root {
            --pc: rgb(24, 24, 24);
            --sc: rgba(63, 63, 63, 0.826);
            --nav: rgba(230, 230, 230, 0.786);
            --text: white;
            --hover: black;

        } */

        * {
            margin: 0;
            padding: 0;
        }

        body {
            background-color: var(--pc);
        }

        nav ul {
            display: flex;
            list-style: none;
            margin: 40px;
            padding: 10px;
            background-color: var(--sc);
            border-radius: 15px;
        }

        nav ul a {
            background-color: none;
            color: var(--nav);
            border: none;
            padding: 0px 20px;
            cursor: pointer;
            text-decoration: none;/*Removes that below line*/
        }

        a:hover {
            color: var(--hover);
        }

        .content{
            border-radius: 10px;
            height: 70vh;
            color: var(--text);
            background-color: var(--sc);
            margin: 40px;
            padding: 10px;
        }


    </style>
</head>

<body>
    <nav>
```

```
        <ul>
            <li><a href="https://google.com">Home</a></li>
            <li><a>Contact</a></li>
            <li><a>About</a></li>
        </ul>
    </nav>

    <div class="content">
        Lorem ipsum dolor sit amet consectetur, adipisicing elit. Aspernatur ea itaque, in exercitationem omnis architecto aliquam, libero nemo minus animi optio distinctio pariatur suscipit recusandae quae autem vero ut ducimus!
    </div>
</body>

</html>
```

17_Float_clear

1_float.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Float and Clear</title>
    <style>

        img{
            float:left;/*This will make the image float left and allows the content on its right side

            >When we are using float if the image is more than its container then it overflows

            > To avoid this we can add the display flow root then the border of the container will be ap
              plied including the image


            > When the image is in float then the content will not be cut it will just continue*/
        }

        .cards{
            border: 2px solid;
            display: flow-root;
        }

        .card{
            border: 3px solid red;
        }
    </style>
</head>
<body>
    <div class="cards">

        <img height="200px" src="https://thumbs.dreamstime.com/b/ripe-strawberry-plant-9284614.jpg" alt=
          "image">
        <div class="card">Lorem ipsum dolor sit amet consectetur adipisicing elit. Beatae modi libero ne
          mo quae ut distinctio a iure corporis! Delectus, nam? Numquam deleniti sit doloribus ullam vel
           consectetur, accusantium eum perferendis.</div>
        <br>
        <div class="card">Card 1</div>
        <div class="card">Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ut provident dolorem
           aliquam aut quasi odit iure ipsa, sapiente tempore nisi in ratione. Hic tempora modi magnam m
          ollitia obcaecati possimus amet cum cumque quis voluptates? Nobis ex impedit ipsam minima fugi
          at, eaque porro dignissimos voluptate eveniet voluptates iusto corrupti, commodi, sapiente rep
          ellendus unde minus? Architecto placeat ipsa, est recusandae beatae qui accusantium praesentiu
          m. Ea perferendis cupiditate corporis, nam tenetur vero, ducimus quisquam, culpa natus aliquid
           expedita laudantium. Dignissimos, tenetur architecto officia est quidem accusamus tempora pra
          esentium libero nostrum esse numquam natus nulla eos. Quisquam quis sint rerum? Laborum evenie
          t consequuntur eaque ipsa labore obcaecati commodi. Sit obcaecati rem eligendi nemo dignissimo
          s nesciunt sint, architecto aliquid quam reiciendis repudiandae eius amet illum sapiente cupid
          itate. Iusto eius maiores dolore! Officia architecto repudiandae facilis officiis minus asperi
          ores porro natus, provident, praesentium corrupti suscipit excepturi velit animi nulla volupta
          tem aspernatur blanditiis nostrum cumque illum? Quae, dolor sunt aspernatur incidunt ipsa inve
          ntore officia magni porro dolore delectus esse, quasi nam illo harum quis. Consequuntur expedi
          ta suscipit nostrum distinctio culpa, maiores reiciendis nihil enim assumenda incidunt? Unde a
           sunt temporibus! Nihil quod corrupti quasi itaque debitis. Quidem et reprehenderit magni face
          re amet voluptatum blanditiis natus assumenda similique!</div>
        <div class="card">Card 3</div>
    </div>
</body>
</html>
```

2_clear.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Clear</title>
    <style>

```

```
        img{
            float:left;
        }

        .cards{
            border: 2px solid;
            display: flow-root;
        }

        .card{
            border: 3px solid red;
        }

        .card:nth-child(4){
            background-color: aquamarine;
            clear: left;/*This will tell that do not allow any element to float on the left but it will
              allow to float on the right

            > it can be made to clear: both were it wont allow to float on both the sides*/
        }
    </style>
</head>
<body>
    <div class="cards">

        <img height="200px" src="https://thumbs.dreamstime.com/b/ripe-strawberry-plant-9284614.jpg" alt=
          "image">

        <div class="card">Lorem ipsum dolor sit amet consectetur adipisicing elit. Beatae modi libero ne
          mo quae ut distinctio a iure corporis! Delectus, nam? Numquam deleniti sit doloribus ullam vel
           consectetur, accusantium eum perferendis.</div>

        <div class="card">Card 1</div>

        <div class="card">Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ut provident dolorem
           aliquam aut quasi odit iure ipsa, sapiente tempore nisi in ratione. Hic tempora modi magnam m
          ollitia obcaecati possimus amet cum cumque quis voluptates? Nobis ex impedit ipsam minima fugi
          at, eaque porro dignissimos voluptate eveniet voluptates iusto corrupti, commodi, sapiente rep
          ellendus unde minus? Architecto placeat ipsa, est recusandae beatae qui accusantium praesentiu
          m. Ea perferendis cupiditate corporis, nam tenetur vero, ducimus quisquam, culpa natus aliquid
           expedita laudantium. Dignissimos, tenetur architecto officia est quidem accusamus tempora pra
          esentium libero nostrum esse numquam natus nulla eos. Quisquam quis sint rerum? Laborum evenie
          t consequuntur eaque ipsa labore obcaecati commodi. Sit obcaecati rem eligendi nemo dignissimo
          s nesciunt sint, architecto aliquid quam reiciendis repudiandae eius amet illum sapiente cupid
          itate. Iusto eius maiores dolore! Officia architecto repudiandae facilis officiis minus asperi
          ores porro natus, provident, praesentium corrupti suscipit excepturi velit animi nulla volupta
          tem aspernatur blanditiis nostrum cumque illum? Quae, dolor sunt aspernatur incidunt ipsa inve
          ntore officia magni porro dolore delectus esse, quasi nam illo harum quis. Consequuntur expedi
          ta suscipit nostrum distinctio culpa, maiores reiciendis nihil enim assumenda incidunt? Unde a
           sunt temporibus! Nihil quod corrupti quasi itaque debitis. Quidem et reprehenderit magni face
          re amet voluptatum blanditiis natus assumenda similique!</div>

        <div class="card">Card 3</div>

    </div>
</body>
</html>
```

18_More_Selectors

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>More on Selectors</title>
    <style>
        /* .box:first-child{
            background-color: aquamarine;
        }
            > If the box is the first child of its parent then the property is applied */

        /* .box:first-line{
            background-color: aqua;
        }
            > Will be applied to all the first line of each box */

        /* .outer *{
            background-color: antiquewhite;
            margin: 20px;
        }
            > This will be applied to all the elements inside the outer */

        /* p, a, .box, [attribute="one"]{
            margin-top: 45px;
            padding-top: 30px;
            background-color: antiquewhite;
        }
            > you can select more tha one element at once */

        /* .box:nth-child(odd){
            background-color:bisque;
        }
            > applied only to odd childs*/

        /* .box:nth-last-child(2){
            background-color: bisque;
        }
            > This will consider the child from last */
        ::selection{
            background-color: black;
            color: bisque;
        }/*This will be applied when any element is selected*/

        .box::first-letter{
            font-size: 30px;
            color: peru;
        }

        input::placeholder{
            color: rgba(0, 0, 255, 0.278);
        }




    </style>
</head>
<body>
    <div class="outer">
        <div attribute="one" class="box">Lorem ipsum dolor sit amet, consectetur adipisicing elit. Iste,
           quae dicta amet eveniet esse consectetur incidunt velit recusandae, voluptate eius quo nihil
          placeat?</div>
        <div class="box">2nd one</div>
        <div class="box">3rd one</div>
        <div>
            <p>This is a paragraph</p>
        </div>

        <div class="ip">
            <input type="text" placeholder="Enter your name">
        </div>
```

```
71
72      </div>
73  </body>
74  </html>
```

19_Exercise_5

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Layout</title>
    <style>

        :root {
            --pc: rgb(178, 178, 178);
            --sc: rgba(211, 211, 211, 0.826);
            --navtext: rgba(62, 62, 62, 0.786);
            --text: black;
            --hover: black;
            --border: gray;

        }

        /* :root {
            --pc: rgb(24, 24, 24);
            --sc: rgba(63, 63, 63, 0.826);
            --nav: rgba(230, 230, 230, 0.786);
            --text: white;
            --hover: black;
            --border: rgb(184, 182, 182);

        } */


        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            color: var(--text);
            /* overflow: auto;
            > Having overflow auto in universal selector makes the elements not to stick to the screen e
              ven if position sticky is applied*/
        }

        body{
            background-color: var(--pc);
        }

        header{
            position: sticky;
            top: 0;
        }

        nav{
            background-color: var(--sc);
            height: 40px;
        }

        .container{
            background-color: var(--sc);
            margin: 3vh 5vw;
            padding:2vh 5vw;
            border-radius: 10px;
            text-align: center;
        }

        .box1, .box2{
            display: inline-block;
            margin: 5vh 5vw;
            width: 26vw;
            height: 70vh;
            background-color:var(--pc);
            border-radius: 10px;
        }

        footer{
            height: 70px;
```

```
            background-color: var(--sc);
        }

        .textbox{
            text-align: center;
            margin: 10vh 10vw;
            padding: 10vh 10vw;
            border-radius: 100px;
            background-color: var(--sc);
            font-size: x-large;
            border: 2px solid var(--border);
        }

        .bottom{
            position: fixed;
            height: 40px;
            width: 40px;
            background-color: orange;
            bottom: 1vw;
            right: 1vw;
            border-radius: 100%;
        }
    </style>
</head>
<body>
    <header>
        <nav>
            Home
        </nav>
    </header>

    <main>
        <div class="container">
            <div class="box1"></div>
            <div class="box2"></div>
        </div>

        <div class="textbox">Welcome to Sigma Web Development Course</div>

        <div class="bottom"></div>
    </main>

    <footer>

    </footer>
</body>
</html>
```

20_Flex_Box

1_index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Flex Box</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;/*Flex box is applied to the container not to the individual items*/
            justify-content: center;/*Main axis centre*/
            align-items: center;/*Vertical center*/

        }

        .item{
            height: 40px;
            width: 40px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
        }

        .c2{
            justify-content: flex-end;
        }

        .c3{
            justify-content: flex-start;
        }
        .c4{
            justify-content: space-around;
        }
        .c5{
            justify-content: space-between;
        }
        .c6{
            align-items: baseline;/*Will be on one line*/
        }
    </style>
</head>
<body>
    <main>
        <div class="container">
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
        </div>
        <div class="container c2">
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
        </div>
        <div class="container c3">
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
        </div>
        <div class="container c4">
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
            <div class="item"></div>
        </div>
        <div class="container c5">
            <div class="item"></div>
            <div class="item"></div>
```

```
            <div class="item"></div>
            <div class="item"></div>
        </div>
        <div class="container c6">
            <div class="item" style="height: 90px; width: 90px;"></div>
            <div class="item" style="height: 30px; width: 30px;"></div>
            <div class="item" style="height: 200px; width: 200px;"></div>
            <div class="item""></div>
        </div>
    </main>
</body>
</html>
```

2_extra.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Flex Box</title>
    <style>
        /*In flex boxes the items are not layed from left to right instead they are alligned in the flex
           direction which can be changed*/
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
            justify-content: center;
            align-items: center;

            /* flex-direction: column;aligns the items in column wise */

            /* flex-direction: column-reverse;  aligns the items in column wise but in reverse order*/

            /*Similarly there is row reverse*/

            /*In a flex box always the contents are aligned across the main axis and on the axis on whic
              h they arent alligned is called the cross axis

            > so by default row is your main axis and column is your cross axis.

            > But when you specify flex-direction as column then your column becomes your main axis and
              row becomes your cross axis

            *> Justify content aligns on the main axis like if you specify justify-contents: start then
              it will align it on the start of the main axis*/
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 40px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
    </div>
</body>
</html>
```

3_wrap.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Wrap</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
            justify-content: center;
            align-content: center;

            flex-wrap: wrap;/*In a flex box it doesnt go to a newline when the content overflows first i
              t decreases the width or height whichever direction in which it is after decresing also if
               the content doesnt fit inside the container then it overflows but when wrap is given then
               the content automatically comes to the newline when it cant fit within the width of the c
              ontainer*/

            /* when you apply wrap and the align items is written then only the middle lines will be in
              the centre of the cross axis and other items will be spaced evenly from the middle ones

            but if you want all the lines to be in the center then align-content is used

            > There is wrap reverse where the content will be wrapped in reverse direction

            > nowrap is the default*/

            flex-flow: column wrap-reverse;/*setting flex direction and wrapping at once*/
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 40px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
```

```
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
    </div>
</body>
</html>
```

4_gap.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Gap</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
            justify-content: center;
            align-content: center;
            flex-wrap: wrap;

            gap: 2px; /*Adds 2px only between items not between the container and item*/

            /* row-gap: 50px;

            column-gap: 10px; */

            gap: 50px 10px;/*row column*/
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 40px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
```

```
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
    </div>
</body>
</html>
```

5_Order.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Gap</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
            justify-content: center;
            align-content: center;
            flex-wrap: wrap;
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 40px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
        }

        .o1{
            order: 1;
        }/*order is a item property the item with more order will appear at the last and with least orde
          r at the first

        > This property can be used to change the order of the elements

        > default order is 0*/

        .o2{
            order: 2;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item o2">1</div>
        <div class="item">2</div>
        <div class="item o1">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
```

```
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
    </div>
</body>
</html>
```

6_grow.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Gap</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 55px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;

        }

        .g{
            flex-grow: 1;/*The item will grow such that the line in which it is there fits the container
               if already the line reaches or exceeds the container and the items in the line are alread
              y conjusted then it wont grow

            > Value given doesnt affect */
        }


    </style>
</head>
<body>
    <div class="container">
        <div class="item g"></div>
        <div class="items1"></div>
        <div class="item"></div>
        <div class="item sh"></div>
```

```
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <!-- <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div> -->

    </div>
</body>
</html>
```

7_grow_2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Gap</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 55px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;
            flex-grow: 2;/*When this is more than the values of the one applied to single items then eac
              h item will take equal space except the one which is given a different value*/
        }

        .g{
            flex-grow: 1;/*If this is 0 then it will take the height and width mentioned if 1 then it wi
              ll take(less space here as the entire one is 2) but as its value increases then its size i
              ncreases but others remain of equal size*/
        }


    </style>
</head>
<body>
    <div class="container">
        <div class="item g"></div>
        <div class="items1"></div>
        <div class="item"></div>
        <div class="item sh"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <!-- <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
```

```
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div> -->

    </div>
</body>
</html>
```

8_shrink.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Shrink</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
        }

        .item{
            text-align: center;
            align-content: center;
            height: 40px;
            width: 55px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;

        }

        .g{
            flex-shrink: 2;/*when this is written then the item will shrink first than the others when t
              he layout size is decreased */
        }


    </style>
</head>
<body>
    <div class="container">
        <div class="item g"></div>
        <div class="items1"></div>
        <div class="item"></div>
        <div class="item sh"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <!-- <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div> -->

    </div>
</body>
</html>
```

10_align_self.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Self align</title>
    <style>
        .container{
            height: 80vh;
            margin: 20px;
            border: 2px solid darkred;
            display: flex;
            justify-content: center;
            align-items: center;
        }

        .item{
            height: 40px;
            width: 55px;
            border: 2px solid;
            background-color: aquamarine;
            margin: 2px;

        }

        .g{
            align-self: flex-end;/*Aligns that particular element at the specified position*/
        }


    </style>
</head>
<body>
    <div class="container">
        <div class="item g"></div>
        <div class="items1"></div>
        <div class="item"></div>
        <div class="item sh"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <!-- <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div>
        <div class="item"></div> -->

    </div>
</body>
</html>
```

21_Grid

1_grid.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Grid</title>
    <style>
        .container{
            border: 3px solid brown;
            padding: 10px;
            display: grid;
            grid-template-columns: [pehla] 50px [doosra] 60px [t]  100px;/*here you are telling there wi
              ll be three columns in the grid one of 50px 2nd of 60px and other of 100px*/

            /*They can be named as you wish*/

            /*If you set even one parameter also row or column then the other will automatically set the
               other equally*/

            grid-template-rows: 100px 100px 100px;/*Creates 3 rows of 100px*/
        }

        .item{
            height: 50px;
            width: 50px;
            border: 2px solid cadetblue;
        }

        .i6{
            grid-row: 1/2;/*It should occupy row 1 to 2 */
            /* or you can write
            grid-row-start: 1;
            grid-row-end: 2;  both mean the same*/
            grid-column: 2/3;/*It should occupy column 2 to 3*/
        }

        .i9{
            grid-row: 2;
            grid-column: pehla;/*as both same 8 and 9 have same coordinates they overlap*/
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item">1</div>
        <div class="item">2</div>
        <div class="item">3</div>
        <div class="item">4</div>
        <div class="item">5</div>
        <div class="item i6">6</div>
        <div class="item">7</div>
        <div class="item i9">8</div>
        <div class="item i9">9</div>
    </div>
</body>
</html>
```

2_grid_area.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Grid Area</title>
    <style>
        .container{
            border: 3px solid brown;
            padding: 10px;
            display: grid;
            grid-template-areas: "nav nav nav"
                                 "article side side"
```

```
                                "footer footer footer";
        }

        .item{
            border: 2px solid chocolate;
            height: 50px;
        }

        .i1{
            grid-area: nav;
            background-color: rgb(120, 105, 85);
        }

        .i2{
            grid-area: article;
            background-color: aquamarine;
        }

        .i3{
            grid-area: side;
            background-color: blueviolet;
        }

        .i4{
            grid-area: footer;
            background-color: rgb(65, 129, 186);
        }

        .i5{
            grid-area: nav;
            background-color: rgba(224, 6, 6, 0.238);
        }/*overlaps on 1*/
    </style>
</head>
<body>
    <div class="container">
        <div class="item i1">1</div>
        <div class="item i2">2</div>
        <div class="item i3">3</div>
        <div class="item i4">4</div>
        <div class="item i5">5</div>
    </div>
</body>
</html>
```

3_fraction_gap.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Grid Area</title>
    <style>
        .container{
            border: 3px solid brown;
            padding: 10px;
            display: grid;

            width: 80vw;
            height: 80vh;
            grid-template-columns: 1fr 3fr 1fr ;/*Divide the  coloumns in the ratio of 1 fraction 3 frac
              tion and 1 fraction*/
            /* grid-template-rows: repeat(2,1fr);is same as the below one repeats two times 1fr */
            /* grid-template-rows: 1fr 1fr; */

            grid-template-rows: repeat(2,minmax(100px,1fr));/*repeats 2 times the minmax which specifies
               that the minimum width should be 100px amd max should be 1fr*/

            gap: 34px;
            row-gap: 34px;/*row and column gaps can be set separately but here there is no change as the
              ir values match to the already applied gap

            > gap is applied only between the items not the item and the container*/
            column-gap: 34px;

            /* justify-content: center do not use this;
```

```
            as the justify content will be applied on the entire grid
            > if at all the grid is smaller than the container and you apply justify content centre then
               the entire grid will be in the center of the container
            > same applies to align content*/

            /* justify-items: center;
            align-items: center;

            both can be applied at once using place-items

            > same applies to content -> place-contents*/

            place-items: center;
        }

        .item{
            border: 2px solid chocolate;
            height: 50px;
            width: 50px;
        }

        .i2{
            align-self: start;
            justify-self: end;/*these are applied with respect to that particular cell not the entire gr
              id */
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="item i1">1</div>
        <div class="item i2">2</div>
        <div class="item i3">3</div>
        <div class="item i4">4</div>
        <div class="item i5">5</div>
    </div>
</body>
</html>
```

22_Exercise_6

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Template</title>
    <style>
        nav {
            display: flex;
            justify-content: space-around;

        }

        .logo img {
            height: 50px;

        }

        nav ul {
            display: flex;
            list-style: none;
            justify-self: right;
            align-items: center;
        }

        nav ul li {
            margin: 0px 20px;
            display: flex;
            align-items: center;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            font-size: 12px;
            cursor: pointer;
        }



        nav ul img {
            height: 20px;
            align-self: center;
        }
    </style>
</head>

<body>
    <nav>
        <div class="logo">
            <img src="https://www.ultraedit.com/wp-content/uploads/2023/01/UE_Site-Logo-1.png" alt="">
        </div>
        <div class="content">
            <ul>
                <li><span>Products </span><svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" wi
                dth="14"
                        height="14" color="currentColor" fill="none" stroke="currentColor" stroke-width=
                "1.5"
                        stroke-linecap="round" stroke-linejoin="round">
                        <path d="M18 9.00005C18 9.00005 13.5811 15 12 15C10.4188 15 6 9 6 9" />
                    </svg></li>
                <li><span>Pricing</span> <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" wid
                th="14"
                        height="14" color="currentColor" fill="none" stroke="currentColor" stroke-width=
                "1.5"
                        stroke-linecap="round" stroke-linejoin="round">
                        <path d="M18 9.00005C18 9.00005 13.5811 15 12 15C10.4188 15 6 9 6 9" />
                    </svg></li>
                <li><span>Resourses</span> <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" w
                idth="14"
                        height="14" color="currentColor" fill="none" stroke="currentColor" stroke-width=
                "1.5"
                        stroke-linecap="round" stroke-linejoin="round">
                        <path d="M18 9.00005C18 9.00005 13.5811 15 12 15C10.4188 15 6 9 6 9" />
                    </svg></li>
```

```
67                 <li><span>About Us</span><svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" wid
                   th="14"
68                         height="14" color="currentColor" fill="none" stroke="currentColor" stroke-width=
                   "1.5"
69                         stroke-linecap="round" stroke-linejoin="round">
70                         <path d="M18 9.00005C18 9.00005 13.5811 15 12 15C10.4188 15 6 9 6 9" />
71                     </svg></li>
72                 <li><img src="https://w7.pngwing.com/pngs/498/901/png-transparent-google-translate-trans
                   lation-google-search-android-android-english-text-logo.png"
73                         alt="image.png" class="i2"></li>
74             </ul>
75         </div>
76     </nav>
77 </body>
78
79 </html>
```

23_Transform

1_rotate.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Rotate</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .container{
            height: 80vh;
            width: 100vw;
            display: flex;
            border: 3px solid brown;
        }

        .box{
            margin: 10px;
            border: 3px solid;
            background-color: aquamarine;
            height: 100px;
            width: 100px;
            display: flex;
            align-items: center;
            justify-content: center;

            /* transform: rotate(45deg); */

            /* transform: rotate(0.25turn); 1 turn = 360 deg so 0.25 turn is 90 deg*/

            /* transform: rotateX(45deg); */

            /* transform: rotateZ(30deg); */

            /* Normally the rotation happens arount 50%,50% of the origin which means centre of the box
              */

            transform-origin: 0% 0%;
            transform: rotate(20deg);/*now as the origin is set at 0 and 0 the box rotates around the st
              art of the box*/
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box</div>
    </div>
</body>
</html>
```

2_scale.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Rotate</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .container{
            height: 80vh;
            width: 100vw;
```

```
            display: flex;
            border: 3px solid brown;
        }

        .box{
            margin: 10px;
            border: 3px solid;
            background-color: aquamarine;
            height: 100px;
            width: 100px;
            display: flex;
            align-items: center;
            justify-content: center;

            /* transform: scale(1.7); Becomes 1.7 times of its size */

            transform: scaleY(2);/* only scales in y axis */
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box</div>
    </div>
</body>
</html>
```

3_skew.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Rotate</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .container{
            height: 80vh;
            width: 100vw;
            display: flex;
            border: 3px solid brown;
        }

        .box{
            margin: 10px;
            border: 3px solid;
            background-color: aquamarine;
            height: 100px;
            width: 100px;
            display: flex;
            align-items: center;
            justify-content: center;

            /* transform: skew(20deg);Is stretched */

            transform: skewY(45deg);

            /*But you cant skew in z axis*/
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box</div>
    </div>
</body>
</html>
```

4_translate.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Rotate</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        .container{
            height: 80vh;
            width: 100vw;
            display: flex;
            border: 3px solid brown;
        }

        .box{
            border: 3px solid;
            background-color: aquamarine;
            height: 100px;
            width: 100px;
            display: flex;
            align-items: center;
            justify-content: center;

            transform: translate(100%) translateY(-50%) skew(10deg);/*100% of box size which is 100px*/
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box</div>
    </div>
</body>
</html>
```

24_Exercise_7

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>UltraEdit</title>
    <style>

        .container{
            padding: 20px;
            margin-top: 10vh;
            display: grid;
            place-self: center;
            grid-template-columns: 1fr;
            gap: 20px;
        }

        .item{
            padding: 10px;
            font-size: 20px;
            text-align: center;
            width: 260px;
            height: 100px;
            background-color: rgba(154, 153, 153, 0.16);
        }

        button{
            display: inline-block;
            margin-top: 43px;
            border: none;
        }

        @media screen and (min-width: 700px){
            .container{
                grid-template-columns: 1fr 1fr;
            }
        }

        @media screen and (min-width:900px){
            .container{
                grid-template-columns: 1fr 1fr 1fr;
            }
        }

        a{
            cursor: pointer;
            text-decoration: none;

            padding: 8px;
            margin: -5px;
            font-weight: bold;
            color: white;
            border-radius: 7px;
        }

        .name{
            font-size: 17px;
            font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
            font-weight: bold;
            margin-top: 4px;
        }

        .c1{
            background-color: rgb(65, 198, 198);
        }

        .c2{
            background-color: rgb(31, 177, 55);
        }
    </style>
</head>
<body>
```

```
    <div class="container">
        <div class="item i1">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i2">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i3">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i4">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i5">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i6">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i7">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i8">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i9">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
        <div class="item i10">
            <div class="name">English</div>
            <button ><a class="c1" href="">Download 32-bit</a></button>
            <button ><a class="c2" href="">Download 64-bit</a></button>
        </div>
    </div>
</body>
</html>
```

25_Transition

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Transition</title>
    <style>
        .container{
            width: 80vw;
            height: 80vh;
            background-color: antiquewhite;
        }

        .box{
            width: 80px;
            height: 50px;
            background-color: aquamarine;
            /* transition-property: transform; or*/
            /* transition-property: all; or */

            /* transition: background-color, transform;
            transition-duration: 3s;
            transition-timing-function: ease-in-out;
            transition-delay: 1s; all this can be written in one line*/

            transition: all 3s ease-in-out 1s;
        }

        .translate{
            transform: translateX(30vw) translateY(30vh) scale(2);
            background-color: brown;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box translate"></div>
    </div>
</body>
</html>
```

26_Animation

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .container {
            width: 80vw;
            height: 80vh;
            background-color: antiquewhite;
        }

        .box {
            width: 80px;
            height: 50px;
            background-color: aquamarine;

            /* animation: name duration timing-function delay iteration-count direction fill-mode; */

            animation: MyAnimation;
            animation-duration: 3s;
            animation-timing-function: ease-in-out;
            animation-delay: 1s;
            animation-iteration-count: 1;
            animation-direction:normal;

            /* animation-direction: alternate-reverse; */

            /* animation-play-state: paused;you can pause or run the animation using js */

            animation-fill-mode: backwards;/*if forwards then the element stays at its final position on
              ly if backwards then it will come back to its first position*/

            animation: MyAnimation 3s ease-in-out 1s 1 , Rotate 3s ease-in-out 1s 1;/*Applying two anima
              tions at once*/
        }

        @keyframes MyAnimation {
            from {
                background-color: blue;
            }

            to {
                transform: translateX(30vw) translateY(30vh) scale(2);
                background-color: brown;
            }
        }
        @keyframes Rotate {
            0%{ transform: rotate(10deg);}
            20%{transform: rotate(80deg);}
            40%{transform: rotate(100de)g;}
            60%{transform: rotate(180de)g;}
            80%{transform: rotate(200de)g;}
            100%{transform: rotate(360de)g;}

        }
    </style>
</head>

<body>
    <div class="container">
        <div class="box"></div>
    </div>
</body>

</html>
```

27_Exercise_8

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Bouncing ball</title>
    <style>
        .container{
            width: 80vw;
            height: 50vh;
            border: 3px solid black;
            background-color: lightgray;
            position: relative;
        }

        .ball{
            position: absolute;
            width: 50px;
            height: 50px;
            background-color: red;
            border-radius: 50%;
            left: 0;
            bottom: 0;
            animation: Bouncex 3s linear 0s infinite ,Bouncey 1s ease-out 0s infinite alternate;/*The ti
              me difference between x bounce and y bounce sould be different x should be more than y*/
        }



        @keyframes Bouncex {
            from{

            }
            to{
                left: 100%;
            }
        }

        @keyframes Bouncey {
            from{
                bottom: 0;
            }
            to{
                bottom: 60%;
            }
        }


    </style>
</head>
<body>
    <div class="container">
        <div class="ball"></div>
    </div>
</body>
</html>
```

28_Object_fit

1_cover.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Cover</title>
    <style>
        .container{
            width: 560px;
            height: 500px;
            border: 2px solid;
            background-color: aquamarine;
        }
        img{
            height: 466px;
            width: 466px;
            /* object-fit: cover; THis zooms the image so that it covers the mentioned height and width
              without changing the aspect ratio */
            object-fit: cover;
            object-position: top right;


        }
    </style>
</head>
<body>
    <div class="container">
        <img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQA-7UIYLcEK1mAwct1XgW8aSMnvm3ZEQ
          BYgQ&s" alt="">
    </div>
</body>
</html>
```

2_contain.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Object fit</title>
    <style>
        .container{
            width: 560px;
            height: 500px;
            border: 2px solid;
            background-color: aquamarine;
        }
        img{
            height: 466px;
            width: 466px;
            /* object-fit: fill; This is the default one  */

            object-fit: contain;
            /*This will maintains its aspect ratio satisfiying only one measure either height or width*/


        }
    </style>
</head>
<body>
    <div class="container">
        <img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQA-7UIYLcEK1mAwct1XgW8aSMnvm3ZEQ
          BYgQ&s" alt="">
    </div>
</body>
</html>
```

29_Filter

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Filters</title>
    <style>
        .img1{
            filter:contrast(4)
        }
        .img2{
            filter: invert(1);
        }
        .img3{
            filter: opacity(50%);
        }
    </style>
</head>
<body>
    <div class="img1">
        <img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQA-7UIYLcEK1mAwct1XgW8aSMnvm3ZEQBYgQ&s" alt="">
    </div>
    <div class="img2">
        <img src="https://i.pinimg.com/736x/2d/95/e5/2d95e5886fc4c65a6778b5fee94a7d59.jpg" alt="">
    </div>
    <div class="img3">
        <img src="https://www.planetware.com/img/gallery/canadas-14-best-destinations-to-snap-beautiful-pictures/intro-1762792954.jpg" alt="">
    </div>
</body>
</html>
```

30_Netflix_Clone

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

    <main>
        <nav>

            <div class="sign">
                <div class="logo"><img src="assets/images/logo.svg" alt="" class="logo"></div>
                <div>

                    <span><label for="Lang"></label>
                        <select name="English" id="Lang" class="select">
                            <option value="English">English</option>
                            <option value="Hindi">Hindi</option>
                        </select>
                    </span>

                    <span class="button">Sign In</span>
                </div>
            </div>
        </nav>

        <div class="centercontent">
            <div class="one">Unlimited movies, shows, and more</div>
            <br>
            <div class="two">Starts at ₹149. Cancel at any time.</div>
            <br>
            <br>
            <div class="three">Ready to watch? Enter your email to create or restart your membership.</d
              iv>
            <br>
            <div class="email">
                <div class="in"><input type="text" placeholder="Email address"></div>

                <div class="button">Get started <img src="assets/images/arrow.svg" alt=""></div>
            </div>
        </div>

        <div class="box"></div>
    </main>

    <footer>
        <div class="mlist"></div>
    </footer>
</body>

</html>
```

style.css

```css
* {
    margin: 0;
    padding: 0;
    font-family: sans-serif;
    ;
}

body {
    background-color: black;
    color: white;
}

main {
```

```
    /* width: 100vw; */
    height: 100vh;
    background-image: url("assets/images/image1.jpg");
    background-position: center center;
    background-size: max(1200px, 100vw);
    background-repeat: no-repeat;
}

main .box {
    width: 100%;
    height: 100vh;
    background-color: black;
    opacity: 0.7;
    position: absolute;
    z-index: 1;
    top: 0%;
}

.logo {
    height: 40px;
}

.sign {
    display: flex;
    position: absolute;
    z-index: 2;
    width: 80vw;
    justify-self: center;
    align-items: center;
    justify-content: space-between;
    margin-top: 20px;
}

.select {
    cursor: pointer;
    text-align: center;
    height: 33px;
    width: 120px;
    font-size: 17px;
    background-color: rgba(18, 18, 18, 0.901);
    color: rgba(255, 255, 255, 0.786);
    border-radius: 4px;
    margin: 5px 4px;
    font-weight: 500;
}

.button {
    cursor: pointer;
    background-color: rgb(196, 0, 0);
    padding: 9px 16px;
    border-radius: 4px;
    margin: 5px 4px;
    font-size: 14px;
    font-weight: 700;
}

.centercontent {
    width: 600px;
    position: absolute;
    justify-self: center;
    z-index: 2;
    text-align: center;
    top: 30vh;
}

.one {
    font-size: 50px;
    font-weight: 900;
}

.two {
    font-size: 21px;
    font-weight: 550;

}

.email {
    display: flex;
```

```
    justify-content: center;
    align-items: center;
}

.email .in input {
    width: 350px;
    height: 53px;
    margin: 0px 5px 0px 10px;
    font-size: 16px;
    padding-left: 15px;
    background-color: rgba(18, 18, 18, 0.901);
    border: 1px solid rgba(255, 255, 255, 0.578);
    color: white;
    border-radius: 4px;
}

.email .button {
    align-content: center;
    width: 200px;
    height: 38px;
    font-size: 25px;
    text-align: left;
    padding-left: 25px;
}

.email .button img {
    position: absolute;
    bottom: 22px;
    padding-left: 10px;
}

@media screen and (max-width:1278px) {
    .centercontent {
        width: 700px;
    }

    .one {
        font-size: 45px;
        font-weight: 800;
    }

    .two {
        font-size: 19px;
        font-weight: 500;
    }

    .email .in input {
        width: 330px;
    }

    .email .button {
        width: 150px;
        font-size: 20px;
    }

    .email .button img {
        bottom: 22px;
        padding-left: 10px;
    }

}

@media screen and (max-width:960px) {
    .logo{
        height: 25px;
    }

    .centercontent {
        width: 450px;
    }

    .one {
        font-size: 35px;
        font-weight: 700;
    }

    .email .in input {
        width: 230px;
```

```
    }

}

@media screen and (max-width:600px) {

    .centercontent{
        width: 95vw;
    }

    .email {
        display: block;
        justify-content: center;
        align-items: center;
    }

    .email .in input{
        width: 90%;
        height: 50px;
    }

    .email .button {
        height: 30px;
        width: 120px;
        font-size: 18px;
        justify-self: center;
        margin-top: 17px;
        padding-left: 20px;
    }

    .email .button img {
        bottom: 18px;
        padding-left: 10px;
    }

    .select{
        width: 50px;
        overflow: hidden;
    }
}

@media screen and (max-width:300px) {
    .sign{
        display: block;
    }
}

@media screen and (max-width:200px) {
    .sign{
        position: absolute;
        right: 10px;
    }

    .sign .button{
        display: block;
        width: 50px;
    }

    .centercontent{
        margin-top: 10px;
    }

    *{
        overflow: hidden;
    }
}
```

3_JavaScript

1_Variables_Datatypes_Objects

1_Variables.js

```
var a=23;
var b=45;
console.log(a+b)

// Vr is lie a global variable it will be available throughout the program irrespective of where it is d
  eclared

var a=100;// var can be redeclared




let c=10;
let d=30;
console.log(c+d);

//let is like local it will be available only in the block where it is declared but if declared ouside a
  ll the blocks acts as a global variable

// let d=1000; let cant be redeclared

if(true){
    console.log(c+d);
    let d=1000;//let can be redeclared inside a block
}




const p=3.14;
console.log(p);

p=35;// a const cant be changed this will give an error
console.log(p);


if(true){
    const p=35;
    console.log(p)//a const can also be redeclared inside a block but cant be changed
    //just p=35 will give an error
}
```

2_Datatypes.js

```
let a=12;
let _b=3.14;//_b valid

// 5n="" invalid
let n="Hello";
console.log(a,_b,n)

let $me="hello"// $ valid
console.log($me)

// SO you can start with _ or letter or $ but cannot start with a number

let a1=23;
let a2=5.3434;
let a3="Hello";
let a4=true;
let a5=null;
let a6=undefined

console.log(a1,a2,a3,a4,a5,a6);
console.log(typeof a1,typeof a2,typeof a3,typeof a4,typeof a5,typeof a6,typeof a7,"\n\nAll these are pri
  mitive data types except the fact that thouogh null is also a primitive data type its type is displaye
  d as object which is a user defined data type");
```

3_Objects.js

```js
let o={
    name:"Ram",
    "job code":5627// name also can be written in Double quotes but the fact that job code is written in
       double quotes is that it has a space
}

console.log(o,typeof o);

o.address="Bengaluru";

// o."job salary"=12345; is not valid
o["job salary"]=120000;
console.log(o)

o["job salary"]="1cr";
console.log(o)

// in object the values are stored as key value pairs where the value can be of any data type even an ob
  ject or a function
```

2_Operators

1_index.js

```js
let a=2;
let b=4;

// Arthimatic Operators
console.log("Arthimatic Operators\n");

console.log("Add -",a+b);
console.log("Subtract -",a-b);
console.log("Multiply -",a*b);
console.log("Divide -",a/b);
console.log("Modulus (returns reminder) -",b%a);
console.log("Exponentiation (b to the power of a) -",b**a);

// Assignment Operators
let c=13;

c += a;//c=c+a
console.log(c)

c -= a;
console.log(c)

c *= a;
console.log(c)

c /= a;
console.log(c)

c %= a;//c=c%a
console.log(c)

c **= a;//c=c**a
console.log(c)
```

2_Comparision_op.js

```js
let a=5;
let b=5.3;
let c="5.3";

console.log(a>b);
console.log(a<b);
console.log(a>=b);
console.log(a<=b);

console.log(c==b);// == checks only the value and as here "5.3" and 5.3 have same value returns true
console.log(c!=b);

console.log(c===b);// === checks both value and type and returns true only if both value and the type ar
  e same
console.log(c!==b);


// ternary operator

console.log(a>b ? "a is greater":"b is greater"); // using this the conditionals can be written in one l
  ine if the condition is true then the first one is executed if false then the second one is executed
```

3_Logical_op.js

```js
let a=23;
let b=45;

console.log(a!=b && a>b)
console.log(a!=b || a>b)
console.log(a!=b)
```

3_Conditionals

index.js

```
let a=6;
let b=4;

if (a<b) console.log("If an exist independently without else");

age =16;

if(age>=18){
    console.log("You can drive");
}

else if(age> 15 && age<18) console.log("wait a lil");

else console.log("Dont enen think of driving");
```

4_Loops

index.js

```js
console.log("\n\nfor loop");

for(let i=0;i<5;i++) console.log(i);


console.log("\n\nwhile loop");

let i=10;
while(i<12){
    console.log(i);
    i++;
}


console.log("\n\ndo while loop");

let j=10;
do{
    console.log(j);
    j++;
} while(j==10);


console.log("\n\nfor in loop");

let o={
    1:'a',
    2:'b',
    3:'c',
    4:'d'
}

for (const key in o) {
    console.log(key + " - " + o[key])
}


console.log("\n\nfor of loop");

for (const element of "Hello World") { //for iterables only and object is not iterable
    console.log(element)
}
```

tempCodeRunnerFile.js

```js
// do{
//     console.log(j);
//     j++;
// }while(j!=10);
```

5_Functions

index.js

```
function sum(a,b,c=0){
    console.log(a,b,c)
    return a+b+c;
}

console.log(sum(12,23));
console.log(sum(12,23,78));
console.log(sum(12));//returns NaN (Not a Number) when you do not pass a argument which is required and
  undefined is passed instead of that


let mul=(a=0,b=0)=>{//arrow fuunction here mul is a function also and a vaiable also
    return a*b;
}

console.log(mul(10,7));
```

6_Exercise_1_9

faulty.js

```
r=Math.random();
console.log(r)

let a=prompt("Enter the number 1");
let o=prompt("Enter the operation");
let b=prompt("Enter the number 2");

let op={
    "+":"-",
    "-":"/",
    '*':'+',
    '/':'**',
    '%':'*',
    '**':'%'
}



if(r>0.1){
    alert(eval(`${a} ${o} ${b}`))
}

else {
    o=op[o];
    alert(eval(`${a} ${o} ${b}`))
}
```

faulty_calculator.js

```
let ch;

console.log("1:Add\n2:Subtract\n3:Multiply\n4:Divide\n5:Exponentiation");

let a, b;

let n=Math.random();

if (n < 0.1) {
    switch (ch) {
        case 1:
            console.log(a + b);
            break;

        case 2:
            console.log(a - b);
            break;

        case 3:
            console.log(a * b);
            break;

        case 4:
            console.log(a / b);
            break;

        case 5:
            console.log(a % b);
            break;

        case 6:
            console.log(a ** b);
            break;

        default:
            console.log("Invalid choice");
    }
}

else{
    switch (ch) {
        case 1:
            console.log(a - b);
```

```
            break;

        case 2:
            console.log(a / b);
            break;

        case 3:
            console.log(a + b);
            break;

        case 4:
            console.log(a ** b);
            break;

        case 5:
            console.log(a * b);
            break;

        case 6:
            console.log(a % b);
            break;

        default:
            console.log("Invalid choice");
    }
}
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="faulty.js"></script>
</body>
</html>
```

7_Strings

index.js

```
let a="HEllo"

console.log(a[0]);
console.log(a[1]);
console.log(a[2]);
console.log(a[3]);
console.log(a[4]);
console.log(a[5]);

console.log(a.length);

b="World"

console.log("printing "+a+" "+b);
console.log(`printing ${a} ${b}`);//string interpolation -> inserting strings

console.log(`This is "double" and this is 'single'`);

console.log("THis is an \" sequence");

console.log(a.toUpperCase());
console.log(a.toLowerCase());//These are functions so there is ()
console.log(a.length);//This is a property so there is no ()

console.log(a.slice(1,4));// 1 is included but 4 is not included

a="llHello"

console.log(a.replace("ll","99"))//only the first occurance is replaced

console.log(a.concat("44"," 6767",78))

b='  hkjhn,khb  '
console.log(b.trim())//removes white spaces

//strings are immutable they cant be changed once created
```

8_Exercise_2_10

name_grnrrator.js

```
/*

Adjectives:
Crazy
Amazing
Fire

name:
Engine
Foods
Garments

Word:
Bros
Limited
Hub
 */

let adj={
    1:"Crazy ",
    2:"Amazing ",
    3:"Fire "
}

let n={
    1:"Engine ",
    2:"Foods ",
    3:"Garments "
}

let word={
    1:"Bros",
    2:"Limited",
    3:"Hub"
}

let choice={
    1:adj,
    2:n,
    3:word
}

b_name=""

for (const key in choice) {
    r=Math.floor(Math.random()*3+1)
    b_name+=choice[key][r];
}

console.log(b_name)
```

9_Array

index.js

```
let arr=[1,2,3,4,"Hello",true];

console.log(arr);
console.log(arr.length);
console.log(arr[4]);

arr[0]="one";

console.log(arr)//arrays are mutable

console.log(typeof arr);// Arrays are objects

console.log(arr.toString());

console.log(arr.join(" and "))

console.log(arr.pop());// removes the last element from the arr and returns it ,it alters the original a
  rray

console.log(arr.push("New"));// It add the specified one at the end of the array and returns the updated
   length of the array

console.log(arr);

console.log(arr.shift());// removes the first element and returns it , it updates the original one

console.log(arr.unshift("Oggy"));//add the element at the beginning and returns the updated length of th
  e array

console.log(arr);

console.log(delete arr[5]);//returns true
console.log(arr);//it only deletes the value but not the memory allocated to it so it appears empty
console.log(arr[5]);//displays undefined

a1=[78,89,987];
console.log(arr.concat(a1,["erfer","wefqwe"]))

console.log(arr.sort());//returns the sorted array and also modifies the original array
//this takes in an additional compare function also and sorts according to it if provided
console.log(arr);

arr=[1,2,3,4,5,6,7,8,9,10];
console.log(arr.splice(4,3));//starting from index 4 removes 3 elements and returns them and updates the
   original array
console.log(arr)

arr=[1,2,3,4,5,6,7,8,9,10];
console.log(arr.splice(4,3,222,333));//removes elements but also adds the 222 and 333 there
console.log(arr)


arr=[0,1,2,3,4,5,6,7,8,9,10];
console.log(arr.slice(4));//removes elements starting from 4 till the end and returns them(as list) and
  does not update the list
console.log(arr);

arr=[0,1,2,3,4,5,6,7,8,9,10];
console.log(arr.slice(4,6));//removes and returns list of elements starting from 4 till index 6 excludin
  g 6th one
console.log(arr);

arr=[1,4,3,27,3,100]
console.log(arr.reverse());//just reverses the order of how you have created them
console.log(arr);
```

loops.js

```js
arr=[1,2,3,4,5];

arr.forEach((value,index,array) => {
    console.log(value,index,array)
});

arr.forEach((v,i)=>{
    console.log(v,i);
})

for (const element of arr) {
    console.log(element);
}
```

mfr.js

```js
let arr=[1,2,3,4,5];

// map
console.log("Map")
let newarr=arr.map(e=>{//or arr.map((e,index,array)=>{})
    return e**2;
})
console.log(newarr);

// Filter
console.log("\n\nFilter")

// const greterthan7=(e)=>{
//     if(e>7) return true;
//     return false;
// }

// console.log(newarr.filter(greterthan7)); or

console.log(newarr.filter(e=>{
    if(e>7) return 1;
    return 0;
}))


// Reduce

console.log("\n\nReduce");

arr=[1,2,3,4,5];

console.log(arr.reduce((a,b)=>{
    return a*b;
}))
console.log(arr.reduce((a,b,c)=>{
    return a*b*c;
}))
console.log(arr.reduce((a,b,c,d)=>{
    return a*b*c*d;
}))
console.log(arr.reduce((a,b,c,d,e)=>{
    return a*b*c*d*e;
}))

/*
EXPLANATION OF reduce() WITH MULTIPLE PARAMETERS
================================================

arr = [1,2,3,4,5];

The reduce() callback receives exactly 4 arguments:
1. Accumulator - the accumulated result
2. Current Value - current array element
3. Current Index - position in array
4. Array - the original array

---

FIRST REDUCE (2 parameters: a, b)
```

```
---------------------------------
arr.reduce((a,b) => a*b)
Output: 120 

- a = accumulator, b = current value
- Iteration:
  - 1st: a=1, b=2 → 1×2 = 2
  - 2nd: a=2, b=3 → 2×3 = 6
  - 3rd: a=6, b=4 → 6×4 = 24
  - 4th: a=24, b=5 → 24×5 = 120

---

SECOND REDUCE (3 parameters: a, b, c)
--------------------------------------
arr.reduce((a,b,c) => a*b*c)
Output: 2880 

- a = accumulator
- b = current value
- c = current index

- Iteration:
  - 1st: a=1, b=2, c=1 → 1×2×1 = 2
  - 2nd: a=2, b=3, c=2 → 2×3×2 = 12
  - 3rd: a=12, b=4, c=3 → 12×4×3 = 144
  - 4th: a=144, b=5, c=4 → 144×5×4 = 2880

Note: First iteration starts at index 1 because reduce uses arr[0] as initial accumulator

---

THIRD REDUCE (4 parameters: a, b, c, d)
----------------------------------------
arr.reduce((a,b,c,d) => a*b*c*d)
Output: NaN 

- a = accumulator
- b = current value
- c = current index
- d = entire array [1,2,3,4,5]

- Multiplying a number by an array gives NaN
- Example: 1 × 2 × 1 × [1,2,3,4,5] = NaN

---

FOURTH REDUCE (5 parameters: a, b, c, d, e)
--------------------------------------------
arr.reduce((a,b,c,d,e) => a*b*c*d*e)
Output: NaN 

- Same reason - multiplying by the array → NaN
- e is undefined (reduce only passes 4 arguments max)

---

KEY TAKEAWAY:
Only the first reduce works as intended for multiplication.
Others include index/array which cause unexpected results!
*/


console.log(Array.from("Hello123"));
console.log(Array.from(123456));//cant be formed = []

let o={
    1:'a',
    2:'b',
    3:'c'
}
console.log(Array.from(o));//cant be formed = []
```

10_Exercise_3_11

factorial.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>

<body>
    <script>
        let n = prompt("Enter a number: ");

        let a = 1;
        for (let i = 1; i <= n; i++) {
            a *= i;
        }

        // alert("The factorial is", a); this doesnt print a
        alert(`The factorial is ${a}`)
        console.log(a)
    </script>
</body>

</html>
```

sc2.js

```js
let a=6;

// function fact(n){
//     let arr=Array.from(Array(n).keys());
//     arr=arr.map(e=>{
//         return e+1;
//     })
//     return arr.reduce((a,b)=>{
//         return a*b;
//     })
// }

function fact(n){
    let arr=Array.from(Array(n+1).keys());

    return arr.slice(1).reduce((a,b)=> a*b)//Arrow function shortcut writing without {} returns a*b
}

console.log(fact(5))
```

sc.js

```js
let arr=Array(6).keys();

console.log(arr)

for (const element of arr) {
    console.log(element)
}

console.log(Array.from(arr))//this works in cmd but not here
```

11_DOM

1_index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM</title>
</head>

<body>
    <div class="box">
        DOM - stands for Document Object Model
    </div>
    <script>
        document.title="Title"//This will be displayed instead of the DOM once the js is executed
        console.log(document.title);
        console.log(document.body);
        document.body.style.backgroundColor="green";//This will add a inline css to the page
    </script>
</body>

</html>
```

2_nodes.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM</title>
    <style>
        .box{
            width: 50px;
            height: 50px;
            border: 2px solid;
            margin: 5px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box1</div>
        <div class="box">Box2</div>
        <div class="box">Box3</div>
        <div class="box">Box4</div>
        <div class="box">Box5</div>
    </div>
    <script src="2_script.js"></script>
</body>
</html>
```

2_script.js

```js
console.log(document.body)
console.log(document.body.childNodes)
console.log(document.body.childNodes[0])
console.log(document.body.childNodes[1]) //Text aooears because the space given in the form of indetatio
  n is considered as a text node
//If a comment is there then a comment node also appears

console.log(document.body.childNodes[1].childNodes)

let cont=document.body.childNodes[1];//now cont is the container

console.log(cont.lastChild)
console.log(cont.firstChild)//All these are text childs if you want to access the first box then

console.log(cont.firstElementChild)//Elements refer to html elements

//Note: text is a child but not a element
```

```

cont.lastElementChild.style.backgroundColor="red"

console.log(cont.lastElementChild.parentElement)

console.log(cont.children)//Gives only element nodes
console.log(cont.children[1])//gives Box2

console.log(cont.children[1].nextElementSibling)//gives Box3
console.log(cont.children[1].previousElementSibling)//gives Box1
```

3_Accessing_using_class.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM</title>
    <style>
        .box{
            width: 50px;
            height: 50px;
            border: 2px solid;
            margin: 20px;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box1</div>
        <div class="box">Box2</div>
        <div class="box">Box3</div>
        <div class="box" id="#123">Box4</div>
        <div class="box">Box5</div>
    </div>
    <script src="3_script.js"></script>
</body>
</html>
```

3_script.js

```
// let boxes=document.getElementsByClassName("box")//now using this boxes variable you cs=an target the elements with box name box

// boxes[2].style.backgroundColor="aqua"

// document.getElementById("#123").style.backgroundColor="red"

// document.querySelector(".box").style.backgroundColor="red" //Selects the first element with this selector (here box1)

// document.querySelectorAll('.box').style.backgroundColor="red" this doesnt work as this returns a node list

// document.querySelectorAll('.box').forEach(e=>{
//     e.style.backgroundColor="peru"
// })

// console.log(document.getElementsByTagName("div"))//returns all div blocks as a html collection

let e=document.querySelector(".container").children

// console.log(e[1].matches(".box"))//specify the css selector in the bracket of matches ,returns true or false

// console.log(e[3].closest(".box"))//returns the closest (next Box4 )element with the class box

// console.log(e[1].closest(".container"))//checks for its neighbouors if not found then checks for its parent then also is not found then checks for its parent's parent and returns if found

console.log(document.querySelector(".container").contains(e[1]))//here it checks whether .container contains e[1] or not if contains then returns true else false
```

12_Exercise_4_12

random.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Random color</title>
    <style>
        *{
            margin: 0;
            padding: 0;
        }
        .container{
            box-sizing: border-box;
            display: flex;
            height: 100vh;
            width: 100vw;
            gap: 20px;
            align-items: center;
            justify-content: center;
            flex-direction: column;
        }

        .box{
            height: 100px;
            width: 100px;
            border: 3px solid;
            display: flex;
            align-items: center;
            justify-content: center;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="box">Box</div>
        <div class="box">Box</div>
        <div class="box">Box</div>
        <div class="box">Box</div>
        <div class="box">Box</div>

        <script src="script.js"></script>
    </div>
</body>
</html>
```

script.js

```js
// function rc(){
//     let r= Math.floor(Math.random()*1000);
//     return "#"+r.toString(16);
// } this doesnt use all the colours as it doesnt go till 16777215(white)


function rc(){
    let r= Math.floor(Math.random()*16777215);
    return "#"+r.toString(16);
}

// document.querySelectorAll(".box").forEach(e=>{
//     e.style.color=rc();
//     e.style.backgroundColor=rc();
// })

document.querySelectorAll(".box").forEach(e=>{
    r=rc();
    console.log(r);
    e.style.color=r;
    r=rc();
    console.log(r);
    e.style.backgroundColor=r;
})
```

13_Inserting_Removing_Elements

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <div class="out">
        <div class="container">
            <div class="box">This is a box</div>
            <div class="box">This is a box</div>
        </div>
        <script src="script.js"></script>
    </div>

</body>
</html>
```

script.js

```js
// console.log(document.querySelector(".container").innerHTML)//displays the html content which is inside the .container in text format

// console.log(document.querySelector(".container").outerHTML)//displays the  html content which is inside and the container itself in text format but doesnt display its parent

// console.log(document.querySelector(".container").innerText)// displays all the text content which is inside the .container

cont=document.querySelector(".container");

// console.log(cont.tagName)//Applicable only to elements
// console.log(cont.nodeName)//Applicable to text and comments also

// console.log(cont.textContent)

// console.log(cont.hidden)//returns true if the container is hidden

// cont.hidden=true//You can hide the container

//Rest refer notes
```

14_Exercise_5_13

index.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Thumbnail</title>
    <style>
        *{
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        .container {
            width: 98vw;
            height: 160px;
            background-color: rgb(0, 0, 0);
            display: flex;
            margin: 1vw;
            padding:20px 20px ;
            border-radius: 10px;
            color: rgba(255, 255, 255, 0.948);
            font-family: Roboto,"Arial",sans-serif;
        }

        .img {
            display: flex;
            width: 200px;
            align-self: center;
            position: relative;
        }

        img{
            width: 200px;
            border-radius: 10px;
            align-self: flex-end;
        }

        .content {
            display: inline-block;
            width: 85%;
            /* background-color: rgb(188, 27, 27); */
            padding: 0px 10px;
            margin: 7px 2px;
            color: rgba(255, 255, 255, 0.948);
            font-family: Roboto,"Arial",sans-serif;
        }

        .title{
            font-size:x-large;
            font-weight: bold;
        }

        .extra{
            margin-top: 15px;
            color: rgba(255, 255, 255, 0.49);
        }

        .dot{
            display: inline-block;
            border-radius: 30%;
            height: 3px;
            width: 3px;
            background-color: rgba(255, 255, 255, 0.49);
            margin: 0px 3px 3px 3px;
        }

        .timestamp{
            background-color: rgba(0, 0, 0, 0.737);
            position: absolute;
            bottom: 5px;
            right: 5px;
```

```
            padding: 3px;
            font-size: small;
            border-radius: 5px;
        }
    </style>

</head>

<body>
    <div class="container">
        <div class="img">
            <img src="Thumbnail.png" alt="image.png">
            <div class="timestamp">00:00</div>
        </div>
        <div class="content">
            <div class="title">
                Title
            </div>

            <div class="extra">
                <span class="cname">Channel Name</span>

                <span><div class="dot"></div></span>

                <span class="views">N views</span>

                <span><div class="dot"></div></span>

                <span class="date">days ago</span>
            </div>

        </div>
    </div>

    <script src="script.js"></script>
</body>

</html>
```

script.js

```
function createCard(thumbnail,title,cname,views,date,duration){
    let cont=document.body.children[0];

    let img=cont.children[0].children[0];
    img.src=thumbnail;

    let content=cont.children[1];
    content.children[0].textContent=title;

    // [span.cname, span, span.views, span, span.date]
    let extra=content.children[1];
    extra.children[0].textContent=cname;

    if(views>=1000 && views<1000000){
        views=String((views/1000).toFixed(2))+" K";
    }
    else if(views>=1000000 && views<1000000000){
        views=String((views/1000000).toFixed(2))+" M";
    }
    else if(views>=1000000000 && views<1000000000000){
        views=String((views/1000000000).toFixed(2))+" B";
    }



    extra.children[2].textContent=views+" views";

    extra.children[4].textContent=String(date)+" ago";

    cont.children[0].children[1].textContent=duration

}

createCard("https://i.ytimg.com/vi/tVzUXW6siu0/hqdefault.jpg?sqp=-oaymwEnCNACELwBSFryq4qpAxkIARUAAIhCGAH
  YAQHiAQoIGBACGAY4AUAB&rs=AOn4CLDf-xrqgUw55JSfKsoykKVhngkzRA","Hello This is an Example","CodeWithHarry
  ",2878,"7 years","23:30");
```

solution.html

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .card {
            display: flex;
            margin: 5px;
            border: 2px solid black;
            width: 900px;
            justify-content: center;
            align-items: center;
            padding: 12px;
        }

        .image img {
            width: 200px;
        }

        .text h1,
        p {
            padding: 2px 0;
        }

        .image {
            position: relative;
        }

        .image .capsule {
            position: absolute;
            bottom: 4px;
            z-index: 10;
            right: 5px;
            background: #242020;
            opacity: 0.9;
            color: white;
            padding: 3px;
            border-radius: 3px;
            font-size: 14px;
        }
    </style>
</head>

<body>
    <div class="container">
        <div class="card">
            <div class="image">
                <img src="https://i.ytimg.com/vi/tVzUXW6siu0/hqdefault.jpg?sqp=-oaymwEcCPYBEIoBSFXyq4qpAw4IARUAAIhCGAFwAcABBg==&rs=AOn4CLACwWOixJVrKLFindK92kYMgTcQbw"
                    alt="">
                <div class="capsule">31:06</div>
            </div>
            <div class="text">
                <h1>Introduction to Backend | Sigma Web Dev video #2</h1>
                <p>CodeWithHarry . 727K views . 2 months ago</p>
            </div>
        </div>
    </div>
    <script src="solution.js"></script>
</body>

</html>
```

solution.js

```
function createCard(title, cName, views, monthsOld, duration, thumbnail) {
    // Finish this function
    let viewStr
    if (views < 1000) {
        viewStr = views;
    }
    else if (views > 1000000) {
        viewStr = views / 1000000 + "M";
    }
    else {
        viewStr = views / 1000 + "K";
    }
    let html = `<div class="card">
    <div class="image">
        <img src="${thumbnail}"
            alt="">
        <div class="capsule">${duration}</div>
    </div>
    <div class="text">
        <h1>${title}</h1>
        <p>${cName} . ${viewStr} views . ${monthsOld} months ago</p>
    </div>
</div>`

    document.querySelector(".container").innerHTML = document.querySelector(".container").innerHTML + ht
      ml
}


createCard("Introduction to Backend | Sigma Web Dev video #2", "CodeWithHarry", 560000, 7, "31:22", "htt
  ps://i.ytimg.com/vi/tVzUXW6siu0/hqdefault.jpg?sqp=-oaymwEcCPYBEIoBSFXyq4qpAw4IARUAAIhCGAFwAcABBg==&rs=
  AOn4CLACwWOixJVrKLFindK92kYMgTcQbw")
```

z.js

```
const date=new Date()
console.log(date)
```

15_Events

1_index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <div class="out">
        Hey Click this below
        <br>
        <button id="b1">Click this</button>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

1_script.js

```js
let button=document.querySelector("#b1");

// button.addEventListener("click" ,()=>{
//     // alert("This was clicked")
//     document.querySelector(".out").textContent="Yayyy! this button was clicked"//You can write anythi
  ng in this and that will be done once the specified action is performed
// })

button.addEventListener("keydown",(e)=>{
    // kewdown is when you press a key on your keyboard
    //e stores the event happend
    alert(e);
    console.log(e)
    //this event happens only when you click the button and press a key on keyboard
})

document.addEventListener("keydown",(e)=>{
    alert(e);
    console.log(e);
})
```

2_eventbubbling.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .child{
            padding: 34px;
            background-color: aliceblue;
        }
        .childcontainer{
            padding: 20px;
            background-color: aquamarine;
        }
        .container{
            padding: 20px;
            background-color: brown;
        }

        .child, .childcontainer, .container{
            cursor: pointer;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="childcontainer">
            <div class="child">
```

```
                This is a child
            </div>
        </div>
    </div>

    <script>
        // document.querySelector(".child").addEventListener("click",()=>{
        //     alert("Child was clicked")
        // })
        // document.querySelector(".childcontainer").addEventListener("click",()=>{
        //     alert("Child Container was clicked")
        // })
        // document.querySelector(".container").addEventListener("click",()=>{
        //     alert("Container was clicked")
        // })

        // //here as child is inside the child container and which is inturn inside container when the c
          hild is clicked the child container and container will also be automatically clicked and the o
          ther two events connected to the childcontainer and container will automatically be bubbled al
          ong with the child event

        //to stop doing from child to its parent you must stop eventpropagation

        document.querySelector(".child").addEventListener("click",(e)=>{
            e.stopPropagation()
            alert("Child was clicked")
        })
        document.querySelector(".childcontainer").addEventListener("click",()=>{
            alert("Child Container was clicked")
        })
        document.querySelector(".container").addEventListener("click",()=>{
            alert("Container was clicked")
        })
        //now when child is clicked only the child event will be executed
        //> But when you click on the child container then the event associated with the container will
          also be executed as the propagationo is not stopped inside the childcontainer


    </script>
</body>
</html>
```

3_interval.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .child {
            padding: 34px;
            background-color: aliceblue;
        }

        .childcontainer {
            padding: 20px;
            background-color: aquamarine;
        }

        .container {
            padding: 20px;
            background-color: brown;
        }

        .child,
        .childcontainer,
        .container {
            cursor: pointer;
        }
    </style>
</head>

<body>
    <div class="container">
```

```
        <div class="childcontainer">
            <div class="child">
                This is a child
            </div>
        </div>
    </div>

    <script>

        function rc() {
            let r = Math.floor(Math.random() * 16777215);
            return "#" + r.toString(16);
        }

        let a=setInterval(()=>{
            document.querySelector(".childcontainer").style.background = rc()
        },500)//repeats the mentioned function after every 500ms

        console.log(a)//the setInterval will return a value which is stored in 'a' if clearInterval(a) i
          s written in the console then the interval function will stop repeating

        // clearInterval(a)
        //or clearInterval(1) will also do the same thing as 1 is only return by the setaINterval


    </script>
</body>

</html>
```

4_timeout.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .child {
            padding: 34px;
            background-color: aliceblue;
        }

        .childcontainer {
            padding: 20px;
            background-color: aquamarine;
        }

        .container {
            padding: 20px;
            background-color: brown;
        }

        .child,
        .childcontainer,
        .container {
            cursor: pointer;
        }
    </style>
</head>

<body>
    <div class="container">
        <div class="childcontainer">
            <div class="child">
                This is a child
            </div>
        </div>
    </div>

    <script>

        function rc() {
            let r = Math.floor(Math.random() * 16777215);
            return "#" + r.toString(16);
```

```
        }

        let a=setTimeout(()=>{
            document.querySelector(".childcontainer").style.background = rc()
        },5000)//It just performs the function once after the specified time here after 5000ms-5s when t
          he page is refreshed or opened it doesnt repeat

        //similar to the setInterval if you want to stop the timeout function before it executed use cle
          arTimeout(a) or (1)


    </script>
</body>

</html>
```

16_Asynchronous_Callbacks_Promises

1_asynchronous.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="1_script.js"></script>
</body>
</html>
```

1_script.js

```js
console.log("Sentence one")
console.log("Sentence two")

setTimeout(()=>{
    console.log("This is timeout")
})
setTimeout(()=>{
    console.log("This is timeout 2")
})

console.log("The end")

//so normally you would be thinking that first one and two would be executed then timeout 1 ans 2 and at last the end but actually wt happens  is that first one and two will be executed but when it comes to timeout it doesnt delay the execution (if delay is specified) but instead it goes to the next and executes it and then comes to timeout though 0ms delay is specified
//this is the asynchronous nature of js

//timeout function executes at last (timeout is a asynchronous function)
```

2_Callbacks.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="2_script.js"></script>
</body>
</html>
```

2_script.js

```js
console.log("Sentence one")
console.log("Sentence two")

setTimeout(()=>{
    console.log("This is timeout")
})
setTimeout(()=>{
    console.log("This is timeout 2")
})

console.log("The end")
const callback=(name)=>{
    console.log("Hello this is "+name)
}

const loadscript =(src,callback)=>{
    let s=document.createElement("script");
    s.src=src;
    s.onload=callback("Name");//here calling a function inside a function is a callback but when there are multiple callbacks i.e.calling a function inside one function and that calling another function
```

```
        inside it like this , it becomes very difficult to debugg the code so for this promises are used
    document.head.append(s);
}

let s="for(let i=0;i<10;i++){console.log(i);}"
loadscript(s,callback)
```

3_promise.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="3_promise.js"></script>
</body>
</html>
```

3_promise.js

```
let prom1=new Promise((resolve,reject)=>{
    setTimeout(()=>{
        console.log("Hey I am done");
        resolve("Info")//so here first if the timeout is executed succcessfully then the prom1 returns "
          Info"

    },5000)
})

// prom1.then((a)=>{//this means after the prom1 is executed successfully do this a has the return value
   of prom1
//     console.log(a)
// })//this kind of promise is used when you send a request to a server and after you get the info you d
  isplay it

let prom2=new Promise((resolve,reject)=>{
    let n=Math.random();
    if(n<0.1){
        reject("Request Timeout")
    }
    else{
        setTimeout(()=>{
            console.log("Hey I am done 2");
            resolve("Request Successful")

        },5)
    }
})

// prom2.then((a)=>{
//     console.log(a)
// }).catch((err)=>{//this is error handling
//     console.log(err);
// })


// let p3=Promise.all([prom1,prom2])//returns an array of the return values of prom1 and prom2 when they
   are resolved, this is not executed if any of the is rejected

// let p3=Promise.allSettled([prom1,prom2])//this returns a array of the status(whether resolved or reje
  cted) and the value if fulfilled or the reason if reected what the both have returned

let p3=Promise.race([prom1,prom2])//the promise which is settled first its result/error is showed then t
  he result/error of the next one is showed
//this race is required when there are multiple links to download a image from the server but you want t
  he one which is the fastest then the this is used

p3.then((a)=>{
    console.log(a);
}).catch((err)=>{
    console.log(err);
})
```

17_async_await

1_index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="1_script.js"></script>
</body>
</html>
```

1_script.js

```javascript
// function getdata(){
//     return new Promise((resolve,reject)=>{
//         setTimeout(() => {
//             resolve(455);
//         }, 3500);
//     })
// }

// console.log("Loading modules");
// console.log("Load data");

// let data=getdata()
// //the below one will not wait for the promise to be fulfilled the promise will be displayed pending but the below one will be already executed if you want the below one to be executed after the promise is fullfilled then one way is using then
// // data.then(()=>{

// //     console.log(data);
// //     console.log("Printing modules");
// // })

// //but this will become a callback approach another way is to use async and await

// //for using async and await the code you want to delay should be inside a async function and the function should also be a async function
//there is no need that the function should also be async


// function getdata(){
//     return new Promise((resolve,reject)=>{
//         setTimeout(() => {
//             resolve(455);
//         }, 3500);
//     })
// }

async function getdata() {
    // let x=fetch('https://jsonplaceholder.typicode.com/todos/1')
    //   .then(response => response.json())
    //   .then(json => console.log(json))

    // let x= fetch('https://jsonplaceholder.typicode.com/todos/1') if you do not put await then the below one which is data=x,json() here x is storing nothing, as fetch takes time js skips it and goes to next

    let x = await fetch('https://jsonplaceholder.typicode.com/todos/1')

    let data = await x.json() // similarly here also if await is not used data will be returned before the data is parsed and converted into json

    //x doesnt store the data it just stores the request and its status so x must be parsed and converted into a suitable format may it be json , txt

    // let data = await x.text();

    return data
}

```

```
async function main() {
    console.log("Loading Modules.............");
    let data = await getdata();//if here await is not written then Promise{pending} will be printed as t
      he promise is not settled still, if await is written then the returned value from the promise is p
      rinted as the lines are executed only after the promise is settled
    console.log(data);
    console.log("Processing data");
}

main()

// the async getdata fuction is in overall returning a promise

// settled means resolved or reject
//resolve means the promise has settled successfully
// reject means the promise has not settled successfully
```

18_Exercise_6_14

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        @import url('https://fonts.googleapis.com/css2?family=Anonymous+Pro:ital,wght@0,400;0,700;1,400;
          1,700&family=Fraunces:ital,opsz,wght@0,9..144,100..900;1,9..144,100..900&display=swap');

        body{
            overflow: hidden;
            background-color: rgb(0, 0, 0);
            position: relative;
            color: rgb(0, 255, 0);
            font-family: "Anonymous Pro", monospace;
        }

         .count{
            display: flex;
            position: absolute;
            top: 50%;
            left: 50%;
            transform: translate(-50%, -50%);
            z-index: 2;            /* Below the image (image is z-index: 3) */
            justify-content: center;
            align-items: center;
            font-size: 400px;
        }

        .cont{
            position: absolute;
            z-index: 1;
        }

        .img{
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            width: 100vw;
            margin: 0;
            padding: 0;
            filter: opacity(0.5);
            overflow: hidden;
        }
        .img img{
            position: relative;
            z-index: 3;
            height: 100%;
            width: 100%;
            object-fit: contain;
            object-position: center;


        }



    </style>
</head>
<body>
    <div class="cont">
        <div class="c_1"></div>
        <div class="c_2"></div>
        <div class="c_3"></div>
        <div class="c_4"></div>
    </div>
    <div class="count"></div>
    <div class="img">
    </div>
        <script src="script.js"></script>
```

```
</body>
</html>
```

script.js

```
async function dot(int,char=".") {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            resolve(char)
        }, int)
    })
}

async function line_1(str,sel,n1,n2=10,int=90) {

    const cont = document.querySelector(sel)
    let c = 0
    for (let i = 0; i < n1; i++) {
        cont.textContent = ">>> "+str
        for (let i = 0; i < n2; i++) {
            cont.textContent += await dot(int)
            c++;
        }
    }
}

async function main() {
    await line_1("Initializing Hacking ",".c_1",4,10,50)
    await line_1("Fetching Files ",".c_2",3)
    await line_1("Exporting Files ",".c_3",3)
    await line_1("Locking your system ",".c_4",3)

    document.querySelector(".count").textContent=await dot(0,"5")
    document.querySelector(".count").textContent=await dot(1000,"4")
    document.querySelector(".count").textContent=await dot(1000,"3")
    document.querySelector(".count").textContent=await dot(1000,"2")
    document.querySelector(".count").textContent=await dot(1000,"1")
    await dot(1000)

    document.querySelector(".count").textContent=""
    document.querySelector(".img").innerHTML="<img src=\"https://www.clker.com/cliparts/f/0/d/7/13174993
      45657265231matrix-stile-skull[1]-md.png\" alt=\"\">"


}

main()
```

19_Errors

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="script.js"></script>
</body>
</html>
```

script.js

```js
let a =prompt("Enter a number")
let b=prompt("Enter other number")

if(isNaN(a) || isNaN(b)){
    throw SyntaxError("You are not allowed to do this")//throw a syntax error in the console if the user enters a non number ,
    // when an error is thrown then the script stops there itself it doesnt go further
}
a=parseInt(a)
b=parseInt(b)
// console.log(a+b*x)//this will throw an error as x is not defined if you want to handle it then trycatch is used

function main(){

    try {
        console.log(a+b*x);
        return true
    } catch (error) {
        console.log(("There was an unexpected error"));
        return false
    }
    finally{
        console.log("Files are being closed");
    }
    //normally once returned from a function the code below it is not executed but,if you want to execute the code even after returning (like closing files and breaking the connection) in such cases finally is used, finally will be executed ven after returning but it wont be executed if an error is throwed
}

let c=main()
```

20_OOPS

1_index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="1_script.js"></script>
</body>
</html>
```

1_script.js

```javascript
let o={
    a:"apple",
    b:"bat"
}//other then these two elements and object will have an additional object element called prototype whic
  h stores additional features of the object

console.log(o)

let animal={
    eats:true,
    sleep:true
}

let rabbit={
    jumps:true
}

// now i want add eats true to rabbit also which i can do now but if there are multiple features like ea
  ts i cant sit adding each feature so instead i can append it to the rabbit prototype

rabbit.__proto__=animal
console.log(rabbit.eats,rabbit.sleep)//this will give true now and not undefined

//prototype is an additional properties of an object

// here you have only one animal rabbit but if you have multiple animals then- this is when class comes
  into picture
```

2_class.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="2_script.js"></script>
</body>
</html>
```

2_script.js

```javascript
class Animal{
    constructor(n){
        this.name=n//this refers to the object .name - you are setting a property and assigning n to it
        console.log("Object is created....");//constructor is function which is executed always
    }

    eats(){
        console.log("I am eating");
    }

    sleeps(){
        console.log("I am sleeping");
```

```
    }
}

let rabbit=new Animal("Bunny")
console.log(rabbit);
console.log(rabbit.name);
rabbit.eats()

//now you want to add lion as a class but it also has some features that of animal so instead of copy pa
  ting you can add animal as a prototype to lion using extends

class Lion extends Animal{
    // constructor(n){
    //     this.name=n;
    //     console.log("Object is created.... and he is a lion");
    // }
    // just writing the above will give an error because when you extend you must call the constructor o
      f the super class also before you run the constructor of the child class

    constructor(n){
        super(n)
        console.log("Object is created.... and he is a lion");
    }

    sleeps(){
        console.log("A lion sleeps");//this is method overriding , sleeps already exists in animal but y
          uo want to modify and you can do that , so when you call the method sleeps the method inside l
          ion will be called not the one inside animal
    }

    l_eats(){
        super.eats()//super keyword when you want to run a super class method
        console.log("I am a carnivore");
    }
}

let l=new Lion("Simba")
l.sleeps()
l.l_eats()

console.log(l instanceof Lion);//checks whether the object is a instance of a particular class or not
console.log(l instanceof Animal);//returns true as the class to which it belongs(Lion) is a child class
  of Animal so inderectly it is an instance of animal
console.log(Lion instanceof Animal);//false
console.log(rabbit instanceof Lion);
console.log(rabbit instanceof Animal);
```

3_getters_setters.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="3_script.js"></script>
</body>
</html>
```

3_script.js

```
class User{
    constructor(n){
        this.name=n;
    }

    get name(){
        // return this.name

        // return this._name//this will work
        //a getter wont work without a settor
        //if there is no setter then there will be an error
        //if there is any error in setter like before(this.name was used so Jhon was not set to this.nam
          e) in such cases the getter will return undefined
```

```
        return `Sir ${this._name}`
    }

    set name(value){
        if(value.length<4)
            console.log("Name is too short")
        else{

            // this.name=value this will give an error
            this._name=value
        }
    }
}

let jhon=new User("Jhon")//here "Jhon" will go to setter and in the setter it is checked whether the length is > 4 if true then name is set
// setter is automatically called when you are setting the name or when you update the name(here applicable to other properties also)

console.log(jhon.name)//get is automatically called when that particualar property specified with get is called (here name)

// jhon=new User("Jhon H")// one way of updating name
jhon.name=""
jhon.name="Jhon H"
console.log(jhon.name)
```

21_Advance_Js

1_IIFE.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script src="1_script.js"></script>
</body>
</html>
```

1_script.js

```js
async function demo() {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            resolve(45)
        }, 1000);
    })
}

// let a = await demo()//This will give an error as this is not inside an async function but everytime y
  ou have to create a main function and again you have to call it so you use IIFE(Immediately Invoked Fu
  nction Expresion) where it is automatically involed no need to call it separately
// console.log(a)

(async function main() {
    let a = await demo()
    console.log(a)
})()// => IIFE
```

2_destructuring.js

```js
let [x,y]=[1,5,3]
console.log(x,y);//1 and 5 will be stored in x and y and 7 will be ignored

var [a,b,...r]=[1,2,3,4,5,6,7,8]
console.log(a,b,r);//first two will be stored in a and b and the rest will be stored in the variable wri
  tten after '...' here r

let o={
    a:10,
    b:23,
    c:45,
    d:67
}

// let {e,n}=o
// console.log(e,n) undefined undefined

var {a,d}=o
console.log(a,d)
```

3_spread.js

```js
function sum(a,b,c){
    return a+b+c
}

let a=[1,2,4]
console.log(a[0]+a[1]+a[2])
console.log(sum(a[0],a[1],a[2]))
console.log(sum(...a))//easy way '...' is called the spread operator it opens the array (here)
//...a returns {0:1,1:2,2:4}
console.log(...a)
```

4_hosting.js

```
// console.log(a)//this would not give an error

// {
//     console.log(a)//though declared afterwards due to hosting(only applicable to var the interpretor moves its declaration(only declaration var a but not its value so it shows undefined not its value) to the top of that particular block) it shows no error
//     console.log(b)//error


//     var a=1;
//     let b=2;
// }


// console.log(a)//this would give an error

function hello(){
    console.log(a)//though declared afterwards due to hosting(only applicable to var the interpretor moves its declaration(only declaration var a but not its value so it shows undefined not its value) to the top of that particular block) it shows no error
    console.log(b)//error


    var a=1;
    let b=2;
}
hello()

//function is automatically hosted but not possible if declared with const or let
```

4_Backend

1_npm_and_node

myserver.js

```
// server.mjs
// import { createServer } from 'node:http';
const create = require("node:http")

const server = create.createServer((req, res) => {
  res.writeHead(200, { 'Content-Type': 'text/plain' });
  res.end('Hello World!\n');
});

// starts a simple http server locally on port 3000
server.listen(3000, '127.0.0.1', () => {
  console.log('Listening on 127.0.0.1:3000');
});

// run with `node server.mjs`
```

package.json

```
{
  "name": "server",
  "version": "1.0.0",
  "description": "this is an example",
  "repository": {
    "type": "git",
    "url": ""
  },
  "license": "ISC",
  "author": "ujwal",
  "type": "commonjs",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "dependencies": {
    "slugify": "^1.6.9"
  }
}
```

server.js

```
console.log("npm - node package manager")
// all the packages required for your node are installed using npm and when you initialize your project as a npm project "npm init" ann include -y for skipping questions
//npm creates a package.json once initialized where all the details abt your project is stored
//and the dependencies are also stored so if the modules get deleted then you can install all those using "npm install"

var slug=require("slugify")

let a=slug("hello world") //a slug is basically the url whih you see in the browsers where there cannot be spaces and some special characters so slugify converts them into a valid slug

console.log(a)

let b=slug("wejbfdskjbh ljfd shnlv khb*&^&^%$%$^&*(*(","_")
console.log(b)
```

2_Commom_js_vs_express_js

1_main.js

```
const create = require("node:http")

const server = create.createServer((req, res) => {
    //   res.writeHead(200, { 'Content-Type': 'text/plain' });//this is plain text
    //   res.end('<h1>Hello World!</h1>\n');//so h1 also will be displayed

    //whenever you make changes in the file for that to reflect on the webpage you have to restart the s
      erver
    //to avoid this you can use nodemon which automatically restarts the server if any changes are made
      in the file
    //to install it => npm install nodemon - this installs nodemon only for this folder but if you want
      it for entire system the
    //npm install --global nodemon / npm i --global nodemon
    //to start this open terminal and open(cd) the folder in which the file is there then type => nodemo
      n file.js

    res.writeHead(200, { 'Content-Type': 'text/html' });//changed to html file type //this content type
      will determine in wt format your output will be given
    res.end('<h1>Hello World!</h1>\n');
});

server.listen(3000, '127.0.0.1', () => {
    console.log('Listening on 127.0.0.1:3000');
});


//nvm => nod version manager
// when you want to use a different node version => nvm use 12 or nvm use version
//to you your version => node -v
//to install a particualar version => nvm install version
```

2_common.js

```
const create = require("node:http")//importing a modeule normal syntax
//commonjs is the default one so according to it the above one is the syntax to import a module

//there is a newer one called => module which is the newer one
//you can change the type in the package.json
```

3_module.js

```
//the main difference between module and commonjs is that

//the modules imported through commonjs load synchronously using blocking code

//the modules imported through module load asynchronously

import http from "http"//this is module syntax
```

4_1_named_export.js

```
export const a=1;// this is named export because you are exporting it using its name
export const b=2;
export const c=3;
export const d=4;
export const e=5;
```

4_2_default_export.js

```
// export default const obj={
//     x:1,
//     y:2
// } wrong



const obj = {
    x: 1,
    y: 2
```

```
}

export default obj


// const obj

// export default obj = {
//     x: 1,
//     y: 2
// } this will show an initializer error in the importing file
```

4_3_export_commonjs.js

```

let c=10;
module.exports=c;

module.exports={
    a:1,
    b:2
}

// when there are two or more exports then the one below will be exported
```

4_4_index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script type="module" src="4_importing.js"></script>
    <!-- You can run the module syntax in html also by using the above line -->
</body>
</html>
```

4_5_commonjs_as_a_function.js

```
//wt ever code you write, commonjs automatically wraps it under the following function

(function(exports, require, module, __filename, __dirname){

});

//so as it wraps we get to use the specified functionalities/variables exports, require .....

console.log(__filename)
console.log(__dirname)
```

4_import_commonjs.js

```
const v=require("./4_3_export_commonjs")
console.log(v)
```

4_importing.js

```
// import {d,a} from "4_1_named_export.js" //this will give an error
import {d,a} from "./4_1_named_export.js"

console.log(a,d);

//default export

//you can use any name

import anyname from "./4_2_default_export.js"
console.log(anyname);
```

package.json

```json
{
  "name": "2_common_vs_module",
  "version": "1.0.0",
  "description": "",
  "main": "2_common.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

3_Files

1_main.js

```
const fs=require("fs")

// console.log(fs)

//=> fs.write

// console.log("starting");
// fs.writeFileSync("file.txt","HEllofdgbc world")//creates a file by blocking code and once done moves further(synchronously)
// console.log("ending");

// console.log("\n\n")

// console.log("starting");
// fs.writeFile("file2.txt","Hello world",()=>{
//     console.log("Done")//.writeFile throws an error without a callback function TypeError cb
// })//works asynchronously
// console.log("ending");

//use .writeFile



//=> .read

// fs.writeFile("file2.txt","Hello world",()=>{
//     console.log("Done")
//     fs.readFile("file2.txt",(error,data)=>{
//         console.log(error)
//         console.log(data);
//     })
// })

//in the above one you will get error as null but
//data you will get <Buffer 48 65 6c 6c 6f 20 77 6f 72 6c 64> so you should convert it into string to make it readable

// fs.writeFile("file2.txt","Hello world",()=>{
//     console.log("Done")
//     fs.readFile("file2.txt",(error,data)=>{
//         console.log(error)
//         console.log(data.toString());
//     })
// })


//=>.append

fs.appendFile("append.txt","Adds content",(e,d)=>{
    // console.log(e)
    console.log(d)//it just appends at the end of the last line but it does not read data so this will give undefined
})
```

2_callback_hell.js

```
fs.writeFile("file2.txt", "Hello world", () => {
    console.log("Done")
    fs.readFile("file2.txt", (error, data) => {
        console.log(error)
        console.log(data.toString());
        fs.writeFile("file2.txt", "Hello world", () => {
            console.log("Done")
            fs.readFile("file2.txt", (error, data) => {
                console.log(error)
                console.log(data.toString());
                fs.writeFile("file2.txt", "Hello world", () => {
                    console.log("Done")
                    fs.readFile("file2.txt", (error, data) => {
                        console.log(error)
                        console.log(data.toString());
                        fs.writeFile("file2.txt", "Hello world", () => {
```

```
                            console.log("Done")
                            fs.readFile("file2.txt", (error, data) => {
                                console.log(error)
                                console.log(data.toString());
                                fs.writeFile("file2.txt", "Hello world", () => {
                                    console.log("Done")
                                    fs.readFile("file2.txt", (error, data) => {
                                        console.log(error)
                                        console.log(data.toString());
                                        fs.writeFile("file2.txt", "Hello world", () => {
                                            console.log("Done")
                                            fs.readFile("file2.txt", (error, data) => {
                                                console.log(error)
                                                console.log(data.toString());
                                                fs.writeFile("file2.txt", "Hello world", () => {
                                                    console.log("Done")
                                                    fs.readFile("file2.txt", (error, data) => {
                                                        console.log(error)
                                                        console.log(data.toString());
                                                    })
                                                })
                                            })
                                        })
                                    })
                                })
                            })
                        })
                    })
                })
            })
        })
    })
})

//to escape from this hell use fs/promises
```

3_mainpromises.js

```
// const fs =require("fs/promises")

// let a = await fs.readFile("append.txt")//as this is already inside a module there is no need for encl
  osing it inside a async function
//but doestnt work for commonjs

// console.log(a.toString())

import fs from "fs/promises"

let a = await fs.readFile("append.txt")
console.log(a.toString());
console.log("\n\nNext file\n\n");

let b= await fs.appendFile("append.txt","\n\n\n\nthis is an example")
console.log(a.toString(),b)
```

4_aboutpath.js

```
import path from "path"

console.log(path);

let mypath ="C:\\Users\\M U\\OneDrive\\Desktop\\My\\Web Development\\4_Backend\\3_Files\\file.txt"

console.log(path.extname(mypath));//gives the extension of the file

console.log(path.dirname(mypath));

console.log(path.basename(mypath));//gives the name of the file

```

```
13 console.log(path.join("c:/","programs\\dsafds")) //though its wrong like one is forward slash and other
     backward path module automatically joins it as per your computer
```

append.txt

```
1  Adds contentAdds contentAdds contentAdds contentAdds content
2
3
4
5  this is an example
6
7
8
9  this is an example
10
11
12
13 this is an example
```

file2.txt

```
1 Hello world
```

file.txt

```
1 HEllofdgbc world
```

package.json

```
1  {
2    "name": "3_files",
3    "version": "1.0.0",
4    "description": "",
5    "main": "main.js",
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1"
8    },
9    "keywords": [],
10   "author": "",
11   "license": "ISC",
12   "type": "module"
13 }
```

4_Expressjs

1_intro/main.js

```
// express is used instead of node for web applications because the sam code written below using http wo
  uld be more complicated but using express makes it simple
//and security issues along with it

const express = require("express")
const app =express()
const port=3000

app.use(express.static('public'))

app.get('/',(req,res)=>{
    res.send("Hello World")
})

app.get('/about',(req,res)=>{
    res.send("This is a sample page")
})

app.get('/contact',(req,res)=>{
    res.send("Ujwal A")
})

app.get('/help',(req,res)=>{
    res.send("there is nothing to help")
})

app.get('/lang',(req,res)=>{
    res.send("Programming languages")
})

// app.get('/lang/js',(req,res)=>{
//     res.send("Java Script")
// })

// app.get('/lang/py',(req,res)=>{
//     // logic to fetch Python from db
//     res.send("Python")
// })
// u acnt sit and make all similar endpoints for all programming langs so use the below

// app.get('/lang/:slug/:second',(req,res)=>{
//     res.send(`Hello this is ${req.params.slug} and ${req.params.second}`);
// })
//when you do this the one you type in the slug in the endpoint will be displayed in the params.slug and
   the one in the second in the endpoint in the params.second
//and you have to type both slug and second for this to display the given thing
//if you just type /lang/java
//it will display cannot GET java
//if you type /lang/java/python
//it will diplay Hello this is java and python

//req.params = {
//   slug: "js",
//   second: "python"
// }
//req.params returns a object that contains the root parameters/url parameters slug and second

app.get('/lang/:slug',(req,res)=>{
    res.send(`Hello this is ${req.params.slug}`)
    console.log(req.params)//contains the slug parameter here
    // console.log(req);//this contains many things including params
    console.log(req.query);
    //query is something which you write after ? after the url
    //like ex you write /lang/java?mode=dark&region=in then the req.query will be displayed as
    //{ mode: 'dark', region: 'in' }
})

app.listen(port,()=>{
    console.log(`Listening at ${port}`)
})

// Endpoint = URL path (/about)
```

```
// Block = Code executed for that endpoint (res.send(...))

// Endpoint = /users
// Block = everything inside { ... }
// Database fetch = db.getUsers() inside the block.

//one more reason why u use express instead of http
//imagine you have a file.txt in your backend folder then the if the backend uses http then you can acce
  ss the file.txt from the browser by /harry.txt and similarly the user can access the backend file also
   from the browser then it will be a blunder

//so we use express
//where the files you want to make public will be saved in separate folder public(can have any name)
//and in your backend you write app.use(express.static('public'(or wtever name you have given)))
//now the user can acccess file.txt by /file.txt only if it is in the public folder

//you can make multiple static directories


//Static files are files that are sent to the browser exactly as they are stored, without any server-sid
  e processing.
// Examples:

// HTML files (index.html)
// CSS files (style.css)
// JavaScript files (script.js)
// Images (.png, .jpg, .svg)
// Fonts (.ttf, .woff)
```

1_intro/package.json

```
{
  "name": "4_expressjs",
  "version": "1.0.0",
  "description": "",
  "main": "main.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^4.22.2"
  }
}
```

1_intro/public/file.txt

```
hello this is example file
```

2_post_request/1_main.js

```
const express=require("express")
const app=express()
const port=3000

app.use(express.static('public'))

app.get('/',(req,res)=>{
    res.send("Hello World")
    console.log("GET request");
})

//wt ever you type in the browser or wtever the browser sends to the backend by default is a get request

//the problem with get is that

//*you cant send sensitive info using get as it will apppear in logs like you want to login and u send /
  login?username=inkjbndfsds&password=122345 and you are sending this as a get request and it appears in
   logs

//* There is a size limit for get where you can send a max of 2048 characters or 8KB

```

```
app.post('/',(req,res)=>{
    console.log("Request recieved successfully");
    res.send("Post request was recieved successfully")
    console.log("Responded for the post request");
})

//similarly you can write for put request or delete request

//and you can club 2 or more requests like below
// app.get('/',(req,res)=>{
//     res.send("Hello World")
//     console.log("GET request");
// }).post('/',(req,res)=>{
//     console.log("Request recieved successfully");
//     res.send("Post request was recieved successfully")
//     console.log("Responded for the post request");
// })

//for testing post requests creating html files everytime is a bit messy so use postman for testing wher
  e you can test all types of requests

app.listen(port,()=>{
    console.log(`listening at ${port}`);
})




// API (APPLICATION PROGRAMMING INTERFACE) - FULL SUMMARY
//
// 1. WHAT IS AN API?
// - API is a server-side route system that allows communication between client and server
// - It consists of endpoint + HTTP method + logic + response
//
// 2. HOW IT WORKS (FLOW)
// - Client (browser) sends request (GET/POST/etc.)
// - Server receives request using Express route (app.get/app.post)
// - Server executes logic inside route handler
// - Server sends response back using res.send/res.json
// - Client receives and uses the response
//
// 3. API STRUCTURE IN EXPRESS
// - app.get('/route', handler)   → GET API
// - app.post('/route', handler)  → POST API
// - app.put('/route', handler)   → PUT API
// - app.delete('/route', handler) → DELETE API
//
// 4. PARTS OF AN API
// - Endpoint (URL like /users)
// - Method (GET, POST, etc.)
// - Request (data sent by client)
// - Logic (code inside route)
// - Response (res.send / res.json)
//
// 5. IMPORTANT POINTS
// - GET → fetch data
// - POST → send data
// - PUT → replace data
// - PATCH → update partial data
// - DELETE → remove data
//
// 6. KEY UNDERSTANDING
// - API is NOT the request
// - API is NOT the response
// - API is the SERVER ROUTE that handles requests and sends responses
//
// 7. EXAMPLE
//
// app.get('/about', (req, res) => {
//     res.send("About page")
// })
//
// → This entire block is an API endpoint

// So when you type localhost:3000/about => this is a request sent to a api

//api recieves it at
```

```
// app.get('/app',.....)
// and the api sends back the response
```

2_post_request/2_main.js

```
//in real peojects you wont send normal text but you send html files as reponse to the requests to the api s so

const express=require("express")
const app=express()
const port=3000

app.use(express.static('public'))

app.get('/home',(req,res)=>{
    res.sendFile("templates/index.html",{root:__dirname})
    //without specifing root the browser wont know which templetes/index.html you are referring to so either use the above or mention the entire path
})

// app.get('/home',(req,res)=>{
//     res.json({1:'a',2:'b',3:'c',4:[100,200,300,400]})
// }) this has same endpoint
//so when two have same endpoints only the first ones response will(which comes first in the code) be shown

app.get('/api',(req,res)=>{
    res.json({1:'a',2:'b',3:'c',4:[100,200,300,400],5:{1:100,2:200,3:300}})
})

app.listen(port,()=>{
    console.log(`listening at ${port}`);
})
```

2_post_request/3_main.js

```
//you are creating a hell number of api endpoints now like ex
// /lang this has some 10 further endpoints including all those in one makes is messy so we use router so that u can define all the endpoints of lang in one folder(lang.js) and then you can import those into your main file

const express=require('express')
const app=express()
const lang=require('./routes/lang')

app.use('/lang',lang)//so now all the requests to the endpoint /lang will be handeled by land
//similarly you can create as many folders and access like this

app.get('/',(req,res)=>{
    res.send("Welcome to main")
})

app.listen(3000,()=>{
    console.log("Listening at 3000");
})
```

2_post_request/package.json

```
{
  "name": "2_post_request",
  "version": "1.0.0",
  "description": "",
  "main": "2_main.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^5.2.1"
  },
  "devDependencies": {}
}
```

2_post_request/public/test_page.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Test Post Request</title>
</head>
<body>
    <script>
        async function testPost() {
            let a=await fetch('/',{method:'POST'})
            let b=await a.text()
            console.log(b);
        }
        testPost()
    </script>
</body>
</html>



<!-- The **HTML is sending the POST request**, not receiving it.

Flow:

1. Browser loads `test_page.html`.
2. `fetch('/', { method: 'POST' })` sends a POST request to the server.
3. Express receives it in:

```js
app.post('/', (req, res) => {
    res.send("Hey this is a post request")
})
```

4. The server sends back `"Hey this is a post request"`.
5. The HTML receives the response:

```js
let a = await fetch('/', { method: 'POST' })
let b = await a.text()
console.log(b)
```

So:

```text
HTML (fetch)  ---> POST Request ---> Express Server

HTML (fetch)  <--- Response ------- Express Server
```

The POST request is **created by the HTML/JavaScript**, and `main.js` **handles it and sends a response back**. -->

<!-- // FULL FLOW (FETCH → EXPRESS → RESPONSE → BACK)
//
// 1. User opens test_page.html in browser
//
// 2. Browser runs JS:
//   fetch('/', { method: 'POST' })
//
// 3. Browser sends HTTP POST request to server at '/'
//
// 4. Express server receives request:
//   app.post('/', (req, res) => { ... })
//
// 5. Inside server block:
//   - onsole.log("Request recieved successfully"); → prints in TERMINAL
//   - res.send("Post request recieved successfully") → sends response back
//   - console.log("Responded for the post request") → prints in TERMINAL
//
// 6. Browser receives response from server
```

```
//
// 7. JS reads response:
//    let b = await a.text()
//
// 8. Browser logs output:
//    console.log(b)
//    → "Hey this is a post request" -->
```

2_post_request/routes/lang.js

```
const express=require('express')
const router=express.Router()

router.get('/',(req,res)=>{
    res.send("Welcome to Languages")
})

router.get('/js',(req,res)=>{
    res.send("Welcome to JavaScript")
})

router.get('/py',(req,res)=>{
    res.send("Welcome to Python")
})

router.get('/c',(req,res)=>{
    res.send("Welcome to C")
})

router.get('/cpp',(req,res)=>{
    res.send("Welcome to C++")
})

router.get('/java',(req,res)=>{
    res.send("Welcome to Java")
})

router.get('//:slug',(req,res)=>{
    res.send(`Welcome to ${req.params.slug}`)
})

module.exports=router
```

2_post_request/templates/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Home</title>
</head>
<body>
    This is <b>Home</b>
</body>
</html>
```

3_Middle_Ware/logs.txt

```
Time - 6/11/2026, 4:34:18 PM Type - GET
Time - 6/11/2026, 4:34:18 PM Type - GET
Time - 6/11/2026, 4:34:23 PM Type - GET
Time - 6/11/2026, 4:34:23 PM Type - GET
```

3_Middle_Ware/main.js

```
const express = require('express')
const app=express()

// app.use((req,res,next)=>{
//     console.log("M1 done");//middleware is a function that takes in request, response and next and pr
  ocesses the request(may or may not)
//     next()//without next() the request will not go forward
//     //for ex you didnot write next the middleware will recieve the request processes it and then it w
  ill be there only it wont g further in the code and browser will be roaming waiting for the respose fo
```

```
    r the request
// })

// const myLogger =(req,res,next)=>{
//     console.log("M2 done");
//     next()
// }

// app.use(myLogger)//middleware can also be written like this

// // app.use((req,res,next)=>{
// //     console.log("M3 done")
// //     res.send("Ends here")//if at all you do res.send here only then there is no use of next once r
    esponded the request is done and an error will be displayed if next() is there
// //     next()
// // })

//middleware are used for storing logs

//ex
const fs=require('fs')
app.use((req,res,next)=>{
    // console.log(`Time - ${Date.now()} Type - ${req.method}`)//the date must be converted

    fs.appendFileSync('./logs.txt',`Time - ${(new Date(Date.now())).toLocaleString()} Type - ${req.metho
      d}\n`)

    console.log(`Time - ${(new Date(Date.now())).toLocaleString()} Type - ${req.method}`)//this can be s
      tored in a file also
    next()
})

app
app.get('/',(req,res)=>{
    res.send("Hello World")
})

app.listen(3000,()=>{
    console.log("Listening at 3000");
})

//Checkout expressjs.com for more info
```

3_Middle_Ware/package.json

```
{
  "name": "3_middle_ware",
  "version": "1.0.0",
  "description": "",
  "main": "main.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^5.2.1"
  }
}
```

5_Exercise_1_15

main.js

```
const fs=require('fs')
const path=require('path')

let dir="C:\\Users\\M U\\OneDrive\\Desktop\\My\\Web Development\\4_Backend\\5_Exercise_1_15\\Sample"

dir=dir+'\\'

let cont=fs.readdirSync(dir)

for (const i of cont) {
    let ext=path.extname(i)
    ext = ext.slice(1)
    let src=dir+i
    let dest=dir+ext+'\\'+i
    ext=dir+ext
    if(!fs.existsSync(ext))
        fs.mkdirSync(ext)
    fs.renameSync(src,dest)


}
```

package.json

```
{
  "name": "5_exercise_1_15",
  "version": "1.0.0",
  "description": "",
  "main": "main.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Sample/txt/file4.txt

```
(empty file)
```

6_Ejs

1_readme.md

```
ejs website

Bootstrap for readymade templates

to use ejs you save your html files as .ejs under views

ejs is basically you are using the same template but for different things
> imagine you have a website in which there are different sections like adidas puma etc
> the navbar,footer and somme elements will be same for all the sections and you cnat sit and write the nav html for all sections so we use ejs to pass variables to a single template

> You can also pass objects, arrays and iterate on them => refer ejs website

> Using ejs u can split your html into parts and at last include them in the main file


> Here in index.ejs when you are including the nav you can override the variables being passed like nprnally the siteName being Passed is 'Adidas' now but in index if you write <%- include('nav', {siteName: 'Puma'}) %>  instead of Adidas Puma will be displayed
```

index.js

```
const express=require('express')
const app=express()
const port=3000

app.set('view engine','ejs')//you are setting the view engine to ejs

app.get('/',(req,res)=>{
    let siteName="Adidas"
    let search="Search Now"
    // res.sendFile("views/index.html",{root:__dirname})
    res.render('index',{siteName,search})//this is same as { siteName: 'Adidas', search: 'Search Now' }
})

app.listen(port,()=>{
    console.log("Listening at 3000");
})
```

package.json

```
{
  "name": "6_ejs",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "ejs": "^6.0.1",
    "express": "^5.2.1"
  }
}
```

views/index.ejs

```
<!doctype html>
<html lang="en">

<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Bootstrap demo</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet"
    integrity="sha384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="ano
```

```
        nymous">
</head>

<body>

  <%- include('nav', {siteName: 'Puma'}) %>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"
    integrity="sha384-FKyoEForCGlyvwx9Hj09JcYn3nv7wiPVlz7YYwJrWVcXK/BmnVDxM+D2scQbITxI"
    crossorigin="anonymous"></script>
  <script src="index.js"></script>
</body>

</html>
```

views/nav.ejs

```
<nav class="navbar navbar-expand-lg bg-body-tertiary">
    <div class="container-fluid">
      <a class="navbar-brand" href="#"><%= siteName  %></a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarSupp
        ortedContent"
        aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarSupportedContent">
        <ul class="navbar-nav me-auto mb-2 mb-lg-0">
          <li class="nav-item">
            <a class="nav-link active" aria-current="page" href="#">Home</a>
          </li>
          <li class="nav-item">
            <a class="nav-link" href="#">Link</a>
          </li>
          <li class="nav-item dropdown">
            <a class="nav-link dropdown-toggle" href="#" role="button" data-bs-toggle="dropdown" aria-ex
              panded="false">
              Dropdown
            </a>
            <ul class="dropdown-menu">
              <li><a class="dropdown-item" href="#">Action</a></li>
              <li><a class="dropdown-item" href="#">Another action</a></li>
              <li>
                <hr class="dropdown-divider">
              </li>
              <li><a class="dropdown-item" href="#">Something else here</a></li>
            </ul>
          </li>
          <li class="nav-item">
            <a class="nav-link disabled" aria-disabled="true">Disabled</a>
          </li>
        </ul>
        <form class="d-flex" role="search">
          <input class="form-control me-2" type="search" placeholder="<%= search %>" aria-label="Search"
            />
          <button class="btn btn-outline-success" type="submit">Search</button>
        </form>
      </div>
    </div>
  </nav>
```

7_Mongodb

1_Intro/data.mongodb.js

```

use('Mongodb');

db.getCollection('courses').insertMany([
  {
    "name": "Java",
    "type": "Programming Language"
  },
  {
    "name": "Python",
    "type": "Programming Language"
  },
  {
    "name": "JavaScript",
    "type": "Programming Language"
  },
  {
    "name": "C",
    "type": "Programming Language"
  },
  {
    "name": "C++",
    "type": "Programming Language"
  },
  {
    "name": "C#",
    "type": "Programming Language"
  },
  {
    "name": "Go",
    "type": "Programming Language"
  },
  {
    "name": "Rust",
    "type": "Programming Language"
  },
  {
    "name": "PHP",
    "type": "Programming Language"
  },
  {
    "name": "Swift",
    "type": "Programming Language"
  }
]);

console.log(`Done Inserting`);
```

2_CURD/curd.mongodb.js

```
use("Curd")
//=> CREATE

db.createCollection('courses')

// db.courses.insertOne({"course":"Web Development","fee":0,"From":"Code With Harry"})

// db.courses.insertMany([
//   {
//     "course": "Web Development",
//     "fee": 0,
//     "from": "Code With Harry"
//   },
//   {
//     "course": "Python Programming",
//     "fee": 499,
//     "from": "Udemy"
//   },
//   {
//     "course": "Java Masterclass",
//     "fee": 799,
//     "from": "Coursera"
```

```
//   },
//   {
//     "course": "Data Structures and Algorithms",
//     "fee": 999,
//     "from": "GeeksforGeeks"
//   },
//   {
//     "course": "Machine Learning",
//     "fee": 1499,
//     "from": "Andrew Ng"
//   },
//   {
//     "course": "React JS",
//     "fee": 699,
//     "from": "freeCodeCamp"
//   },
//   {
//     "course": "Node.js Backend Development",
//     "fee": 899,
//     "from": "Udemy"
//   },
//   {
//     "course": "Cyber Security Fundamentals",
//     "fee": 1299,
//     "from": "Cisco Networking Academy"
//   },
//   {
//     "course": "Cloud Computing with AWS",
//     "fee": 1999,
//     "from": "AWS Training"
//   },
//   {
//     "course": "Docker and Kubernetes",
//     "fee": 1199,
//     "from": "KodeKloud"
//   }
// ])//you must insert an array of elements [{...},{....},...]

//If you run this code again then these elements will be inserted again though they exist already


//=> READ

// let a=db.courses.find({"fee":0})
// console.log(a);
// // console.log(a.toArray());

// let b=db.courses.findOne({"fee":799})//returns the first element with that value
// console.log(b);


// => UPDATE

// db.courses.updateOne({"fee":0},{$set:{"fee":100}})
// db.courses.updateMany({"fee":899},{$set:{"price":1000}})


// => DELETE

// db.courses.deleteOne({"fee":0})
// db.courses.deleteMany({"price":1000})

//for operators refer mongodb website docs
```

8_Mangoose

main.js

```
import mongoose, { get, set } from "mongoose";
import express from 'express';
import { Todo } from "./models/todo.js";

const app =express()
const port=3000

let connect= await mongoose.connect('mongodb://localhost:27017/todos')//todos will be the db name

app.use(express.static('public'))

app.get('/',(req,res)=>{
    // const todo =new Todo({title:"1st todo",desc:"This is a todo",status:false})
    const todo =new Todo({title:1,desc:"This is a todo",status:false,days:7})//even if its a number it w
      ill be inserted but it will be converted into a string and then inserted same applies for objects
      and arrays converted into a string and then inserted
    //but if at all days is a string then the mongoose will not add it and the app will crash
    todo.save()
    res.send("Hello World")
})


app.get('/json', async(req,res)=>{
    let todo= await Todo.findOne({})
    res.json({entire:todo,title:todo.title,desc:todo.desc});
})



app.listen(port,()=>{
    console.log(`Listening at ${port}`)
})
```

models/todo.js

```
import mongoose from "mongoose";

const TodoSchema = new mongoose.Schema({
    title:String,
    desc:String,
    status:Boolean,
    days:Number,
    example : {type:String,required:true,default:"Username"}//this says that the type is string and this
       is required and the default value will be taken if the field is not provided
})

//this is the advantage of mongoose you can define how the data is stored directly like here title can h
  ave only string and cannot be a number, therefore it makes inducing security easy

//but the same task with mongodb you would have to write if else statements and your code will be filled
   with if else statements, so mongoose makes it easy

export const Todo = mongoose.model('List',TodoSchema)//lists (not list and not capital) will be the coll
  lection name
```

package.json

```
{
  "name": "8_mangoose",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "type":"module",

  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "nrun":"nodemon main.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
```

```
  "dependencies": {
    "express": "^5.2.1",
    "mongoose": "^9.7.0"
  }
}
```

6_React

1_Why_react

without_react.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <div class="count">0</div>
    <div class="button"><button>Click me</button></div>

    <script>
        document.querySelector('.button').addEventListener("click",()=>{
            let n=parseInt(document.querySelector('.count').textContent)
            document.querySelector('.count').textContent=n+1
        })
    </script>
</body>
</html>
```

2_Components_Props

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href=""/>
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>basic-component-props</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "basic-component-props",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.card_box{
  display: flex;
  margin: 0px 20px;
  gap: 40px;
  justify-content: space-evenly;
  flex-wrap: wrap;
}
```

src/App.jsx

```
import { useState } from 'react'
import './App.css'
import Navbar from './components/navbar'
import Footer from './components/footer'
import Card from './components/card'

function App() {
  const [count, setCount] = useState(0)

```

```
  return (
    <>
      <Navbar/>

      <main className='card_box'>
        <Card title='Card 1' description = 'This is the description of Card 1'/>
        <Card title='Card 2' description = 'This is the description of Card 2'/>
        <Card title='Card 3' description = 'This is the description of Card 3'/>
        <Card title='Card 4' description = 'This is the description of Card 4'/>
        <Card title='Card 5' description = 'This is the description of Card 5'/>
        <Card title='Card 6' description = 'This is the description of Card 6'/>
        <Card title='Card 7' description = 'This is the description of Card 7'/>
        <Card title='Card 8' description = 'This is the description of Card 8'/>
        <Card title='Card 9' description = 'This is the description of Card 9'/>
        <Card title='Card 10' description = 'This is the description of Card 10'/>
        <Card title='Card 11' description = 'This is the description of Card 11'/>
        <Card title='Card 12' description = 'This is the description of Card 12'/>
      </main>

      <Footer/>

    </>
  )
}

export default App
```

src/components/card.css

```
.card{
    background-color: rgba(78, 78, 78, 0.252);
    width: 190px;
    height: 230px;
    border: 2px solid ;
    border-radius: 10px;
    padding: 10px;
    text-align: center;
    display: flex;
    flex-direction: column;
    gap: 50px;
}
```

src/components/card.jsx

```
import React from 'react'
import './card.css'

const card = (props) => {
  return (
    <div className="card">
        <div className="title" style={{fontSize:30}}>{props.title}</div>{/* inline css will be applied l
          ike this style={{fontSize:30}} */}
        <div className="description">{props.description}</div>
    </div>
  )
}

export default card
```

src/components/footer.css

```
.line{
    margin: 0px 10px;
    width: calc(100vw - 20px);
    border-radius: 10px;
    border: 2px solid rgba(91, 91, 91, 0.181);
}

.cont{
    position: fixed;
    bottom: 0px;
}
.footer{
    /* background-color: rgba(91, 91, 91, 0.181); */
    display: flex;
```

```
    gap: 20px;
    margin: 20px;
    width: calc(100vw - 40px);
    justify-content: center;
}

li{
    list-style: none;
}
```

src/components/footer.jsx

```
import React from 'react'
import './footer.css'

const footer = () => {
    return (
        <div className='cont'>
        <div className="line"></div>
            <div className='footer'>
                <ul>
                    <li>Copyright</li>
                    <li>NP2U.com</li>
                    <li>| All Rights Reserved</li>
                </ul>
            </div>
        </div>
    )
}

export default footer
```

src/components/navbar.css

```
nav ,ul{
    display: flex;
}

li{
    list-style: none;
}

ul{
    gap: 20px;
}

nav{
    margin: 20px;
    background-color: rgba(91, 91, 91, 0.181);
    padding: 20px;
    border-radius: 10px;
}
```

src/components/navbar.jsx

```
import React from 'react'
import './navbar.css'

const navbar = () => {
  return (
    <div>
        <nav>
            <ul>
                <li>Home</li>
                <li>About</li>
                <li>Contact</li>
            </ul>
        </nav>
    </div>
  )
}

export default navbar
```

src/index.css

```css
*{
  margin: 0;
  padding: 0;
}

body{
  background-color: black;
  color: white;
  font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
}
```

src/main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```js
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

3_Hooks_States

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
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import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>3-hooks-states</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "3-hooks-states",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
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    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
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```

src/App.css

```
(empty file)
```

src/App.jsx

```
import { useState } from 'react'
import './App.css'

function App() {
  const [count, setCount] = useState(23) //This means that create a state count and a function setCount to update the value of count and the value of the state count is set to 0 by useState(0)
  // these are called as Hooks and the above one is State Hook

  return (
    <>
      <div>The Count is {count}</div>
      <button onClick={()=>{setCount(count**2)}}>Click Me</button>
    </>

    // We cant use it like
```

```
    // <>
    // let a=0;
    //   <div>The Count is {a}</div> //Because this will already be rendered
    //   <button onClick={()=>{a=a+1}}>Click Me</button>// So even if the value of a is updated the abov
      e one will not change as it is alredy rendered
    // But useState and setState make it possible
    // </>
  )
}

export default App
```

src/index.css

```
(empty file)
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

4_useEffect

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>4-useeffect</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "4-useeffect",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState, useEffect } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'
import Navbar from './components/Navbar'
import Footer from './components/Footer'

function App() {
  const [count, setCount] = useState(0)
  const [color,setColor]=useState(0)
  const [filter,setFilter]=useState(0)

  // useEffect(()=>{
  //   alert("Welcome to the page")//this will be displayed whenever the app.jsx will be rendered
  //   //this will appear twice because in the main.jsx there is .Strictmode which will run the code twi
    se once for checking and another time really so it appears twice
  // },[])

  useEffect(() => {
    alert("count was changed")//This will appear when count is updated
    setColor(color+1)
    // setFilter(filter+1)
  }, [count])

  useEffect(()=>{
    alert("I will run on each render main")
  })


  /*
  > First when the page is opened everything will be rendered so navbar will be rendered and hence Welco
    me will run
  > Next the each render nav will run
    {
      The reason why the main render will not run is that in that as the rendering of the code is not co
        mpleted still
      }
  > next as the navbar is rendering the color is set(initialized) (note its state is not set and the col
    or changed is not running because of usestate) the color changes will run
  > Next after navbar is completly rendered it comes to the main code where Count is set Count changed w
    ill run
  > As the count changed will run, the color is updated
  > As a state is updated (need not be only color it can be a state like filter which is not used any wh
    ere - the point is setState(Basically when the satate changes) causes the entire code to be rendered
     again) the entire section is rendered again so first the main will be run and then as in the sectio
    n the navbar is also rendered the each nav render will run
  > Color was changed so Color changed will run
  > Now as the rendering of the entire code is done the each main render will run and stop
  */

  /*
  > Add a footer at the top
  > Now the Footer is rendered before Navbar so first the Filter changed will run
  > Then the useEffects in the navbar will run
  > Though the components are at the last
  > First the components will be rendered then the rest code and in the components the ine which appears
     first will be rendered first
  */


     return (
       <>
      <section id="center">
        <div className="hero">
```

```
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>
          <h1>Get started</h1>
          <p>
            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
          </p>
        </div>
        <button
          type="button"
          className="counter"
          onClick={() => setCount((count) => count + 1)}
          >
          Count is {count}
        </button>
      </section>

      <div className="ticks"></div>

      <section id="next-steps">
        <div id="docs">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#documentation-icon"></use>
          </svg>
          <h2>Documentation</h2>
          <p>Your questions, answered</p>
          <ul>
            <li>
              <a href="https://vite.dev/" target="_blank">
                <img className="logo" src={viteLogo} alt="" />
                Explore Vite
              </a>
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                  >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                  >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
```

```
                  role="presentation"
                  aria-hidden="true"
                  >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                  >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
      <Footer/>
      <Navbar color={color}/>

    </>
  )
}

export default App
```

src/components/Footer.jsx

```
import {useEffect} from 'react'

const Footer = ({filter}) => {
    useEffect(()=>{
        alert("Filter was changed")
    },[filter])
  return (
    <div>

    </div>
  )
}

export default Footer
```

src/components/Navbar.jsx

```
import {React,useEffect} from 'react'

const Navbar = ({color}) => {

    useEffect(()=>{
        alert('Welcome to the page')
        return(()=>{
            alert("This Navbar was removed or commented out")//imagine you use if else statements to rem
              ovw this component so this alert is shown when the component is removed or to experience h
              ow this happens you could just comment it out
        })
    },[])

    useEffect(()=>{
        alert('I will run on each render')//this will run twice when count is changed because first when
           the count was changed then the entire comonent <></> containing the count will be rendered ag
          ain (basically the entire page) so it will be displayed

        //second time as the count was changed the color will also be changed so the navbar component wi
          ll be rendered again so as the effect is in navbar itself it will run again
    })

```

```
    useEffect(()=>{
        alert('Color was changed')
    },[color])

  return (
    <div>Navbar</div>
  )
}

export default Navbar
```

src/index.css

```
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
```

```
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  // <StrictMode>
    <App />
  // </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

5_useRef

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>5-useref</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "5-useref",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState, useEffect, useRef } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

/*
useRef is used for
> Persisting the value of a variable over re-renders
> Accessing a DOM element

For more understanding refer react/useRef docs
*/

function App() {
  const [count, setCount] = useState(0)

  const btnRef=useRef()

  useEffect(()=>{
    console.log("First Render")
    btnRef.current.style.backgroundColor='red'//refrencing the button via btnRef
    // so here we mostly use this only and we dont use DOM manipulation => document.queryselector('...')
      .backgroundColor='red'
    btnRef.current.style.color='white'
  },[])

  return (
    <>
      <section id="center">
        <div className="hero">
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>
          <h1>Get started</h1>
          <p>
            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
          </p>
        </div>
        <button
          ref={btnRef} //now this button can be referenced in this code as btnRef
          type="button"
          className="counter"
          onClick={() => setCount((count) => count + 1)}
        >
          Count is {count}
        </button>
      </section>

      <div className="ticks"></div>

      <section id="next-steps">
        <div id="docs">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#documentation-icon"></use>
          </svg>
          <h2>Documentation</h2>
          <p>Your questions, answered</p>
          <ul>
            <li>
              <a href="https://vite.dev/" target="_blank">
                <img className="logo" src={viteLogo} alt="" />
                Explore Vite
```

```
              </a>
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/App_1.jsx

```jsx
import { useState, useEffect, useRef } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {
  const [count, setCount] = useState(0)

  // let a=0
  // useEffect(()=>{
  //   a=a+1
  //   console.log("rendering...")
  //   console.log(`The value of a is ${a}`)//here each time you do this the value of a will be 1 only
  //   // Because each time when the count value is updated the entire code is rendered again so a is se
    t to 0 again and then incremented so if if want a value to be retained then for that useRef is used
  // },[count])

  const a = useRef(0)
  useEffect(()=>{
    a.current=a.current+1// Updating a State variabe causes rerendering but of a Ref variable doesnt cau
      se rerendering
    console.log('rendering...')
    console.log(`The value of a is ${a.current}`)//note a.current should only be used
    })

  return (
    <>
      <section id="center">
        <div className="hero">
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>
          <h1>Get started</h1>
          <p>
            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
          </p>
        </div>
        <button
          type="button"
          className="counter"
          onClick={() => setCount((count) => count + 1)}
        >
          Count is {count}
        </button>
      </section>

      <div className="ticks"></div>

      <section id="next-steps">
        <div id="docs">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#documentation-icon"></use>
          </svg>
          <h2>Documentation</h2>
          <p>Your questions, answered</p>
          <ul>
            <li>
              <a href="https://vite.dev/" target="_blank">
                <img className="logo" src={viteLogo} alt="" />
                Explore Vite
              </a>
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
```

```
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/index.css

```
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
```

```
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
```

```
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  // <StrictMode>//Remove this only during development to get production time experience
    <App />
  // </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

7_Conditional_Rendering

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>7-conditional-rendering</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "7-conditional-rendering",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "@tailwindcss/vite": "^4.3.1",
    "react": "^19.2.6",
    "react-dom": "^19.2.6",
    "tailwindcss": "^4.3.1"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
```

```
}

.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
```

```
}

#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
```

```
  }

  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState, useRef, useEffect } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {
  const [count, setCount] = useState(0)
  const [btn, setbtn] = useState(0)
  const [todolist, setTodo] = useState([
    {
      id: '1',
      title: "T1",
      desc: "This is T1"
    },
    {
      id: '2',
      title: "T2",
      desc: "This is T2"
    },
    {
      id: '3',
      title: "T3",
      desc: "This is T3"
    }
  ])

  // const Todo = ({ todo }) => {
  //   return (<>

  //     {/* <div>I am a Todo</div> */}
  //     <div className="m-5 border-4">

  //       <div className="todo">{todo.title}</div>
  //       <div className="todo">{todo.desc}</div>

  //     </div>
  //   </>)
  // }

  return (
    <>
      <section id="center">
        <div className="hero">
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>

          {/* <Todo/> */}

          {/* {todolist.map(item => {
            return <Todo todo={item} />
          })} 

          {/* For the above one you will get an error saying
          react-dom_client.js?v=902f3ded:12936 Each child in a list 'should have a unique "key" prop.'
          Check the render method of `div`. It was passed a child from App. See https://react.dev/link/w
            arning-keys for more information.

          > This error appears because whenever a list is rendered we have to provide a key which is uni
            que to each element
```

```
62
63          > For the error to go you must give a key and the key should be unique for each element so ide
              ally choose id here as title is also different for each you can use it but always go for a i
              d which will be always unique (Like for databases also)  /} */}
64
65          {/* {todolist.map(item => {
66            return <Todo key={item.id} todo={item} />
67          })} */}
68
69          {/* Mostly you do not follow the above approach in which you create a separate a function and
              return it and here you access it instead you directly return the part wt you returned in the
               function like the below one */}
70
71          {todolist.map(item => {
72            return (<div key={item.id}>
73
74              {/* <div>I am a Todo</div> */}
75              <div  className="m-5 border-4">
76
77                <div className="todo">{item.title}</div>
78                <div className="todo">{item.desc}</div>
79
80              </div>
81            </div>)
82          })}
83
84          <h1>Get started</h1>
85          <p>
86            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
87          </p>
88        </div>
89        {/* { One way} */}
90        {/* {btn?<button>Shown only on a particular condition</button>:""} */}
91
92        {/* Second Way  */}
93        {btn && <button>Shown only on a particular condition</button>}
94
95        <button
96          type="button"
97          className="counter"
98          onClick={() => { setCount((count) => count + 1); setbtn(!btn) }} //this means if btn is true t
              hen set false , if false set true so=> on click the btn will be shown on second click it wil
              l dissapear
99        >
100         Count is {count}
101       </button>
102     </section>
103
104     <div className="ticks"></div>
105
106     <section id="next-steps">
107       <div id="docs">
108         <svg className="icon" role="presentation" aria-hidden="true">
109           <use href="/icons.svg#documentation-icon"></use>
110         </svg>
111         <h2>Documentation</h2>
112         <p>Your questions, answered</p>
113         <ul>
114           <li>
115             <a href="https://vite.dev/" target="_blank">
116               <img className="logo" src={viteLogo} alt="" />
117               Explore Vite
118             </a>
119           </li>
120           <li>
121             <a href="https://react.dev/" target="_blank">
122               <img className="button-icon" src={reactLogo} alt="" />
123               Learn more
124             </a>
125           </li>
126         </ul>
127       </div>
128       <div id="social">
129         <svg className="icon" role="presentation" aria-hidden="true">
130           <use href="/icons.svg#social-icon"></use>
131         </svg>
132         <h2>Connect with us</h2>
133         <p>Join the Vite community</p>
```

```
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/index.css

```
@import 'tailwindcss';

:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;
```

```
15
16    --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
17    --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
18    --mono: ui-monospace, Consolas, monospace;
19
20    font: 18px/145% var(--sans);
21    letter-spacing: 0.18px;
22    color-scheme: light dark;
23    color: var(--text);
24    background: var(--bg);
25    font-synthesis: none;
26    text-rendering: optimizeLegibility;
27    -webkit-font-smoothing: antialiased;
28    -moz-osx-font-smoothing: grayscale;
29
30    @media (max-width: 1024px) {
31      font-size: 16px;
32    }
33  }
34
35  @media (prefers-color-scheme: dark) {
36    :root {
37      --text: #9ca3af;
38      --text-h: #f3f4f6;
39      --bg: #16171d;
40      --border: #2e303a;
41      --code-bg: #1f2028;
42      --accent: #c084fc;
43      --accent-bg: rgba(192, 132, 252, 0.15);
44      --accent-border: rgba(192, 132, 252, 0.5);
45      --social-bg: rgba(47, 48, 58, 0.5);
46      --shadow:
47        rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
48    }
49
50    #social .button-icon {
51      filter: invert(1) brightness(2);
52    }
53  }
54
55  body {
56    margin: 0;
57  }
58
59  #root {
60    width: 1126px;
61    max-width: 100%;
62    margin: 0 auto;
63    text-align: center;
64    border-inline: 1px solid var(--border);
65    min-height: 100svh;
66    display: flex;
67    flex-direction: column;
68    box-sizing: border-box;
69  }
70
71  h1,
72  h2 {
73    font-family: var(--heading);
74    font-weight: 500;
75    color: var(--text-h);
76  }
77
78  h1 {
79    font-size: 56px;
80    letter-spacing: -1.68px;
81    margin: 32px 0;
82    @media (max-width: 1024px) {
83      font-size: 36px;
84      margin: 20px 0;
85    }
86  }
87  h2 {
88    font-size: 24px;
89    line-height: 118%;
90    letter-spacing: -0.24px;
91    margin: 0 0 8px;
92    @media (max-width: 1024px) {
```

```
    font-size: 20px;
  }
}
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react(),tailwindcss()],
})
```

8_Exercise_18_1

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>8-exercise-18-1</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "8-exercise-18-1",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {

    "react": "^19.2.6",
    "react-dom": "^19.2.6"

  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@tailwindcss/vite": "^4.3.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12",
    "tailwindcss": "^4.3.1"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) us
  es [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react
  -swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To
   add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules
   enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/te
  mplate-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://type
  script-eslint.io) in your project.
```

src/App.css

```
(empty file)
```

src/App.jsx

```
import { useState,useEffect } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {
  const [card, setCard] = useState([])

  async function fetchData() {
      let x= await fetch('https://jsonplaceholder.typicode.com/posts')
```

```
      let y=await x.json()
      setCard(y)
      console.log(y)
    }

  useEffect(()=>{
    fetchData()
  },[])

  return (
    <div className='text-white bg-black'>
      <div className="flex bg-black flex-wrap justify-between gap-y-20">
        {card.map((item)=>{
          return(<div key={item.id} className="border-3 w-80 h-100">
              <div>UID :- {item.userId}</div>
              <div><h2 className='text-center mt-10'>{item.title}</h2></div>
              <div className='mt-10 text-center'>{item.body}</div>
          </div>)
        })}
      </div>
    </div>
  )
}

export default App
```

src/index.css

```
@import 'tailwindcss';
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react(),tailwindcss()],
})
```

9_Events

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>9-events</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "9-events",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@tailwindcss/vite": "^4.3.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "tailwindcss": "^4.3.1",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
(empty file)
```

src/App.jsx

```
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {
  const [count, setCount] = useState(0)
  // const [name, setname] = useState("Name")
  // const [form, setForm] = useState({email:"",phone:""})
  const [form, setForm] = useState({})// if you set it to empty then you should use the ternary operator or else you will get error (see the input tag)

```

```
  const handleClick = () => {
    alert("Button was Clicked")
  }

  const handleMouseOver = () => {
    alert("Mouse on div")
  }

  // const handleInput = (e) => {
  //   setname(e.target.value)
  // }

  // const handleInput = (e) => {
  //   setForm(e.target.name)
  // }//Doing this for the email phone whould replace wt ever you type inplace of the entire object instead of updating the object

  const handleInput = (e) => {
    setForm({...form,[e.target.name]:e.target.value})// ...form evaluates into { } where into you pass name:value

    // U cant write e.target.name: e.target.value, [] are must
    // {...form} returns { email: 'dsfcawfs', phone: '34253234567' }

    console.log(form)//type and see in the console => Reason for that is setForm is asynchronos so console.log runs before it
  }

  return (
    <div className='flex h-screen bg-neutral-800 justify-center items-center text-white flex-col'>
      <div>
        <button onClick={handleClick} className='px-5 py-2 bg-black rounded-[10px]'>Click Me</button>
      </div>

      <div onMouseOver={handleMouseOver} className="h-50 w-50 text-center bg-red-400"> This is a div</div>

      {/* <input type="text" className='bg-black m-3' value={name} /> When you set this value the Name will appear in the textbox and you cannot change it neither add or remove wt is there so you have to handle this event to allow to type in the input */}
      {/* <input type="text" className='bg-black m-3' value={name} onInput={handleInput}/> */}
      {/* <input type="text" className='bg-black m-3' value={name} onChange={handleInput}/>Both are same */}

      {/* <input type="text" className='bg-black m-3' name='email' value={form.email} onChange={handleInput}/>
      <input type="text" className='bg-black m-3' name='phone' value={form.phone} onChange={handleInput} /> */}

      <input type="text" className='bg-black m-3' name='email' value={form.email?form.email:""} onChange={handleInput}/>
      <input type="text" className='bg-black m-3' name='phone' value={form.phone?form.phone:""} onChange={handleInput}/>
    </div>
  )
}

export default App
```

src/index.css

```
@import 'tailwindcss';
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react(),tailwindcss()],
})
```

10_React_Router

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>10-react-router</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "10-react-router",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6",
    "react-router-dom": "^7.18.0"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@tailwindcss/vite": "^4.3.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "tailwindcss": "^4.3.1",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

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- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
(empty file)
```

src/App.jsx

```
import { useState } from 'react'
import './App.css'
import { createBrowserRouter, RouterProvider } from 'react-router-dom'
import Navbar from './components/Navbar'
import Home from './components/Home'
import Login from './components/Login'
import About from './components/About'
import User from './components/User'

// When you have multiple pages in your website and you use anchor tag to access from your main dashboard, the problem which comes into picture is that each time you click on the tag your page reloads so to avoid reloading of the page react router is used
```

```

function App() {
  const [count, setCount] = useState(0)
  const router = createBrowserRouter([
    {
      path: '/',
      element: <><Navbar /><Home /></>
    },
    {
      path: '/login',
      element: <><Navbar /><Login /></>
    },
    {
      path: '/about',
      element: <><Navbar /><About /></>
    },
    {
      path: '/user/:username',// This like u have to use /user/ujwal
      element: <><Navbar /><User /></>
    }
  ])

  return (
    <>
      <RouterProvider router={router} />
    </>
  )
}

export default App
```

src/components/About.jsx

```
import React from 'react'

const About = () => {
  return (
    <div>I am About</div>
  )
}

export default About
```

src/components/Home.jsx

```
import React from 'react'

const Home = () => {
  return (
    <div>
      I am Home
    </div>
  )
}

export default Home
```

src/components/Login.jsx

```
import React from 'react'

const Login = () => {
  return (
    <div>
      I am Login
    </div>
  )
}

export default Login
```

src/components/Navbar.jsx

```jsx
import React from 'react'
// import { Link } from 'react-router-dom'
import { NavLink } from 'react-router-dom' // We use this so that when a particualr page is active we ca
  n bring some classes into effect

const Navbar = () => {
  return (
    <div>
        {/* <a href="/">Home</a> instead of this the below is used */}
        <nav>
        {/* <Link to='/'><li>Home</li></Link>
        <Link to='/login'><li>Login</li></Link>
        <Link to='/about'><li>About</li></Link> */}

        <NavLink to='/' className={(e)=>{return e.isActive?"red":""}}><li>Home</li></NavLink>
        <NavLink to='/login' className={(e)=>{return e.isActive?"red":""}}><li>Login</li></NavLink>
        <NavLink to='/about' className={(e)=>{return e.isActive?"red":""}}><li>About</li></NavLink>
        </nav>
        {/* Link is not a new html tag it is also converted into a anchor tag at runtime */}
    </div>
  )
}

export default Navbar
```

src/components/User.jsx

```jsx
import React from 'react'
import { useParams } from 'react-router-dom'

const User = () => {
    const use=useParams()
  return (
    <div>
      I am user {use.username}
    </div>
  )
}

export default User
```

src/index.css

```css
*{
  margin: 0;
  padding: 0;
}
nav{

  list-style-type: none;
  display: flex;
  background-color: black;
  /* color: white; */
}

li{
  padding: 24px;
  color: white;
}

.red{
  background-color: rebeccapurple;
}
```

src/main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
```

```
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

11_useContext_hook

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>11-usecontext-hook</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```json
{
  "name": "11-usecontext-hook",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```md
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```css
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'
import Navbar from './components/Navbar'
import { counterContext } from './context/context'

function App() {
  const [count, setCount] = useState(0)

  return (
    <>
      {/* <counterContext.Provider value={count}> what if you want to make setCount also available in th
        e Button so that even if you click on that button the count changes, then pass a object*/}
      <counterContext.Provider value={{count,setCount}}>


        <Navbar count={count} />
        <section id="center">
          <div className="hero">
            <img src={heroImg} className="base" width="170" height="179" alt="" />
            <img src={reactLogo} className="framework" alt="React logo" />
            <img src={viteLogo} className="vite" alt="Vite logo" />
          </div>
          <div>
            <h1>Get started</h1>
            <p>
              Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
            </p>
          </div>
          <button
            type="button"
            className="counter"
            onClick={() => setCount((count) => count + 1)}
          >
            Count is {count}
          </button>
        </section>

        <div className="ticks"></div>

        <section id="next-steps">
          <div id="docs">
            <svg className="icon" role="presentation" aria-hidden="true">
              <use href="/icons.svg#documentation-icon"></use>
            </svg>
            <h2>Documentation</h2>
            <p>Your questions, answered</p>
            <ul>
              <li>
                <a href="https://vite.dev/" target="_blank">
                  <img className="logo" src={viteLogo} alt="" />
                  Explore Vite
                </a>
              </li>
              <li>
                <a href="https://react.dev/" target="_blank">
                  <img className="button-icon" src={reactLogo} alt="" />
                  Learn more
                </a>
              </li>
            </ul>
          </div>
          <div id="social">
```

```
            <svg className="icon" role="presentation" aria-hidden="true">
              <use href="/icons.svg#social-icon"></use>
            </svg>
            <h2>Connect with us</h2>
            <p>Join the Vite community</p>
            <ul>
              <li>
                <a href="https://github.com/vitejs/vite" target="_blank">
                  <svg
                    className="button-icon"
                    role="presentation"
                    aria-hidden="true"
                  >
                    <use href="/icons.svg#github-icon"></use>
                  </svg>
                  GitHub
                </a>
              </li>
              <li>
                <a href="https://chat.vite.dev/" target="_blank">
                  <svg
                    className="button-icon"
                    role="presentation"
                    aria-hidden="true"
                  >
                    <use href="/icons.svg#discord-icon"></use>
                  </svg>
                  Discord
                </a>
              </li>
              <li>
                <a href="https://x.com/vite_js" target="_blank">
                  <svg
                    className="button-icon"
                    role="presentation"
                    aria-hidden="true"
                  >
                    <use href="/icons.svg#x-icon"></use>
                  </svg>
                  X.com
                </a>
              </li>
              <li>
                <a href="https://bsky.app/profile/vite.dev" target="_blank">
                  <svg
                    className="button-icon"
                    role="presentation"
                    aria-hidden="true"
                  >
                    <use href="/icons.svg#bluesky-icon"></use>
                  </svg>
                  Bluesky
                </a>
              </li>
            </ul>
          </div>
        </section>

        <div className="ticks"></div>
        <section id="spacer"></section>
      </counterContext.Provider>{/* Now all the components wrapped under this tag will be able to access
         the value passed to this */}
    </>
  )
}

export default App
```

src/components/Button.jsx

```jsx
import React from 'react'
import Component1 from './component1'
import { useContext } from 'react'
import { counterContext } from '../context/context'

const Button = () => {
    const value=useContext(counterContext)
  return (
    <div>
      <button onClick={() => value.setCount((count) => count + 1)}><span><Component1/></span>I am a Butt
        ton</button>
    </div>
  )
}

export default Button
```

src/components/component1.jsx

```jsx
import React from 'react'
import { counterContext } from '../context/context'
import { useContext } from 'react'

const component1 = () => {
    const value =useContext(counterContext)
  return (
    <div>
      {value.count}
    </div>
  )
}

export default component1
```

src/components/Navbar.jsx

```jsx
import React from 'react'
import Button from './Button'

const Navbar = () => {
  return (
    <div>
        This is Navbar
      <Button/>
    </div>
  )
}

export default Navbar
```

src/context/context.js

```jsx
import { createContext } from "react";

export const counterContext =createContext(0)
```

src/index.css

```css
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
```

```
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
```

```
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

src/Without_Context/components/Button.jsx

```
import React from 'react'
import Component1 from './component1'

const Button = ({count}) => {
  return (
    <div>
      <button><span><Component1 count={count}/></span>I am a Buttton</button>
    </div>
  )
}

export default Button
```

src/Without_Context/components/component1.jsx

```
import React from 'react'

const component1 = ({count}) => {// Using {count} directly without using props(this is a object) and the
  n props.count is called destructuring

  return (
    <div>
      {count} {/* To get this count from App.jsx you are prop drilling from App => Navbar => Button => c
        omponent1 , instead of drilling it from these many pages u can directly access it from App with
        the help of useContext hook */}
    </div>
  )
}

export default component1
```

src/Without_Context/components/Navbar.jsx

```
import React from 'react'
import Button from './Button'

const Navbar = ({count}) => {
  return (
    <div>
        This is Navbar
      <Button count={count}/>
```

```
    </div>
  )
}

export default Navbar
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

12_useMemo_hook

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>12-usememo-hook</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```json
{
  "name": "12-usememo-hook",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

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This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

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## React Compiler

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## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```css
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```
import { useState, useMemo } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {

  /*
  > [10,20,30,40,50].map((e,i)=>{return {element:e,index:i}})
[
  { element: 10, index: 0 },
  { element: 20, index: 1 },
  { element: 30, index: 2 },
  { element: 40, index: 3 },
  { element: 50, index: 4 }
  ]
  > Array(30_000).fill(0)
  [
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 0,
    ... 29900 more items
    ]
    > Array(30_000).fill(0).length
    30000
    >
    */

  // const nums = new Array(30_000_000).fill(0).map((_, i) => {
  //   return {
  //     index: i,
  //     isMagical: i === 29_000_000 // so this will run till the 29 million th index, its a very heavy computation
  //   }
  // })

  /*
  Type ".help" for more information.
  > Array(30_000_000).fill(0).map((_,i)=>{
    | return {
      | index: i,
      | isMagical:i===29_000_000
      | }})
      [
        { index: 0, isMagical: false },
        { index: 1, isMagical: false },
        { index: 2, isMagical: false },
        { index: 3, isMagical: false },
        { index: 4, isMagical: false },
        { index: 5, isMagical: false },
        { index: 6, isMagical: false },
        { index: 7, isMagical: false }, .....so on
        */
  const [count, setCount] = useState(0)


  // const magical = number.find(item => item.isMagical === true)// or can be directly written as item.isMagical
```

```
64
65  // The above heavy computation will take place again and again as the count changes so the entire page
       re renders again even though the answer is same everytime, when you open the page itself it takes t
    ime to open and when you click on the count the count increasing will also take time sometimes gets
    stuck, so basically your page doesnt slow or (its responsiveness decreases)
66  // So to tackle this problem we use useMemo
67
68  // const magical = useMemo(()=>number.find(item => item.isMagical),[])// This says that memoise the fi
    nal return of the function provided, so that the function is not executed again and again between re
    -renders, So useMemo is used when there is a heavy computation which you do not want to run again an
    d again between re-renders
69
70  // Even after using the above thing the website is still slow because
71  // 1=> on each render the const num reuns again and again so use useMemo on that
72
73  //   return new Array(30_000_000).fill(0).map((_, i) => ({
74  //     index: i,
75  //     isMagical: i === 29_000_000
76  //   }))
77  // }, [])
78  // const [number, setNumber] = useState(nums)
79  // const magical = useMemo(()=>number.find(item => item.isMagical),[])
80
81  // the above one is still slow because
82  // 2=> the const [number, setNumber] = useState(nums) also runs again and again generating array again
       and again
83
84  const magical = useMemo(() => {
85    return new Array(30_000_000)
86      .fill(0)
87      .map((_, i) => ({
88        index: i,
89        isMagical: i === 29_000_000
90      }))
91      .find(item => item.isMagical)
92  }, []) // the square bracket one is the dependence variable whre if the provided one changes the useMe
    mo will run and the magical will be updated, like here store 30_000_000 or 29_000_000 in a variable
    provide any one and if it changes, it will be claculated again
93
94  return (
95    <>
96      <div>The magical number is {magical.index}</div>
97      <section id="center">
98        <div className="hero">
99          <img src={heroImg} className="base" width="170" height="179" alt="" />
100         <img src={reactLogo} className="framework" alt="React logo" />
101         <img src={viteLogo} className="vite" alt="Vite logo" />
102       </div>
103       <div>
104         <h1>Get started</h1>
105         <p>
106           Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
107         </p>
108       </div>
109       <button
110         type="button"
111         className="counter"
112         onClick={() => setCount((count) => count + 1)}
113       >
114         Count is {count}
115       </button>
116     </section>
117
118     <div className="ticks"></div>
119
120     <section id="next-steps">
121       <div id="docs">
122         <svg className="icon" role="presentation" aria-hidden="true">
123           <use href="/icons.svg#documentation-icon"></use>
124         </svg>
125         <h2>Documentation</h2>
126         <p>Your questions, answered</p>
127         <ul>
128           <li>
129             <a href="https://vite.dev/" target="_blank">
130               <img className="logo" src={viteLogo} alt="" />
131               Explore Vite
132             </a>
```

```
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/index.css

```
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
```

```
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

13_useCallback_hook

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>13-usecallback-hook</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "13-usecallback-hook",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.6",
    "react-dom": "^19.2.6"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState, useCallback } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'
import Navbar from './components/Navbar'

function App() {
  const [count, setCount] = useState(0)
  const [adjective, setAdjective] = useState("good")
  // Refer video

  // const getAdj=()=>{
  //   return "another"
  // }
  // When you passed only the adjective the navbar was not rendering again and again
  // But when you pass the function it is rendering again and again as here the function each time it ru
    ns js considers to be a different function so each time as the function passed is being considered d
    iff the navbar is rendering again and again
  // So we use callback to tell js that the function is same so that navbar is not rendered again and ag
    ain

  const getAdj=useCallback(()=>{
    return "another"
  },[])// []=> Dependence variable whn you want the navbar to be rendered , you can add more variables i
    n it if you want
  return (
    <>
    <Navbar adjective={adjective} getAjective={getAdj}/>
      <section id="center">
        <div className="hero">
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>
          <h1>Get started</h1>
          <p>
            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
          </p>
        </div>
        <button
          type="button"
          className="counter"
          onClick={() => setCount((count) => count + 1)}
        >
          Count is {count}
        </button>
      </section>

      <div className="ticks"></div>

      <section id="next-steps">
        <div id="docs">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#documentation-icon"></use>
          </svg>
          <h2>Documentation</h2>
          <p>Your questions, answered</p>
          <ul>
            <li>
              <a href="https://vite.dev/" target="_blank">
                <img className="logo" src={viteLogo} alt="" />
                Explore Vite
              </a>
```

```
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/components/Navbar.jsx

```jsx
import React,{memo} from 'react'

const Navbar = ({adjective, getAjective}) => {
    console.log("Navbar rendered")
  return (
    <div>
      I am a good {adjective}
      <button onClick={()=>{getAjective()}}>Get</button>
    </div>
  )
}

export default memo(Navbar)
```

src/index.css

```css
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
```

```
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
p {
  margin: 0;
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(

    <App />

)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

14_Forms

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

backend/script.js

```
import express from 'express';
import cors from 'cors'
import bodyParser from 'body-parser';// not required express parser does the work
const app =express()
const port=3000

app.use(express.static('public'))
app.use(cors())
app.use(express.json())

app.get('/',(req,res)=>{
    res.send("Hello World")
})

app.post('/',(req,res)=>{
    console.log(req.body)
    res.send(req.body)
})


app.listen(port,()=>{
    console.log(`Listening at ${port}`)
})
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
```

```
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>14-forms</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "14-forms",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "body-parser": "^2.3.0",
    "cors": "^2.8.6",
    "express": "^5.2.1",
    "react": "^19.2.6",
    "react-dom": "^19.2.6",
    "react-hook-form": "^7.80.0"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.1",
    "eslint": "^10.3.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.6.0",
    "vite": "^8.0.12"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules
```

```
 enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```css
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

.red{
  color: red;
  font-size: 10px;
}

.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

```

```
  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
```

```
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```
import { useState } from 'react'
import './App.css'
import { useForm } from 'react-hook-form'

function App() {
  const [count, setCount] = useState(0)
  const {
    register,
    handleSubmit,
    formState: { errors,isSubmitting },
    setError

  } = useForm()

  const delay=(d)=>{
    return new Promise((resolve,reject)=>{
      setTimeout(()=>{
        resolve()
      },d*1000)
    })
  }// Function to stimulate the delay when data submitted to backend

  const Submit = async data => {
    // await delay(3)// Here there is a delay of 3 set wt if the user submits the form multiple times th
      inking its not submitted(not logged in the console), to avoid this isSubmitting is used

    let r = await fetch("http://localhost:3000",{method:"POST",headers:{"Content-Type": "application/jso
      n"},body:JSON.stringify(data)})
    // you cant do b.toString() cause it gives '[object Object]' headers is mandatory so that the expres
      s parser knows that the content-type is application or json so that it changes accordingly

    let res=await r.text()
    console.log(res)

```

```
    // if(data.username!=="Shubam")
    //   setError("myform",{message:"Invalid Credentials"})
    // if(data.username==="Rohan")
    //   setError("blocked",{message:"Access Denied"})
  }
  return (
    <>
      {/* <div>
        <form action="">
          <input type="email" name="" id="" />
          <input type="password" name="password" id="" />
          <input type="submit" value="Submit" />
        </form>
      </div>
      Making validations with traditiona forms using if else statement will make the script longer and c
        omplex so to make it easy use react-hook-form */}

      <div>
        <form action="" onSubmit={handleSubmit(Submit)}>
          {/* <input type="text" {...register('username', { required: true, minLength: 3, maxLength: 20
            })} />
          {errors.username && <div className='red'>There was an error</div>} */}
          {/* Here above there constraints are put that 1=>it is required, you cant submit it without fi
            lling the field 2=>minlength and 3=>maxlength

          Here if any one is not satisfied the error will be displayed if you want a specific message fo
            r each case then follow the below one */}

          {isSubmitting && <div>Submitting...</div>}
          <input type="text" {...register('username', { required: {value:true, message:"This field is re
            quired"}, minLength: {value:3, message:"Enter a min of 3 characters"}, maxLength: {value:8,
            message:"You can enter a max of 8 characters"} })} />
          {errors.username && <div className='red'>{errors.username.message}</div>}
          <br />
          <input type="password" {...register('password', { required: {value:true, message:"This field i
            s required"}, minLength: {value:3, message:"Enter a min of 8 characters"}})} />
          {errors.username && <div className='red'>{errors.username.message}</div>}
          {errors.myform && <div className='red'>{errors.myform.message}</div>}
          {errors.blocked && <div className='red'>{errors.blocked.message}</div>}
          <br />
          <input disabled={isSubmitting} type="submit" value="Submit" />
        </form>
      </div>
    </>
  )
}

export default App
```

src/index.css

```
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
```

```
27
28    @media (max-width: 1024px) {
29      font-size: 16px;
30    }
31  }
32
33  @media (prefers-color-scheme: dark) {
34    :root {
35      --text: #9ca3af;
36      --text-h: #f3f4f6;
37      --bg: #16171d;
38      --border: #2e303a;
39      --code-bg: #1f2028;
40      --accent: #c084fc;
41      --accent-bg: rgba(192, 132, 252, 0.15);
42      --accent-border: rgba(192, 132, 252, 0.5);
43      --social-bg: rgba(47, 48, 58, 0.5);
44      --shadow:
45        rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
46    }
47
48    #social .button-icon {
49      filter: invert(1) brightness(2);
50    }
51  }
52
53  body {
54    margin: 0;
55  }
56
57  #root {
58    width: 1126px;
59    max-width: 100%;
60    margin: 0 auto;
61    text-align: center;
62    border-inline: 1px solid var(--border);
63    min-height: 100svh;
64    display: flex;
65    flex-direction: column;
66    box-sizing: border-box;
67  }
68
69  h1,
70  h2 {
71    font-family: var(--heading);
72    font-weight: 500;
73    color: var(--text-h);
74  }
75
76  h1 {
77    font-size: 56px;
78    letter-spacing: -1.68px;
79    margin: 32px 0;
80    @media (max-width: 1024px) {
81      font-size: 36px;
82      margin: 20px 0;
83    }
84  }
85  h2 {
86    font-size: 24px;
87    line-height: 118%;
88    letter-spacing: -0.24px;
89    margin: 0 0 8px;
90    @media (max-width: 1024px) {
91      font-size: 20px;
92    }
93  }
94  p {
95    margin: 0;
96  }
97
98  code,
99  .counter {
100   font-family: var(--mono);
101   display: inline-flex;
102   border-radius: 4px;
103   color: var(--text-h);
104 }
```

```

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(

    <App />

)
```

vite.config.js

```js
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

15_Redux

.gitignore

```
# Logs
logs
*.log
npm-debug.log*
yarn-debug.log*
yarn-error.log*
pnpm-debug.log*
lerna-debug.log*

node_modules
dist
dist-ssr
*.local

# Editor directories and files
.vscode/*
!.vscode/extensions.json
.idea
.DS_Store
*.suo
*.ntvs*
*.njsproj
*.sln
*.sw?
```

eslint.config.js

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      globals: globals.browser,
      parserOptions: { ecmaFeatures: { jsx: true } },
    },
  },
])
```

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>15-redux</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

package.json

```
{
  "name": "15-redux",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.2.7",
    "react-dom": "^19.2.7"
  },
  "devDependencies": {
    "@eslint/js": "^10.0.1",
    "@types/react": "^19.2.17",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.2",
    "eslint": "^10.5.0",
    "eslint-plugin-react-hooks": "^7.1.1",
    "eslint-plugin-react-refresh": "^0.5.3",
    "globals": "^17.6.0",
    "vite": "^8.1.0"
  }
}
```

README.md

```
# React + Vite

This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.

Currently, two official plugins are available:

- [@vitejs/plugin-react](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react) uses [Oxc](https://oxc.rs)
- [@vitejs/plugin-react-swc](https://github.com/vitejs/vite-plugin-react/blob/main/packages/plugin-react-swc) uses [SWC](https://swc.rs/)

## React Compiler

The React Compiler is not enabled on this template because of its impact on dev & build performances. To add it, see [this documentation](https://react.dev/learn/react-compiler/installation).

## Expanding the ESLint configuration

If you are developing a production application, we recommend using TypeScript with type-aware lint rules enabled. Check out the [TS template](https://github.com/vitejs/vite/tree/main/packages/create-vite/template-react-ts) for information on how to integrate TypeScript and [`typescript-eslint`](https://typescript-eslint.io) in your project.
```

src/App.css

```
.counter {
  font-size: 16px;
  padding: 5px 10px;
  border-radius: 5px;
  color: var(--accent);
  background: var(--accent-bg);
  border: 2px solid transparent;
  transition: border-color 0.3s;
  margin-bottom: 24px;

  &:hover {
    border-color: var(--accent-border);
  }
  &:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
}

```

```
.hero {
  position: relative;

  .base,
  .framework,
  .vite {
    inset-inline: 0;
    margin: 0 auto;
  }

  .base {
    width: 170px;
    position: relative;
    z-index: 0;
  }

  .framework,
  .vite {
    position: absolute;
  }

  .framework {
    z-index: 1;
    top: 34px;
    height: 28px;
    transform: perspective(2000px) rotateZ(300deg) rotateX(44deg) rotateY(39deg)
      scale(1.4);
  }

  .vite {
    z-index: 0;
    top: 107px;
    height: 26px;
    width: auto;
    transform: perspective(2000px) rotateZ(300deg) rotateX(40deg) rotateY(39deg)
      scale(0.8);
  }
}

#center {
  display: flex;
  flex-direction: column;
  gap: 25px;
  place-content: center;
  place-items: center;
  flex-grow: 1;

  @media (max-width: 1024px) {
    padding: 32px 20px 24px;
    gap: 18px;
  }
}

#next-steps {
  display: flex;
  border-top: 1px solid var(--border);
  text-align: left;

  & > div {
    flex: 1 1 0;
    padding: 32px;
    @media (max-width: 1024px) {
      padding: 24px 20px;
    }
  }

  .icon {
    margin-bottom: 16px;
    width: 22px;
    height: 22px;
  }

  @media (max-width: 1024px) {
    flex-direction: column;
    text-align: center;
  }
}

```

```
#docs {
  border-right: 1px solid var(--border);

  @media (max-width: 1024px) {
    border-right: none;
    border-bottom: 1px solid var(--border);
  }
}

#next-steps ul {
  list-style: none;
  padding: 0;
  display: flex;
  gap: 8px;
  margin: 32px 0 0;

  .logo {
    height: 18px;
  }

  a {
    color: var(--text-h);
    font-size: 16px;
    border-radius: 6px;
    background: var(--social-bg);
    display: flex;
    padding: 6px 12px;
    align-items: center;
    gap: 8px;
    text-decoration: none;
    transition: box-shadow 0.3s;

    &:hover {
      box-shadow: var(--shadow);
    }
    .button-icon {
      height: 18px;
      width: 18px;
    }
  }

  @media (max-width: 1024px) {
    margin-top: 20px;
    flex-wrap: wrap;
    justify-content: center;

    li {
      flex: 1 1 calc(50% - 8px);
    }

    a {
      width: 100%;
      justify-content: center;
      box-sizing: border-box;
    }
  }
}

#spacer {
  height: 88px;
  border-top: 1px solid var(--border);
  @media (max-width: 1024px) {
    height: 48px;
  }
}

.ticks {
  position: relative;
  width: 100%;

  &::before,
  &::after {
    content: '';
    position: absolute;
    top: -4.5px;
    border: 5px solid transparent;
  }

```

```
  &::before {
    left: 0;
    border-left-color: var(--border);
  }
  &::after {
    right: 0;
    border-right-color: var(--border);
  }
}
```

src/App.jsx

```jsx
import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'

function App() {
  const [count, setCount] = useState(0)

  return (
    <>
      <section id="center">
        <div className="hero">
          <img src={heroImg} className="base" width="170" height="179" alt="" />
          <img src={reactLogo} className="framework" alt="React logo" />
          <img src={viteLogo} className="vite" alt="Vite logo" />
        </div>
        <div>
          <h1>Get started</h1>
          <p>
            Edit <code>src/App.jsx</code> and save to test <code>HMR</code>
          </p>
        </div>
        <button
          type="button"
          className="counter"
          onClick={() => setCount((count) => count + 1)}
        >
          Count is {count}
        </button>
      </section>

      <div className="ticks"></div>

      <section id="next-steps">
        <div id="docs">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#documentation-icon"></use>
          </svg>
          <h2>Documentation</h2>
          <p>Your questions, answered</p>
          <ul>
            <li>
              <a href="https://vite.dev/" target="_blank">
                <img className="logo" src={viteLogo} alt="" />
                Explore Vite
              </a>
            </li>
            <li>
              <a href="https://react.dev/" target="_blank">
                <img className="button-icon" src={reactLogo} alt="" />
                Learn more
              </a>
            </li>
          </ul>
        </div>
        <div id="social">
          <svg className="icon" role="presentation" aria-hidden="true">
            <use href="/icons.svg#social-icon"></use>
          </svg>
          <h2>Connect with us</h2>
          <p>Join the Vite community</p>
          <ul>
            <li>
              <a href="https://github.com/vitejs/vite" target="_blank">
```

```
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#github-icon"></use>
                </svg>
                GitHub
              </a>
            </li>
            <li>
              <a href="https://chat.vite.dev/" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#discord-icon"></use>
                </svg>
                Discord
              </a>
            </li>
            <li>
              <a href="https://x.com/vite_js" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#x-icon"></use>
                </svg>
                X.com
              </a>
            </li>
            <li>
              <a href="https://bsky.app/profile/vite.dev" target="_blank">
                <svg
                  className="button-icon"
                  role="presentation"
                  aria-hidden="true"
                >
                  <use href="/icons.svg#bluesky-icon"></use>
                </svg>
                Bluesky
              </a>
            </li>
          </ul>
        </div>
      </section>

      <div className="ticks"></div>
      <section id="spacer"></section>
    </>
  )
}

export default App
```

src/index.css

```
:root {
  --text: #6b6375;
  --text-h: #08060d;
  --bg: #fff;
  --border: #e5e4e7;
  --code-bg: #f4f3ec;
  --accent: #aa3bff;
  --accent-bg: rgba(170, 59, 255, 0.1);
  --accent-border: rgba(170, 59, 255, 0.5);
  --social-bg: rgba(244, 243, 236, 0.5);
  --shadow:
    rgba(0, 0, 0, 0.1) 0 10px 15px -3px, rgba(0, 0, 0, 0.05) 0 4px 6px -2px;

  --sans: system-ui, 'Segoe UI', Roboto, sans-serif;
  --heading: system-ui, 'Segoe UI', Roboto, sans-serif;
  --mono: ui-monospace, Consolas, monospace;

```

```
  font: 18px/145% var(--sans);
  letter-spacing: 0.18px;
  color-scheme: light dark;
  color: var(--text);
  background: var(--bg);
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;

  @media (max-width: 1024px) {
    font-size: 16px;
  }
}

@media (prefers-color-scheme: dark) {
  :root {
    --text: #9ca3af;
    --text-h: #f3f4f6;
    --bg: #16171d;
    --border: #2e303a;
    --code-bg: #1f2028;
    --accent: #c084fc;
    --accent-bg: rgba(192, 132, 252, 0.15);
    --accent-border: rgba(192, 132, 252, 0.5);
    --social-bg: rgba(47, 48, 58, 0.5);
    --shadow:
      rgba(0, 0, 0, 0.4) 0 10px 15px -3px, rgba(0, 0, 0, 0.25) 0 4px 6px -2px;
  }

  #social .button-icon {
    filter: invert(1) brightness(2);
  }
}

body {
  margin: 0;
}

#root {
  width: 1126px;
  max-width: 100%;
  margin: 0 auto;
  text-align: center;
  border-inline: 1px solid var(--border);
  min-height: 100svh;
  display: flex;
  flex-direction: column;
  box-sizing: border-box;
}

h1,
h2 {
  font-family: var(--heading);
  font-weight: 500;
  color: var(--text-h);
}

h1 {
  font-size: 56px;
  letter-spacing: -1.68px;
  margin: 32px 0;
  @media (max-width: 1024px) {
    font-size: 36px;
    margin: 20px 0;
  }
}
h2 {
  font-size: 24px;
  line-height: 118%;
  letter-spacing: -0.24px;
  margin: 0 0 8px;
  @media (max-width: 1024px) {
    font-size: 20px;
  }
}
p {
  margin: 0;
```

```
}

code,
.counter {
  font-family: var(--mono);
  display: inline-flex;
  border-radius: 4px;
  color: var(--text-h);
}

code {
  font-size: 15px;
  line-height: 135%;
  padding: 4px 8px;
  background: var(--code-bg);
}
```

src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```